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Master Thesis

The Role of Battery Energy Storage Systems in Germany's Energy Transition Toward Climate Neutrality by 2045

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Abstract

The transition of the German energy system towards climate neutrality is driven by the expansion of renewable energy sources such as photovoltaics (PV) and wind power. However, these technologies introduce challenges due to their weather-dependent generation patterns, increasing the need for flexibility in the energy system. Battery storage has emerged as a key technology to provide such flexibility, yet uncertainties remain regarding its deployment, cost development, used technologies, market participation and impact on the energy system. Existing studies present divergent assumptions and methodologies, particularly regarding capacity expansion and investment optimization and the role of different battery storage technology types. This thesis refines projections of utility-scale battery storage deployment and assesses the competition and synergies between various storage types such as utility-scale batteries, decentralized prosumer batteries and mobile electric vehicle (EV) batteries as well as investigating the key deployment drivers for utility-scale battery storage deployment. By using the open-source capacity expansion and dispatch model PyPSA-DE, the study provides a comprehensive and transparent analysis to support future grid planning and energy system design. The results show that batteries serve as key short-term flexibility assets, becoming the main source for balancing intra-day fluctuations from 2035 onward. The share of utility-scale batteries depends strongly on the scenario assumptions, as well as the substitution through prosumer home batteries and vehicle-to-grid (V2G)-enabled mobile battery storage. Cost sensitivities indicate a strong elasticity of utility-scale battery deployment with respect to capital cost changes, while increased V2G participation significantly reduces the total system costs. The findings demonstrate that substantial battery deployment can smooth extreme price peaks and significantly reduce market price volatility.

Zusammenfassung

Kernbestandteil der deutschen Energiewende hin zur Klimaneutralität ist der Ausbau erneuerbarer Energien wie Photovoltaik (PV) und Windkraft. Diese Technologien stellen das Stromsystem jedoch vor neue Herausforderungen, da ihre wetterabhängige Erzeugung ein höheres Maß an Flexibilität im Stromsystem erfordert. Batteriespeicher haben sich als Schlüsseltechnologie zur Bereitstellung dieser Flexibilität etabliert, dennoch bestehen weiterhin große Unsicherheiten hinsichtlich ihres zukünftigen Ausbaus, der Kostenentwicklung, der eingesetzten Batterietypen, der Marktintegration und ihrer Auswirkungen auf das Energiesystem. Bestehende Studien weisen teils stark divergierende Annahmen, Ergebnisse und methodische Ansätze auf, insbesondere in Bezug auf Investitionsoptimierung, Kapazitätserweiterung und die Rolle verschiedener Batteriespeichertechnologien. Diese Arbeit erweitert das bestehende Spektrum an Projektionen zum Ausbau von Großbatteriespeichern und untersucht die Konkurrenz und Synergien zwischen unterschiedlichen Speichertechnologien, darunter Großbatterien, dezentrale Prosumer Batterien und Vehicle-to-Grid (V2G)-fähige Elektroauto Batterien, sowie die zentralen Einflussfaktoren für die zukünftige Entwicklung von Großbatteriespeichern. Mithilfe des Open-Source-Modells PyPSA-DE zur Kapazitätserweiterungs- und Einsatzoptimierung wird eine umfassende und transparente Analyse durchgeführt, um zukünftige Netzplanungs- und Energiesystementscheidungen zu unterstützen. Die Ergebnisse zeigen, dass Batterien in erster Linie als Kurzzeit-Flexibilitätsoptionen dienen und ab 2035 zur wichtigsten Technologie werden, um Tagesschwankungen auszugleichen. Der Anteil der Großbatterien hängt dabei stark von den jeweiligen Szenarioannahmen sowie von möglichen Substitutionseffekten durch Prosumer-Heimspeicher und V2G-fähige Elektroauto Batterien ab. Kostensensitivitätsanalysen zeigen eine hohe Elastizität des Batteriespeicherbaus gegenüber Änderungen der Investitionskosten, während eine zunehmende V2G-Durchdringung die Gesamtkosten des Systems signifikant senkt. Die Ergebnisse zeigen zudem, dass ein umfassender Ausbau von Batteriespeichern Preisspitzen glätten und die Preisvolatilität am Strommarkt deutlich reduzieren kann.

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List of Acronyms

aFRR	automatic frequency restoration reserve
BEV	battery electric vehicle
BMS	battery management system
BNetzA	Federal Network Agency
CTO	Chief Technology Officer
DSM	demand-side management
E2P	energy-to-power
EV	electric vehicle
FCR	frequency containment reserve
LFP	lithium-iron-phosphate cathode
M&O	maintenance & operation
NEP	German Grid Development Plan
NMC	nickel–manganese–cobalt cathode
mFRR	manual frequency restoration reserve
PHS	pumped hydro storage
PV	photovoltaic
PyPSA	Python for Power System Analysis
RTE	round-trip efficiency
SoC	state of charge
TSO	Transmission System Operator
TYNDP	Ten-Year Network Development Plan
TB	top-bottom
V2G	vehicle-to-grid
VRE	variable renewable energy

1. Introduction

The German energy system is undergoing a fundamental transformation, driven by the urgent need to mitigate climate change. This shift necessitates a transition from conventional fossil fuel power plants to renewable energy sources. Germany reached a renewable electricity share of more than 60% for the first time in 2024 and has set the targets of achieving an 80% renewable share by 2030 and climate neutrality by 2045 (Luderer et al. 2025). The vast majority of these renewables will be provided by variable renewable energy (VRE) sources, primarily PV and wind power. However, this transition also brings new challenges. Unlike traditional power plants, which can be dispatched according to demand, variable renewable energy generation is highly dependent on weather conditions. As a result, the need for flexibility in the power system is increasing significantly (Luderer et al. 2025).

1.1. Problem Statement

Battery storage systems have emerged as a key solution for providing flexibility. They are widely considered an essential component of a future climate-neutral energy system, particularly one that relies heavily on PV and wind power. In recent years, the demand for battery storage has surged, driven by declining costs and the increasing need for system flexibility. This trend is reflected in the current exponential rise in grid connection requests for utility-scale battery storage (Enkhardt 2025) with a planned commissioning year before 2030, which now far exceed the projections until 2030 outlined in various studies, as illustrated in Figure 2.4.

However, existing studies on battery storage deployment present divergent assumptions and results, which are discussed in more detail in Chapter 2.2. Some studies lack transparency regarding their cost and technology assumptions, making it difficult to compare findings. Most of the studies focus on the entire sector-coupled energy system and therefore are not placing battery storage as the central subject of investigation which results in no comprehensive in-depth analysis regarding battery storage deployment and their key deployment driver. Different types of battery storage, such as stationary utility-scale storage, home PV battery storage, and mobile vehicle-to-grid-enabled storage are often not all included or clearly distinguished or comprehensively analyzed. Furthermore, these studies assume a purely market-oriented optimization and neglect potential additional revenue streams for battery storage, such as future capacity markets or ancillary system services. Moreover, key performance indicators such as the installed power capacity, en-

ergy capacity, or annual discharge are often reported inconsistently, which further hinders comparison.

The scenario framework for the 2023 German Grid Development Plan (NEP) does not rely on studies which include investment modeling for battery storage but instead applies a different methodological approach which relies mainly on the expansion of PV and wind power (NEP Szenariorahmen 2023). The 2025 draft scenario framework (NEP Szenariorahmenentwurf 2025) for the upcoming NEP notes that its assumptions regarding utility-scale battery storage are not based on top-down capacity optimization either, instead focusing on existing grid connection requests and market surveys. The report highlights that the assumed deployment of power and energy capacities is not supported by any further detailed analyses from the transmission system operators (TSOs) and that, from today's perspective, the future deployment of utility-scale batteries remains highly uncertain. This further underscores the necessity of deeper investigation into the key drivers influencing battery development and possible deployment pathways. Comparing these approaches with investment modeling and capacity expansion techniques might be helpful complementing the bottom-up reasoning in the NEP.

1.2. Research Question & Objectives

The objective of this thesis is to systematically investigate the role of utility-scale battery energy storage systems in Germany's transition toward a climate-neutral energy system by 2045.

1.2.1. Research Question

The central research question that is guiding this thesis is the following:

What role will utility-scale battery energy storage systems play in Germany's energy transition toward climate neutrality by 2045 and what drives their deployment?

This question is further specified through the objectives mentioned below.

1.2.2. Objectives

The primary goal of this thesis is to outline possible pathways and scenarios for the deployment of utility-scale battery energy storage systems. It aims to explore various battery-related metrics in depth, highlight their interdependencies, and investigate cost sensitivities as well as the elasticity of battery demand. Further, the interplay and substitution effects through expansion of other battery technology types, especially prosumer home batteries and V2G-enabled EVs will be analyzed.

This thesis addresses four interrelated objectives:

1. **Quantify deployment pathways for different battery storage technologies.**
Develop scenarios for the expansion of utility-scale, decentralized, and mobile (V2G-enabled) battery systems using the open-source capacity expansion model PyPSA-DE. The analysis focuses on the evolution of installed power and energy capacities, annual electricity supply through batteries and other key battery performance indicators while also determining the system's flexibility needs across target years.
2. **Analyze substitution effects between battery types and other flexibility options.**
Investigate the extent to which other battery types, such as prosumer home batteries and V2G-enabled EVs, can substitute utility-scale battery storage. Furthermore, evaluate the contribution of other energy storage technologies and flexibility options to providing flexibility needs.
3. **Assess cost sensitivities and the resulting elasticity of utility-scale battery deployment.**
Examine how variations in investment costs (CAPEX per kW and per kWh) affect the endogenous deployment of battery storage and identify cost thresholds that determine whether battery expansion becomes elastic or inelastic. Additionally, analyze which technologies substitute utility-scale batteries under high-cost conditions and which are substituted by them in low-cost scenarios.
4. **Evaluate system-level impacts of battery deployment.**
Analyze how changes in battery costs and deployment levels influence renewable curtailment, electricity price volatility, total system costs and the endogenous expansion of wind and PV capacities, with a particular focus on how price volatility affects the economic viability and investment attractiveness of utility-scale storage systems.

Scenarios for battery storage development in the German energy system up to 2045 will be created and analyzed using scientifically robust assumptions and the established energy system modeling framework PyPSA-DE. In addition to these quantitative objectives, the thesis provides a structured review and comparison of existing German energy system studies, highlighting differences in assumptions and methodological approaches regarding battery modeling. Further, key deployment driver for utility-scale batteries are developed to systematically evaluate the results of the model and the model limitations. By combining the qualitative assessment with model-based scenario analysis, this thesis aims to provide a transparent and reproducible foundation for future grid- and energy system planning regarding battery storage.

1.3. Structure of the Thesis

The structure of this thesis is designed to gradually build up from a general introduction towards a detailed analysis and discussion of the model results. Chapter 1 introduces the topic by formulating the problem statement and research objectives that guide this study. Chapter 2 provides an overview of the current status of battery deployment in Germany followed by an overview of existing studies that describe possible future deployment pathways for battery storage in Germany, which are systematically reviewed and compared. Afterwards, several key driver for utility-scale battery deployment are identified and examined. Chapter 3 outlines the methodological framework, focusing on the PyPSA-DE modeling environment and how the key driver are included in the model. Further, the developed scenarios, together with the modeling assumptions that form the basis of the analysis, are presented. Subsequently, Chapter 4 introduces the model results and highlights the key findings. These findings are further interpreted in Chapter 5, where they are compared to existing literature and discussed in a broader context. Finally, the last Chapter 6 derives a conclusion, provides an outlook, and formulates future research topics. Additional results and figures are included in the appendix to complement the main analysis.

2. State of the Art and Key Deployment Drivers

This chapter provides the theoretical and empirical foundation for the subsequent model-based analysis. It begins with an overview of the current status of battery storage deployment in Germany, highlighting recent trends and developments across different storage types. This is followed by a structured review of state-of-the-art studies that examine future battery deployment pathways, aiming to identify research gaps and establish a consistent data basis for the modeling framework. Finally, the chapter outlines and discusses the key drivers influencing the deployment of utility-scale battery storage, which serve as the conceptual link between the literature review and the modeling approach applied in the following chapters.

2.1. Current Battery Deployment in Germany

As shown in Figure 2.1, approximately 13.7 GW of stationary battery storage is currently installed in Germany. The majority of this capacity consists of home storage systems (11.2 GW), while utility-scale batteries (2.1 GW) and commercial/industrial systems (0.5 GW) account for a substantially smaller share. Figure 2.2 further shows the presently installed battery energy capacity of 21.5 GWh in total. Here too, home storage dominates with 17.6 GWh, whereas utility-scale batteries (3 GWh) and commercial/industrial storage (0.9 GWh) contribute significantly less. From this follows that home storage systems are currently installed with an average energy-to-power (E2P) ratio of about 1.6 hours, and utility-scale batteries with an average of 1.4 hours. When also including EV batteries, they constitute by far the largest share: at present, roughly 1.84 million EVs with a storage capacity of 115 GWh are registered in Germany (Figgenger et al. 2025a).

Currently, lithium iron phosphate (LFP) batteries dominate the stationary storage market with roughly 90% market share, as shown in Figure 2.3. LFP batteries are durable, safe, and cost-effective compared with nickel manganese cobalt (NMC) lithium-ion battery cells, making them well-suited for stationary applications (Geladen Podcast 2025b). NMC lithium-ion batteries are currently only used to a small degree in stationary systems. Their primary use case remains as Electric Vehicle Batteries due to their higher energy density, which is important for the vehicles range. Looking ahead, to around 2030, sodium-ion batteries are expected to see broader deployment in stationary storage, as also indicated in the figure. In China, the first utility-scale, grid-forming sodium-ion storage project was completed this year (Shaw 2025). Sodium-ion systems could prove

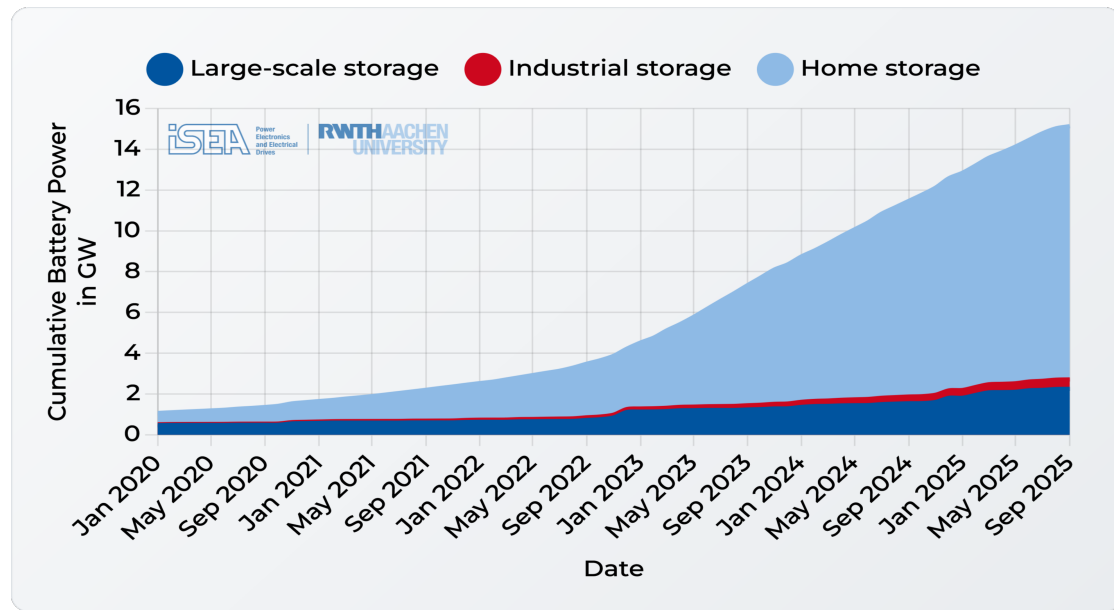


Figure 2.1.: Deployment of installed battery power capacity in Germany

Source: Figgenger et al. 2025a

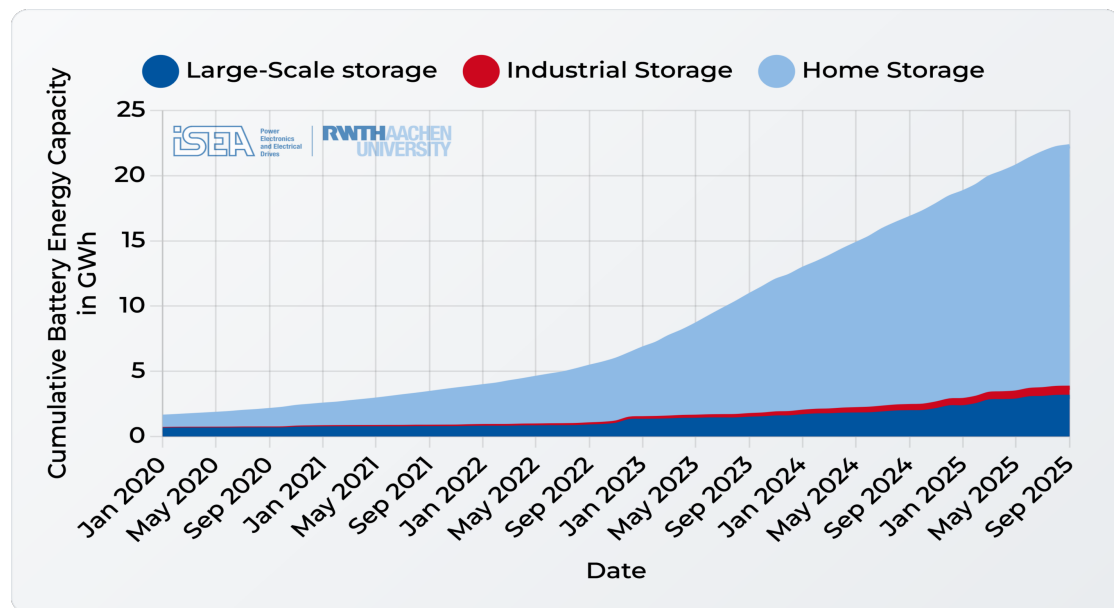


Figure 2.2.: Deployment of installed battery energy capacity in Germany

Source: Figgenger et al. 2025a

significantly cheaper and offer favorable cycle-life characteristics for stationary use, which would further strengthen their competitiveness (Maisch 2024).

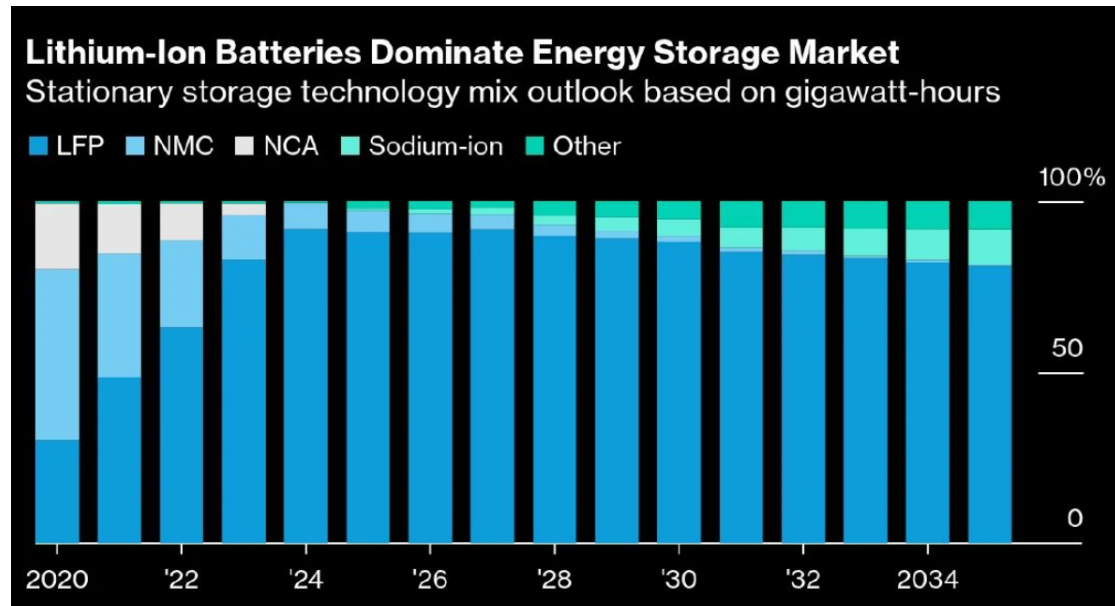


Figure 2.3.: Global lithium-ion battery cell chemistries for stationary battery storage

Source: BloombergNEF 2025

2.2. Overview of Existing Studies and Research on Battery Storage Expansion

In this part of the thesis different studies which have investigated deployment pathways of battery storage in Germany are shown, classified and compared.

2.2.1. Energy System Studies

Several studies have explored capacity expansion and investment optimization with endogenous expansion of battery storage, including:

- Agora Energiewende - Klimaneutrales Deutschland (Agora Energiewende 2021)(Agora Energiewende 2024)
- Langfristszenarien für die Transformation des Energiesystems in Deutschland (Langfristszenarien 2024)
- Fraunhofer ISE - Wege zu einem klimaneutralen Energiesystem (Fraunhofer ISE 2021)(Fraunhofer ISE 2024)

- Ariadne-Report - Deutschland auf dem Weg zur Klimaneutralität 2045 (Kopernikus-Projekt Ariadne 2021)
- Ariadne-Report - Die Energiewende kosteneffizient gestalten - Szenarien zur Klimaneutralität 2045 (Luderer et al. 2025)
- e.venture - Stromverbrauch bis 2045 (e.venture 2024)
- What-where-when: Investigating the role of storage for the German electricity system transition (Thimet and Mavromatidis 2023)

These studies share a common approach: they apply capacity expansion and dispatch models to explore the future development of the German energy system, with endogenous determination of key variables, including installed battery power and energy capacity. The results regarding battery deployment are illustrated in Figure 2.4. Each study develops a set of scenarios to explore different development paths, usually focused on key target years leading up to 2045, in line with Germany’s climate neutrality goal. While the ultimate target year is the same across all studies, the intermediary milestone years and scenario assumptions vary. A sector-coupled energy system is modeled, rather than a pure electricity system, with the overall energy system typically divided into the sectors of electricity, industry, heating and transport, though the specific classification may vary slightly between studies.

Both open-source and commercial models are used across the literature. An overview of the models used in the studies mentioned above for battery deployment optimization is presented in Table 2.1. The cost assumptions used as exogenous model inputs, which have a significant impact on model results, are only partially published, and the detailed model outputs and the model codebase are often not publicly available. Missing information on cost assumptions and detailed model results related to battery storage was requested from the study authors. However, no additional data were provided. This lack of transparency makes the results difficult to reproduce and hinders meaningful comparison across studies.

The studies assume an energy-only market framework with system-cost minimization and do not incorporate other possible revenue streams for battery storage systems. Other potential battery markets such as for ancillary services, future capacity markets, or industrial peak shaping are excluded from consideration. This limits the scope of analysis regarding the full potential and application diversity of battery storage technologies.

A categorization of the different types of battery storage systems as well as the mentioned units in the results for battery deployment used across the studies can be found in Table 2.2. The units in the results are inconsistent, and the classification of battery systems varies, making direct comparison of the results difficult.

Table 2.1.: Model overview and publication status

Study	Model (Open-Source or Commercial)	Cost Assumptions Published?	As- sumptions Published?	Detailed Model Re- sults Pub- lished?
Agora 2024	Commercial (Prognos)	Not found (re- quested)		No
Fraunhofer ISE 2021/2024	Partially (REMod)	Open	Partially (re- quested)	No
Langfristszenarien O45	Commercial (En- ertile)	Open	Partially (re- quested)	No
Ariadne Report 2021	Partially (mainly REMod)	Open	Yes	Yes
Ariadne Report 2025	Open-Source (PyPSA)		Yes	Yes
e.venture – Consumption 2045	Commercial (e.venture)		Not found (re- quested)	No
Thimet and Mavromatidis 2023	Partially (MANGOelec)	Open	Yes	Yes, mostly

Table 2.2.: Reported model result metrics and classification of battery storage system types in various studies. (x) = parentheses mean indirectly stated through other parameters.

Study	Reported Units				Classification
	GW	GWh	TWh/yr	E2P-ratio	
Agora 2024	×		×		Utility-scale, pumped hydro & V2G combined; home-PV excluded
Fraunhofer ISE 2021/24		×			Stationary & mobile combined
Langfristszenarien O45	×	(x)		×	Home-PV & utility- scale separated
Ariadne Report 2021	×	(x)		×	Stationary & mobile separated

Study		Reported Units				Classification
		GW	GWh	TWh/yr	E2P-ratio	
Ariadne	Report 2025	×	×	×	×	Utility-scale, home, mobile separated (V2G not modelled)
e.venture	2045		×			Home-PV, utility-scale & V2G mobile (no V2G results)
Thimet & Mavromatidis	2023	×	×		×	Total Battery Storage
NEP SzR	2023	×	(x)		×	Stationary large-scale and decentralized battery storage
NEP SzR	2025	×	(x)		×	Stationary large-scale and decentralized battery storage
NEP SzR BNetzA Approval	2025	×	(x)		×	Stationary large-scale and decentralized battery storage

Table 2.3 presents exogenous model input assumptions for utility-scale and home battery storage. These include investment costs per installed kW and kWh, maintenance and operation (M&O) costs per stored kilowatt-hour, as well as assumptions on lifetime and round-trip efficiency (RTE). This comparison serves as the data basis for the battery input cost assumptions applied in this thesis. The cost assumptions are further evaluated against current data from utility-scale battery projects under construction in Germany, as discussed later in this chapter. Among the reviewed studies, the Ariadne Reports published in 2021 and 2025 apply the most optimistic input data for utility-scale batteries, featuring the lowest investment costs and the highest lifetime and efficiency values.

Table 2.3.: Exogenous model input assumptions for utility-scale battery storage

Study	Invest. (€/kWh)	Invest. (€/kW)	M&O (€/kWh)	Lifetime	Efficiency (RTE)
Langfristszenarien O45 – Heimspeicher	2025: 350 2045: 252	-	2025: 14 2045: 10.084	15 Jahre	90%
Langfristszenarien O45 – Großspeicher	2025: 250 2045: 158	-	2025: 10 2045: 6.3	15 Jahre	90%
Ariadne Projekt 2021 (Utility-scale)	2025: 187 2030: 142 2035: 118 2040: 94 2045: 84,5 2050: 75	2025: 215 2030: 160 2035: 130 2040: 100 2045: 80 2050: 60	0,25% of the Invest. Costs in 2025 and 0,9% in 2050	Battery inverter: 10 years Battery storage: 22,5 years in 2025 and 30 years in 2050	96 %
Ariadne Projekt 2025 (Utility-scale)	2025: 198 2030: 150 2035: 125 2040: 100 2045: 89 2050: 80	2025: 228 2030: 169 2035: 138 2040: 106 2045: 85 2050: 64	0,25% of the Invest. Costs in 2025 and 0,9% in 2050	Battery inverter: 10 years Battery storage: 22,5 years in 2025 and 30 years in 2050	95,5%

Table 2.3.: Exogenous model input assumptions for utility-scale battery storage

Study		Invest. (€/kWh)	Invest. (€/kW)	M&O (€/kWh)	Lifetime	Efficiency (RTE)
Ariadne 2025 battery)	Projekt (Home- battery)	2025: 280 2030: 215 2035: 180 2040: 144 2045: 130 2050: 115	2025: 321 2030: 241 2035: 197 2040: 153 2045: 123 2050: 93	0,25% of the Invest. Costs in 2025 and 0,9% in 2050	Battery inverter: 10 years Battery storage: 22,5 years in 2025 and 30 years in 2050	95,5% in 2025, 96% ab 2030
Fraunhofer 2021 – Battery	ISE Stationary Battery	2020: 400 2025: 275 2030: 150 2035: 138 2040: 125 2045: 113 2050: 100	-	1% der install. Kosten z.B. 2025 = 2.75 €/kWh	2020 bis 2025: 10 Jahre ab 2030: 15 Jahre	2020: 85.8% 2050: 90%
Thimet 2023	and Mavromatidis	2025: 393.8 2030: 238.4 2035: 163.7 2040: 136.2 2045: 122.9 2050: 112	-	–	15 years	2025: 85.5%, 2050: 94%
Agora 2024		–	–	-	–	–
e.venture Studie		–	–	-	–	–

2.2.2. Scenario Framework of the German Grid Development Plan

The most recent scenario frameworks of the German grid development plan are further evaluated as they present alternative approaches for pathways of utility-scale battery deployment in Germany until 2045, including:

- Szenariorahmen zum Netzentwicklungsplan Strom 2037 mit Ausblick 2045, Version 2023 (NEP Szenariorahmen 2023)
- Szenariorahmen zum Netzentwicklungsplan Strom 2037/2045, Version 2025 (NEP Szenariorahmenentwurf 2025)
- Genehmigung des Szenariorahmens für den Netzentwicklungsplan Strom 2025-2037/2045 (Bundesnetzagentur 2025)

The scenario framework draft for the Grid Development Plan uses a significantly different methodology for assessing battery storage development pathways than energy system studies, since it does not apply endogenous investment modeling. This section briefly presents the 2023 and 2025 scenario framework drafts of the Grid Development Plans in regard to battery storage. The drafts of the scenario framework were chosen because the methodology used is explained there in the greatest detail. The NEPs from 2023 and 2025 are discussed because they are the most recent, and earlier NEPs focused on operation in the ancillary services market rather than market-oriented operation for battery storage. In addition to the two drafts, the current approval of the NEP scenario framework 2025 by the Federal Network Agency (BNetzA) is presented, as the build-out figures for utility-scale battery storage were significantly adjusted there.

In the scenario framework draft of the NEP 2023, battery storage is divided into residential PV storage and utility-scale battery storage. The development of installed power capacity is tied to the prescribed expansion of wind and PV capacity. An E2P ratio of 2 is assumed. The build-out of utility-scale batteries is aligned with the expansion of ground-mounted PV plants. Assuming a linear PV build-out, from 2030 roughly one third (30%) of the newly added installations will be equipped with utility-scale batteries. From 2035 the share rises to 70%. From 2040 every newly added ground-mounted PV plant is equipped with a utility-scale battery. Existing assets and planned installations are taken into account in all scenarios additionally. Residential PV battery storage is linked to the build-out of building-integrated PV systems. While in the next few years only just over half of all new rooftop PV systems will be installed together with an associated storage system, this share increases to 100% by 2035. As a result, the installed power capacity of decentralized residential PV storage amounts, in all scenarios, to between 38–47% of the installed power capacity of rooftop PV systems.

In the NEP 2025 scenario framework draft, the methodology for assessing battery-storage development pathways changes. The division of battery storage into utility-scale and small-scale storage (called residential PV storage in the NEP 2023 scenario framework)

remains. The assessment of utility-scale batteries for 2037 and 2045 is based largely on projects that have been registered with the TSOs or captured through a market survey. A large share of the reported utility-scale batteries is planned with an energy-to-power ratio of 2 hours and is scheduled to go into operation by 2028. Given the expectation of falling costs for battery storage capacity and market-driven flexibility needs, it is assumed that from 2037 installed power increases only slightly, while the energy capacity of utility-scale batteries continues to expand. The average energy-to-power ratio for utility-scale batteries increases to 4 hours by 2045, aligning with assumptions from the Ten-Year Network Development Plan (TYNDP) 2024 (ENTSO-E and ENTSO-G 2025). However, in the BNetzA's approval of the NEP scenario framework, this view is no longer shared: there, energy capacity does not increase further between 2037 and 2045, so the E2P ratio remains at two hours. Existing assets and the reported projects are, as described above, considered with site-specific detail in all scenarios. Apart from the available grid-connection requests and the market survey, the TSOs have not provided more in-depth analyzes as a basis for the assumptions about the development of power and energy capacity of battery storage. Starting from the existing stock, a further build-out of small-scale battery storage is assumed. Their expansion and spatial distribution are tied exclusively to the installation of new rooftop PV systems. By 2037, rooftop PV systems will be added together with batteries capable of storing roughly one hour of the PV system's peak output (1 kWh/kWp). Thereafter, significantly more storage capacity is added up to 2045, so that the ratio rises to 1.3 kWh/kWp in Scenarios A and B. In Scenario C for 2045, by contrast, more rooftop PV is added than small-scale batteries, causing the ratio to fall below 1 kWh/kWp. It is assumed that the average ratio of storage capacity to installed power for small-scale batteries is 2.5 kWh/kW. In the scenarios, the operating strategy of small-scale batteries is assumed to be either oriented toward maximizing self-consumption or increasingly market-oriented.

2.2.3. Meta Study

The expected deployment pathways of battery storages until 2045 of different studies are shown in Figure 2.4. The results differ significantly across studies.

The Ariadne Report (2021) assumes an E2P-ratio of 1 hour, resulting in an estimated 150 GW of installed power and 150 GWh of energy capacity by 2045 for stationary battery storage as well as for mobile battery storage. In contrast, the Ariadne Report (2025) presents significantly different results: in the "Technology Mix" scenario for 2045, 55 GW of installed power and 450 GWh of energy capacity of utility-scale batteries are expected, reflecting a substantial change in the assumed E2P-ratio. Almost no Home Batteries are to be installed in the scenario and the V2G option was not available. The Agora 2024 report, on the other hand, expects about 60 TWh per year of discharge from battery storage and 102 GW of installed power capacity by 2045. Utility-scale batteries, pumped hydro storage (PHS) and mobile V2G batteries are combined as total battery storage power capacity. No further information is provided on the E2P-ratio or installed energy

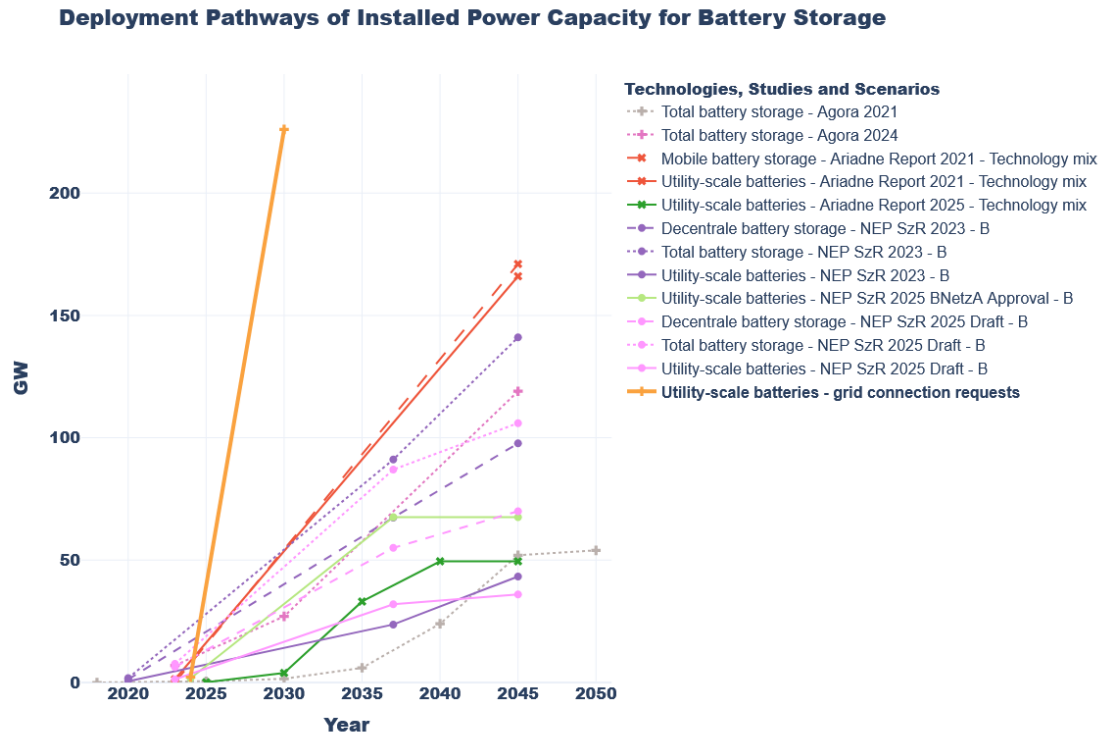


Figure 2.4.: Selection of studies with development pathways for battery storage in Germany. Categorization into total battery storage, decentral battery storage, utility-scale battery storage, and mobile battery storage. Additionally the current grid connection requests for utility-scale batteries in 2025 are added.

capacity. In the Agora 2021 report, 52 GW of total installed battery power capacity were expected in 2045. The e.venture study reports 202 GWh (low scenario) and 275 GWh (high scenario) of installed capacity by 2045, without specifying an E2P-ratio or installed power. The Langfristszenarien report assumes a E2P-ratio of 2 hours, but does not provide clear quantitative results. It specifies an exogenous addition of 4 GW of utility-scale battery storage starting in 2031, with little further endogenous expansion. The research paper from Thimet & Mavromatidis from 2023 assumes an E2P ratio of 3.3 Hours in 2045 and around 48 GW of installed power capacity as well as 158.4 GWh of installed energy capacity of total battery storage. No further distinction between large/small or stationary/mobile battery storage was being made. The Fraunhofer ISE Study (2024) estimates a need for short-term flexible electricity storage of approximately 350 GWh to 600 GWh by 2045, which is expected to be provided by stationary battery storage and V2G systems.

According to the NEP 2023 scenario framework, Scenario B/C results in 97.8 GW of

residential PV battery systems and 57.1 GW of utility-scale battery storage in 2045. In the NEP 2025 scenario framework, Scenario B projects 70 GW / 175 GWh of small-scale battery storage and 36 GW / 143 GWh of utility-scale battery storage by 2045. According to the approval of the NEP scenario framework 2025 by the BNetzA, the expected installed power capacity of utility-scale batteries in 2045 is 67.6 GW and 134.5 GWh of installed energy capacity in Scenario B.

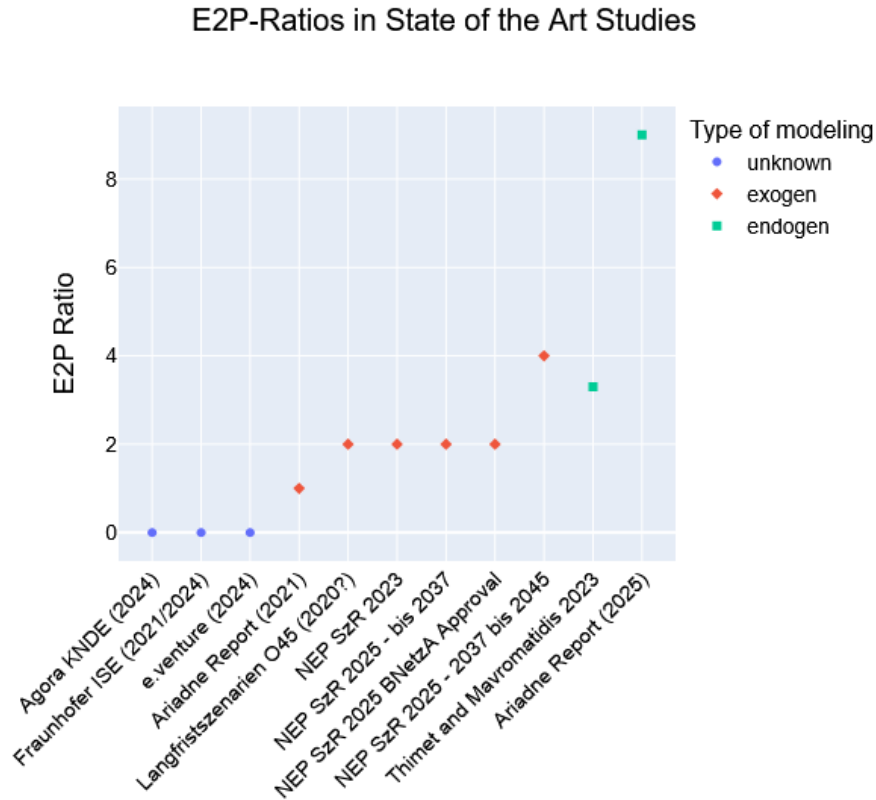


Figure 2.5.: Assumed E2P-ratios in the studies

The assumed E2P-ratios and their type of modeling in the studies are shown in Figure 2.5. Most studies use an exogenous approach by setting a fix E2P-ratio based on external data and assumptions. The endogenous optimized E2P-ratios tend to be higher than the exogenous. For the endogenous optimized E2P-ratios, it is unclear if the E2P-ratio changes in the different target years. Further, for most studies, there is no differentiation between decentralized storages and utility-scale storages regarding the E2P-ratio (exception are the NEPs).

2.3. Key Drivers for Utility-scale Battery Deployment

To develop pathways for utility-scale battery storage deployment, a range of influencing factors and interdependencies must be considered. Based on a comprehensive review of the current literature, eleven key driver for utility-scale battery deployment were identified and grouped into four overarching categories.

The first category concerns the need for system flexibility. Historically, conventional power plants such as coal and gas units were capable of adjusting their electricity generation to match demand, thereby reducing the requirement for energy storage. In contrast, variable renewable energy sources such as solar PV and wind are intermittent and weather dependent, which significantly increases the need for storage to ensure that demand can be reliably met at all times. Flexibility needs can be distinguished across different temporal dimensions like daily, weekly, and annual, allowing a clearer differentiation between the roles of short-term and long-term storage technologies. Batteries are generally classified as short-term storage solutions as further addressed later in this chapter.

When assessing the potential role of batteries in this context, it is essential to consider competing flexibility options. These include other types of batteries, such as prosumer home batteries and V2G solutions, as well as alternative storage technologies like PHS, hydrogen, or thermal/heat storage. DSM options, such as smart charging of EVs, flexible operation of heat pumps, or hydrogen production through electrolysis, also contribute to system flexibility. Moreover, cross-border electricity trade plays a role in balancing supply and demand.

The third key dimension is economic drivers for battery deployment. The competitiveness of batteries relative to other flexibility technologies depends on their profitability and business case. Thus, cost trajectories and technological maturity for utility-scale batteries must be evaluated alongside the market frameworks in which they operate and the specific services they can provide. Particular attention must be paid to the phenomenon of self-cannibalization, where increasing battery penetration leads to diminishing returns and reduced profitability for individual assets. Technical parameters such as lifetime, round-trip efficiency, and the E2P ratio also play a critical role, as they determine both the economic attractiveness and the potential applications of battery storage.

Finally, the regulatory framework and policy environment must be taken into account. Current and future regulatory decisions are pivotal in shaping the economic viability of different technologies and can either accelerate or hinder the deployment of utility-scale batteries. In this sense, regulatory frameworks not only influence investment decisions but also guide the overall trajectory of energy system transformation.

Each of the following key drivers is explained in more detail below:

1. **The Need for System Flexibility:**
 - Expansion of VRE, especially wind and PV
2. **Competitive Technologies:**
 - The development and expansion of other battery types
 - Other storage options
 - Other flexibility options
 - Import/export capacities
3. **Economic Drivers:**
 - Cost developments / trajectories
 - Markets & use cases
 - Economic viability & self-cannibalization effects
 - Technical parameters
4. **Policy and Regulatory Framework**
 - Grid fees, taxes and levies
 - Smart grid infrastructure

2.3.1. The Need for System Flexibility

Expansion of variable renewables (wind, PV)

This subsection briefly outlines how the increasing share of VRE, drives the need for flexibility in the German energy system. Several studies have highlighted important thresholds for the deployment of storage technologies. Both Cebulla et al. 2018 and Thimet and Mavromatidis 2023 argue that once the VRE share surpasses approximately 65–67%, investment in energy storage grows disproportionately, with capacity expansion accelerating rapidly beyond this point. Specifically, Thimet and Mavromatidis 2023 shows that utility-scale battery storage begins to scale significantly once VRE penetration exceeds 65% in the modeled system. This observation is in line with current developments in Germany, where a large number of grid connection requests for storage projects are concentrated in the period between 2026 and 2030, suggesting that the markets already anticipate passing this threshold in the near future. The type of VRE mix also has a decisive influence on the required amount of storage. Thimet and Mavromatidis 2023 suggests that PV-dominant systems demand higher levels of energy storage capacity compared to wind-dominant systems, since short-term storages such as batteries are particularly well-suited to balancing fluctuations between day and night. In parallel, increasing variability on the supply side, coupled with limited flexibility on the demand side, leads to higher market price volatility. Energy storage systems, by providing flexibility, can mitigate these effects and contribute to more stable electricity market prices. This dual role, enabling higher shares of VRE and stabilizing market outcomes, emphasizes the growing importance of battery storage in the German energy transition.

2.3.2. Competitive Technologies

The development and expansion of other battery types

Beyond utility-scale batteries, two other battery types are expanding and will be analyzed in more detail in this thesis: First, prosumer home batteries, which are often paired with rooftop PV and second, mobile battery storage in EVs, increasingly with the prospect of bidirectional charging (V2G). Both can provide short-duration flexibility and, if effectively integrated into markets and tariffs, may substitute part of the services otherwise delivered by stationary utility-scale batteries. Current policy documents (Bundesministerium für Wirtschaft und Energie 2023) emphasize the need to enable system-oriented operation of small PV batteries and EVs (especially with V2G) through appropriate technical standards and regulatory incentives (e.g., time-variable tariffs and limiting levies to storage losses), precisely because these resources interact with and can partially replace other flexibility options. The policies are more detailed below in the policy framework part.

Prosumer home batteries as well as EV Batteries and V2G are discussed in more detail at the end of this chapter.

Other storage options

Utility-scale batteries compete within a broader portfolio of electricity, heat, and hydrogen storage options who are often divided into short-term and long-term storages. In line with recent system overviews, “short-term” denotes discharge durations of a few hours, whereas “long-term” covers multi-week to seasonal shifts (Schill et al. 2025). This distinction maps closely to technologies and their cost structures (Cebulla et al. 2018). An overview of the different storage types classified by their duration time was recently published by an Ariadne background report on energy storages as depicted in Figure 2.6. Electrical energy storages like batteries are usually considered short-term storages (Bundesministerium für Wirtschaft und Energie 2023). Further, pumped hydro storage is usually considered a short-term storage, which was the biggest storage capacity in Germany until this year. But further capacity expansion potential is limited in Germany and therefore only a small expansion is expected in the coming years (Bundesministerium für Wirtschaft und Energie 2023). Thermal- and Hydrogen Storages are generally considered long-term storages, as they tend to have lower costs per kWh stored energy but also lower efficiency (higher losses) compared to battery storage. But there are certain applications of them which are considered short-term storages as is shown in Figure 2.6.

In Thimet and Mavromatidis 2023, battery storage and PHS are operated extensively as daily flexibility resources, whereas hydrogen salt-cavern storage acts exclusively as seasonal storage. This study projects substantial hydrogen-based storage energy, approximately 2.7–5.9 TWh by 2050 across scenarios, with storage build-out accelerating after 2030 as VRE rise. Prior studies have considered wider portfolios of storage technologies, including multiple battery chemistries, PHS, adiabatic compressed air energy

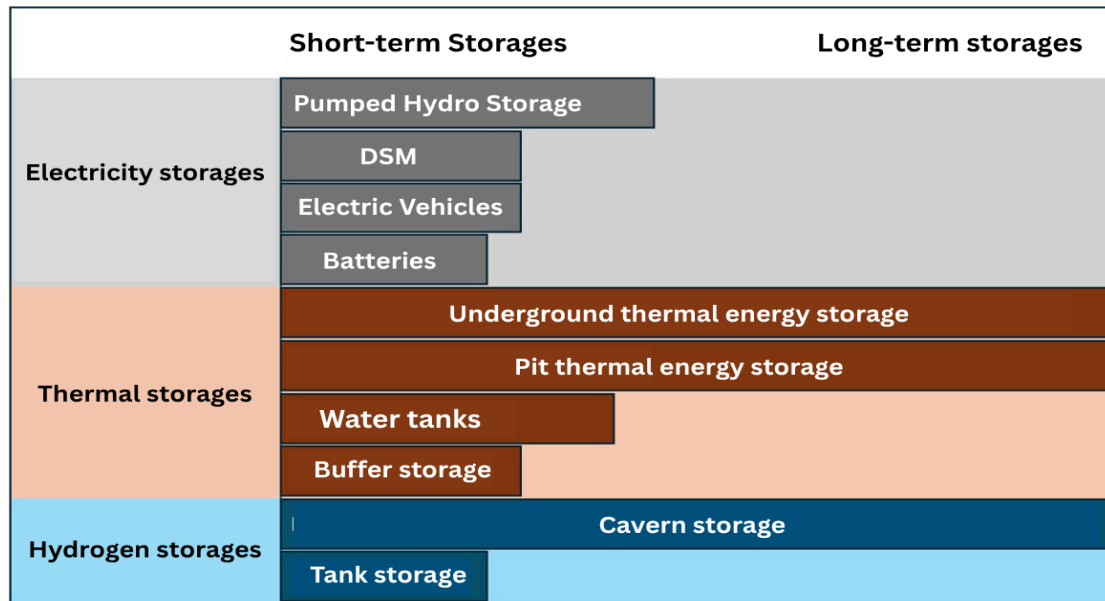


Figure 2.6.: Systematic overview of energy storages classified into their duration time

Source: Schill et al. 2025, translated into english

storage, and power-to-gas hydrogen pathways (Thimet and Mavromatidis 2023).

Other flexibility options

Further, batteries are not only in direct competition with alternative storage technologies but also with non-storage flexibility, especially DSM, flexible power plants on the supply side and cross-border exchange, with the optimal allocation determined by costs, technical parameters, and market design. In this thesis, “flexibility” is defined as the capability of the energy system to balance the residual load across timescales from minutes to seasons. In line with Cebulla et al. 2018, flexibility can be provided by networks (transmission and distribution grid), supply-side measures (flexible plants and purposeful curtailment of VRE), demand-side management (including electrified loads such as electro-heating/cooling, EV smart-charging, and power-to-gas), and electrical energy storage. Schill et al. 2025 adopts an encompassing definition that includes “all technologies and measures that help align supply and demand,” which is consistent with the residual-load framing and provides a common language for comparing options in this study. On the supply side, short-notice flexibility can be delivered by gas-, coal- and, prospectively, hydrogen-power plants, while controlled VRE curtailment reduces extreme peaks and eases integration at low cost. On the demand side, DSM shifts consumption to periods of abundant renewables and low prices. Next to the aforementioned options, DSM technologies include smart charging of EVs, heat pumps, electrolyzers for green hydrogen production and flexible industrial processes which are thereby dampening price

volatility and reducing storage requirements. Flexibility needs rise markedly with higher VRE shares. Therefore a portfolio spanning storage, DSM, sector coupling and cross-border exchange is required to balance both intraday and seasonal patterns, especially in prolonged “Dunkelflaute” episodes (Jürgens et al. 2024; Luderer et al. 2025).

Import/export capacities

As already mentioned above also cross-border electricity exchange provides an important source of system flexibility by smoothing generation and demand peaks across larger geographic areas, thereby reducing curtailment, mitigating price spikes, and lowering the need for domestic balancing resources (Luderer et al. 2025). The effectiveness of this mechanism depends primarily on interconnector capacities and on the technology mix and flexibility portfolio of neighboring countries (e.g., shares of VRE, hydro/PHS, nuclear, demand response, and storage).

2.3.3. Economic Drivers

Cost developments / trajectories:

In recent years, the cost of batteries has experienced a steep decline, significantly improving their competitiveness in energy systems. Looking ahead to 2045, however, the future cost trajectory of batteries remains highly uncertain. As shown in Table 2.3, cost assumptions in different energy system studies vary widely, but all agree on the expectation of further reductions. Energy system studies typically consider the total project costs of installing one kilowatt-hour or one kilowatt of battery capacity. In contrast, most other technology-focused studies report either battery cell or module costs. Since the battery modules represent the largest share of total project costs, the latter can still serve as a useful benchmark for assessing broader cost developments.

According to a recent market overview by PV Magazine, investment costs for utility-scale battery projects scheduled in 2025 are broadly in line with the assumptions of the major energy system studies (Lichner 2025). For the cell costs, BloombergNEF, reports considerably lower global market prices in 2024, with battery pack costs averaging 115 USD/kWh and prices as low as 94 USD/kWh in China (Catsaros, 2024). Moreover, a recent large-scale auction in China for 16 GWh of storage capacity resulted in an average bid price of 66 USD/kWh (Shaw, 2024). LFP battery cell costs are currently dominated by developments in China, as the majority of LFP batteries are either imported from China or manufactured by Chinese companies as shown in Figure 2.7. China holds an estimated 98% share of the production of cathode active materials (CAM) for LFP cells. Panel (a) depicts the geographic distribution of key steps in the lithium-ion battery value chain, while panel (b) illustrates ownership concentration. This heavy dependence on Chinese manufacturing introduces risks related to geopolitical tensions and the possibility of a trade conflict, including the imposition of tariffs, which could significantly increase the cost of battery projects in Germany. Conversely, continued innovation in

battery technologies and manufacturing processes could accelerate cost reductions beyond current expectations.

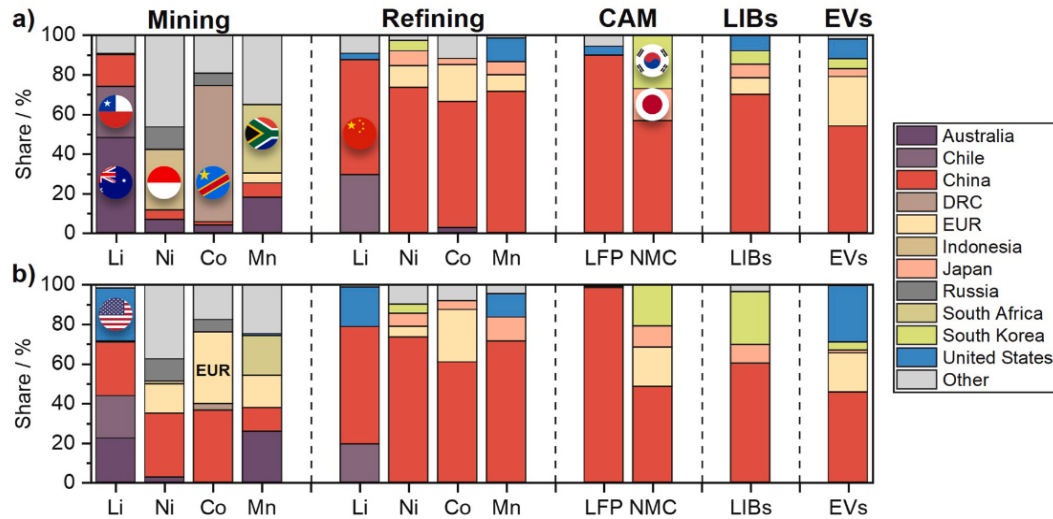


Fig. 1. a) Geographical distribution (revisualized from Cheng et al.) [29], and b) Ownership distribution of the LIB supply chain.

Figure 2.7.: Supply chain analysis of lithium-ion batteries by geographic distribution and ownership concentration

Source: Greitemeier et al. 2025

The current structure of project costs in Germany was recently described by the Chief Technology Officer (CTO) of EcoStor, one of the leading developers of utility-scale battery projects, in the podcast *Batterie Geladen* (Geladen Podcast 2025a). According to this source, a complete battery container imported from China, including the battery management system (BMS) and cooling costs approximately 100 EUR/kWh. Battery modules account for around 35–45% of total project costs, while inverters and transformers represent approximately 15–20%. For larger projects (exceeding 10 MW), the share of transformer costs is even higher. The remaining costs are attributable to construction, cabling, and external services. As the cost of batteries has decreased so substantially in recent years, these external cost components have become increasingly important in determining overall project economics. The same source reported that EcoStor relies on LFP batteries, which are expected to have a lifetime of about 15 years and a cycle life of 7,000–8,000 cycles. The inverters typically last between 15 and 20 years. With a targeted operational strategy of 1.5 cycles per day equivalent to roughly 8,200 cycles over 15 years, their utility-scale projects currently operate across multiple markets, including frequency containment reserve (FCR), automatic frequency restoration reserve (aFRR), and spot markets. By pursuing this cross-market strategy, the company expects to achieve an average annual return of about 15% of project costs, resulting in an amortization period

of approximately six to seven years. In the short term, returns are expected to be even higher, while a gradual decline is anticipated in the longer term as market competition increases.

Battery markets

An overview of different markets on which utility-scale batteries operate, as well as a description of their various use-cases, is detailed further below. The markets and use-cases fundamentally shape utility-scale battery deployment by determining the available revenue streams, operational strategies, and investment signals.

Economic viability & self-cannibalization effects

Profitability shapes investment attractiveness in battery storage and therefore possible battery deployment. The business cases for utility-scale battery storage are described in more detail later in this chapter. In Bundesministerium für Wirtschaft und Energie 2023 as well as in Agora Energiewende 2024 it is stated that utility-scale batteries are already profitable without subsidies. The huge amount of grid connection requests further shows the current economic attractiveness for utility-scale batteries. Further, in the above mentioned interview with the Ecoster CEO the current low expected amortization rate shows their profitability. The Ecoster CEO expects the profitability to be high in the near future but to diminish due to market saturation effects in the more long-term future (Geladen Podcast 2025a). As installed battery capacity grows, price signals (e.g., on spot markets) weaken, compressing margins for new storage projects and slowing further expansion which is meant here by the self-cannibalization effect.

Technical parameters

Further, different technical parameters of batteries are important in deciding their economic viability as well as their use-cases and possible markets to operate on. These include especially the round-trip-efficiency, battery lifetime and degradation and the E2P-ratio. The RTE, comprising cell, inverter, transformer, auxiliary, and thermal losses, directly scales the monetizable energy shifted and thus the arbitrage and balancing value.

Battery lifetime and degradation consist of two main components: calendar aging (capacity fade over time, mainly temperature-dependent) and cycle aging (throughput-dependent). The cycle life of current LFP batteries is around 4 000–10 000 cycles, while for NMC batteries it typically ranges between 2 000–5 000 cycles (Claire 2025). As most EVs in the EU currently use NMC chemistry (International Energy Agency 2025), they may have a disadvantage regarding degradation compared to stationary battery systems that predominantly employ LFP technology. However, recent studies highlight that for LFP-based EV batteries, the impact of cycle aging may be less critical in practical applications, since calendar aging is expected to dominate overall degradation under typical driving and charging conditions. As shown by Zhang et al. 2025, calendar aging

can account for up to two-thirds of total degradation costs in vehicle operation. This implies that additional cycling through V2G services may only marginally affect battery lifetime while enabling additional revenue streams, effectively providing an incentive to utilize the battery's remaining cycle capacity more efficiently. Therefore, understanding the interplay between calendar and cycle degradation is essential for assessing lifetime, amortization, and profitability in both stationary and mobile battery applications.

The E2P ratio is a key design parameter that defines the suitable use-case of a storage system: low E2P ratios (e.g., 1–2 h) are typical for applications requiring high power and fast response such as frequency regulation, while higher ratios (e.g., 4+ h) are characteristic of energy-shifting and arbitrage applications. Therefore, the E2P ratio directly influences both investment costs and the market segments in which a utility-scale battery can operate profitably.

2.3.4. Policies and Regulatory Framework

As the policy framework represents an important key driver and is included here to provide conceptual context, it is not further analyzed in the modeling and results sections. Therefore, only the most relevant aspects for battery storage deployment are briefly outlined.

Different policies and the regulatory framework are essential for reducing or building market barriers and establishing market-rules where batteries can operate on. The BMWK published the electricity storage strategy in end of 2023, underlining the importance of battery storage and going into detail about planned policies and the regulatory framework (Bundesministerium für Wirtschaft und Energie 2023). The document emphasizes the importance of a regulatory framework that enables batteries to be optimally integrated into the energy system, working in a grid-supporting way. They further stress the importance of enabling bidirectional charging and prosumer home batteries as a possible key component of batteries in the future German energy system. In recent years there has been a discussion about a bidding zone division in Germany as its currently one unilateral bidding zone. A separation of the bidding zone could lead to an increase in the need for battery storage as the price differences between zones could be used for arbitrage from batteries. The current Government has emphasized their commitment to keep a unilateral bidding zone in Germany in the coalition agreement (CDU/CSU and SPD 2025). Another possible policy that could have a significant impact on the amount of batteries needed in the system could be to establish a capacity market, similar to the one in the US or different EU states. A description of capacity markets and a possible implementation in Germany is detailed further below in this chapter.

The wide-spread build-out of Smart-Grid infrastructure, the grid expansion as well as grid fees, taxes and levies play an important role for the deployment of utility-scale batteries and especially for the competitiveness and enabling of prosumer home batteries and EV flexibility through smart-charging and V2G. Further, standards, incentives and

subsidies shape the framework but are not discussed further.

Grid fees, taxes and levies

For large-scale storage, special rules largely or fully waive network charges (including grid losses) as a transitional full exemption, currently extended until 2029. The regime aims to incentivize grid-relieving operation and reduce economic barriers for the expansion of utility-scale storage (Bundesministerium für Wirtschaft und Energie 2023). These exemptions apply only to new or expanded large-scale storage facilities and do not extend to home batteries or EV batteries. Decentralised home storage as well as V2G-capable EVs benefit instead via § 14a EnWG (controllable loads) and the roll-out of time-variable tariffs, which started in early 2025. Under § 14a EnWG, operators of controllable loads, including private home batteries, bidirectional charging points, and heat pumps, can opt into one of several reduction models for grid fees: a flat reduction, a percentage-based reduction, or a time-variable tariff. This regulation was introduced to support operation in a grid-supportive manner, for example by shifting charging times during periods of low grid stress (Energiewirtschaftsgesetz 2024; Bundesnetzagentur 2023). Under § 21 EnFG, bidirectionally used storage systems benefit from the netting of inflows and outflows for levies like the KWKG levy and the offshore grid levy. This mechanism explicitly aims to prevent double charging and thus encourages market participation of both utility-scale and decentralised storage, including V2G-capable EVs (Energiefinanzierungsgesetz 2025). Furthermore, stationary battery storage systems benefit from electricity tax relief when re-injecting electricity into the grid. According to § 5 StromStG, electricity that is temporarily stored and subsequently fed back into the public grid is exempt from electricity tax, provided that no consumption takes place during the storage process (Stromsteuergesetz 2022).

Smart grid infrastructure

Smart-grid infrastructure will be necessary to enable the effective integration of battery storage into the energy system, especially for decentralised home batteries and V2G-capable EVs. In 2024, Germany's smart-meter rollout still lagged significantly behind many other European countries: As of March 2025, the installation rate of intelligent metering systems in mandatory cases stood at only 15.1 % (with overall household penetration still below 3 %) (FfE - Forschungsstelle für Energiewirtschaft e. V. 2025).

To accelerate progress, the government enacted the Gesetz zum Neustart der Digitalisierung der Energiewende (GNDEW) in April 2023. The law sets binding rollout quotas for smart meters in "mandatory installation cases": at least 20 % by the end of 2025 and around 90 % by 2032. These mandatory cases include, among others, households operating controllable consumption devices under § 14a EnWG such as home battery systems, EV charging points, or heat pumps, which are technically required to be equipped with smart meters for all new installations, to enable remote control and dynamic pricing (Deutscher Bundestag 2023; FfE - Forschungsstelle für Energiewirtschaft e. V. 2025). In parallel, from 1 January 2025 all electricity suppliers must offer dynamic tariffs (EUPD

Group 2025).

2.4. Battery Markets

The major markets where utility-scale batteries currently operate on are ancillary services markets and electricity wholesale markets like the spot markets (Figgenger et al. 2025a).

Ancillary services markets

This section refers to markets for different grid services. Operators of battery storage systems can participate in tenders for providing capacity in such service markets. Currently, utility-scale battery storage is primarily been deployed for ancillary services, most notably for Frequency Containment Reserve (FCR). Looking ahead, battery storage systems are expected to play an increasingly important role in additional service areas. These include, in particular, further segments of ancillary services such as Automatic Frequency Restoration Reserve (aFRR) and Manual Frequency Restoration Reserve (mFRR), as well as voltage control, inertia provision, and black start capabilities (Bundesministerium für Wirtschaft und Energie 2023; Grimm 2024b). Since the latter three constitute comparatively small markets, they will not be examined in greater detail here.

Figure 2.8 distinguishes the three types of balancing reserves in Germany: FCR, aFRR, mFRR as well as the balancing process, by their activation speed and duration. FCR is the fastest product, fully delivered within 30 seconds, activated automatically and locally based on measured frequency, and procured symmetrically (no separate up/down products). It halts the frequency deviation system-wide and is jointly procured in Europe via the FCR Cooperation. aFRR restores the balance within each control area. It has positive and negative products, must be fully available within 5 minutes, and is dispatched automatically via setpoints from the TSO. European activation is coordinated through PICASSO/IGCC. mFRR addresses larger imbalances, with positive/negative products, manual activation via MOLS, and a full activation time of 12.5 minutes. European coordination runs through MARI. The maximum activation duration for FCR is 15 min and for aFRR and mFRR 60 min.

The tender volumes of the balancing service markets in 2023 are shown in Table 2.4. The average tendered capacity is by far the largest for aFRR, at almost 4 GW, followed by mFRR at just around 1 GW, while FCR is smallest at 0.57 GW. While utility-scale batteries already dominate the FCR market, with around 750 MW pre-qualified Battery Capacity in 2024, Batteries start to participate in the aFRR market as well. From 2023 to 2024 the pre-qualified Batters Capacity in aFRR market quadrupled from around 150 MW to 600 MW. 2025 is the first year that batteries participate in the mFRR market as well. The market share of batteries is expected to grow further in the coming years in the balancing markets (Antunes 2024). As the batteries fleet grows, the risk of cannibalization grows as well as heightened competition compresses prices and margins.

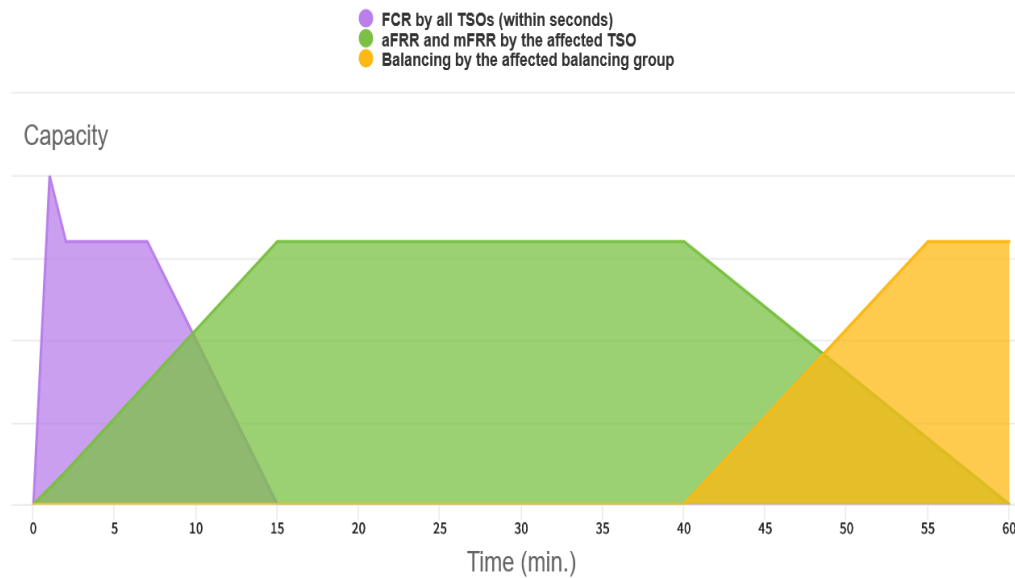


Figure 2.8.: Balancing services in Germany

Source: Regelleistung.net 2025

This saturation effect can already be seen in the FCR market as Battery capacity already outstrips FCR demand (750 MW prequalified vs. 570 MW procured) in 2024 (Williams 2025).

Spot markets - day ahead and intraday

Spot markets are short-term exchanges where electricity is traded close to real time on an exchange. In addition to spot trading, there are bilateral contracts, forward/futures markets and power purchase agreements (PPAs). These long-term instruments shape risk management but are generally less central for short-duration battery storage than spot and balancing markets. Germany operates a single bidding zone, so a uniform market-clearing price applies across the zone for each traded period. For batteries, spot markets are the primary venue for energy arbitrage.

Day ahead market

The Day-Ahead Market is a blind auction where participants submit hourly bids and offers for each hour of the following day. Auctions close at 12:00 PM on D-1, and a single Market Clearing Price (MCP) is determined by matching aggregated supply and demand curves via the EUPHEMIA algorithm. This mechanism ensures transparency and gives generators, retailers, and large consumers a fixed schedule and price for delivery the next day (Next Kraftwerke 2025a).

Table 2.4.: Balancing services capacity market in Germany

Source: Bundesnetzagentur and Bundeskartellamt 2024

Market	Direction	Average tendered capacity 2023
FCR (symmetric)	$\pm$	570 MW
aFRR	positive	1 923 MW
aFRR	negative	1 842 MW
mFRR	positive	681 MW
mFRR	negative	372 MW
Total Sum		≈ 5.39 GW

Intraday market

The Intraday Market allows continuous trading of energy after the Day-Ahead auction up until a few minutes before physical delivery (as late as 5 minutes prior to real-time). It operates across Europe under the Single Intraday Coupling (SIDC), enabling cross-border adjustment of positions to correct forecast errors or changing load patterns. Intraday trading thus provides participants with high-resolution flexibility to balance portfolios close to real time (Next Kraftwerke 2025b).

Capacity market

A capacity market pays resources for firm capacity availability, separate from energy delivered, to ensure security of supply during scarcity events. It complements the energy-only market and can be open to generators, battery storage, and demand response. Many EU countries as well as the US already use such capacity mechanisms (European Commission 2025). For example in the UK, Italy, Belgium and Poland have capacity markets been launched. Battery technologies have won increasingly large shares of the long-term revenue contracts they provide. In some places, like Poland, it even provides the bedrock of the battery business case (Murray 2025).

Germany historically operated an energy-only market (with strategic reserves), therefore, no broad market-wide capacity payments existed. However, a capacity market is planned to be introduced around 2028, with design work under way. The capacity market is planned as technology-neutral, therefore batteries could theoretically qualify and participate (Rühs 2025), (Bundesministerium für Wirtschaft und Energie 2024). If the current design would actually be technology-agnostic is in debate as industry voices are concerned about gas power plants being favored in the current design (Murray 2025). In such a scheme, batteries could participate by offering accredited firm capacity (e.g., ELCC-based), therefore adding a new revenue stream alongside energy arbitrage and

ancillary services.

2.5. Business Cases & Use Cases

In Figure 2.9 the current potential revenue Streams for Utility-scale batteries in Germany are listed as well as possible future revenue streams. They consist of energy arbitrage on the spot-market, ancillary service participation as well as a possible future capacity market.

The German revenue stack

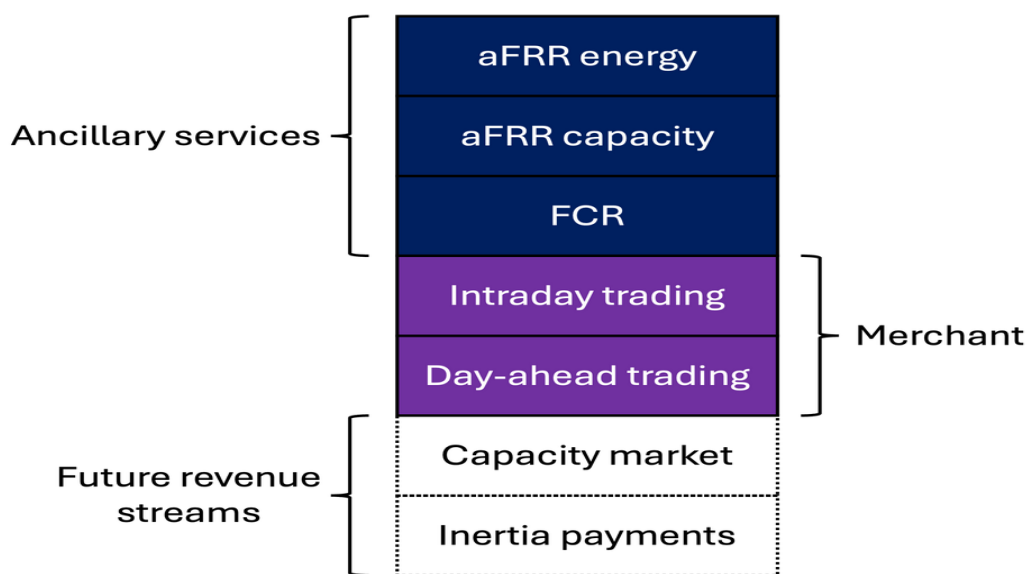


Figure 2.9.: Revenue streams for utility-scale batteries in Germany

Source: Williams 2025

Energy arbitrage

Energy arbitrage exploits price differentials between markets or time periods. A trader or an battery operator, will first buy low-priced electricity, often in the Day-Ahead auction when prices dip. Then store that energy in a battery and then sell it back into the market at higher prices. Either later in the Day-Ahead market, on the Intraday market, or even in Balancing markets. This strategy smooths out supply-demand imbalances and can generate significant revenue when price spreads are large. Therefore in the day ahead market the expected daily volatility in the auction is the main revenue stream

for the batteries. The average daily spreads as well as the top-down spreads are indicators for the volatility and profitability of the batteries. A detailed calculation method is presented in Figgenger et al. 2025b. In the intraday market, short-term price differentials arise primarily from forecast errors in supply and demand. For example, due to unexpected weather deviations affecting wind and PV output or load. Battery storage can respond rapidly by discharging to cover unforeseen demand or charging to absorb surplus generation, thereby reducing imbalance, mitigating price spikes, and monetizing the resulting spreads. Currently the revenue from the intraday market are exceeding the day-ahead market revenues as is detailed in Figgenger et al. 2025b.

Ancillary service participation

Many utility-scale battery projects currently target ancillary services markets because daily revenues are highest there, as illustrated in Figgenger et al. 2025a. However, as discussed above, market saturation and cannibalization effects are expected to emerge in these balancing markets, which would reduce per-asset daily revenues and likely reallocate battery dispatch toward the day-ahead (and intraday) market where energy-arbitrage opportunities can be optimized.

Cross market

Currently, utility-scale batteries are often marketed across multiple markets in parallel to maximize revenue. The specifics of such strategies are described in Figgenger et al. 2025b. A classic approach to cross-market operation is a parallel, co-optimized marketing of capacity on the FCR and aFRR markets combined with intraday trading.

Additional use-cases

There are further possible use-cases and business cases for utility-scale batteries which could become more relevant in the near future:

- Peak Shaving for Businesses and Industry and optimizing the energy management at big industry facilities.
- Other ancillary services like Voltage Support, Inertia response and black start capabilities.
- "Grid booster" - for grid operation management
- Integration of large PV and wind farms as part of the "Innovation-Tenders" in Germany
- Redispatch participation

These options are further discussed e.g. in Bundesministerium für Wirtschaft und Energie 2023, Grimm 2024a and Pant, Hamacher, and Perić 2025.

2.6. Other Battery Storage Types

While there is broad consensus that prosumer home batteries and V2G-enabled electric vehicles can theoretically provide system flexibility and partially substitute utility-scale batteries, the actual scale, time horizon, and market readiness of these technologies remain uncertain. The following section discusses key aspects related to these uncertainties.

2.6.1. Prosumer Home Battery Storage

In this thesis, residential and commercial battery systems are jointly referred to as home battery storage for simplicity, following the terminology used in the PyPSA-DE model for decentralized batteries. Currently, most home batteries in Germany are primarily installed in combination with a PV system to increase self-consumption and reduce household electricity costs rather than to participate in wholesale or ancillary-service markets. Consequently, their system-level flexibility potential remains largely untapped. For example, surpluses from wind generation are not utilized by home battery storage systems even when they are not fully charged. The current scenario framework of the NEP describes that unlocking the flexibility potential of home batteries is hindered by several factors. Fundamental prerequisites include the ability of households to perceive price signals, control devices, and measure charging power precisely. In this context, the rollout of smart meters plays a crucial role. However, as of 2022, smart-meter penetration in Germany was still below one percent of all metering systems. Further barriers include the absence of standardized dynamic tariffs, limited interoperability among devices, regulatory hurdles for aggregators that take over flexibility optimization for consumers, and the general willingness of consumers to adapt their electricity usage flexibly (NEP Szenarioahmentwurf 2025).

Nevertheless, the transition toward a digitalized energy system is progressing. As is shown in more detail above in the Smart-Grid infrastructure subsection, Germany has ambitious plans for the Smart-Meter-rollout as well as mandatory offerings of dynamic-tariffs since 2025. Key enabler for market-oriented prosumer home batteries is the emergence of virtual power plants, which aggregate thousands of distributed assets, such as PV systems, home batteries, electric vehicles, and heat pumps, into collective flexibility pools that can participate directly in energy markets. A prominent example is Flexa, a joint venture founded in 2024 by Enpal and Entrix, which already connects around 42 000 households with approximately 400 MW of aggregated capacity (Solarserver 2024; flexa GmbH 2025). Flexa's goal is to build Europe's largest household-based virtual power plant, capable of providing several gigawatts of flexibility within a few years, offering participants tangible reductions in electricity costs and thereby creating strong incentives for market participation. As these digital and market-oriented frameworks mature, prosumer home batteries are expected to evolve from self-consumption tools into active system flexibility providers. With over 18.5 GWh home batteries currently installed and continuous growth, the potential for home batteries to supply a significant share of the total flexibility provided by batteries is substantial.

2.6.2. EV Batteries and V2G

The current scenario framework of the German Grid Development Plan characterizes the role of EVs as an energy storage and flexibility option as follows: Mobility needs have priority, charging behavior and charge status must primarily serve transportation requirements. But because passenger EVs are often connected to charging infrastructure for extended periods, there is flexibility with respect to the timing and quantity of charging. In addition, EV batteries could, in principle, feed electricity back into the grid via V2G. However, since V2G-capable vehicles and charging points are not yet market-ready and are more expensive than smart-charging alternatives, the scenarios in the NEP assume that most flexible EVs will not return electricity to the grid. In the BNetzA approval of the NEP 2025 scenario framework, V2G-capable EVs are counted within the category of “small battery storage” without a more detailed breakdown or differentiation between stationary and mobile small-scale storage.

The federal electrical storage strategy (Bundesministerium für Wirtschaft und Energie 2023) estimates V2G technology as almost market ready and highlights an active policy track to activate the potential of bidirectional charging, with concrete steps underway to create non-discriminatory conditions and a roadmap toward deployment, pointing to an emerging but still nascent resource. Further, a recent study published for the NEP 2025 estimates a significant ramp-up of V2G-capable EVs to 65% of the EV fleet by 2035, 70% by 2040, and 75% by 2045 (Bonin et al. 2024). Regarding the market readiness, BYD and Octopus-Energy just launched their first "V2G-Bundle" in 2025 in the UK. This all-in-one package includes a V2G-capable BYD Dolphin, a bidirectional charger and a smart tariff for a leasing cost of around 300 british pounds per month. Thereby offering drivers completely free home charging and turning EVs into flexible energy assets (Westerheide 2025), (Octopus Electric Vehicles 2025). In a recent interview in 2025, the CEO of Ambibox, a producer of bidirectional chargers for EV wallboxes, said in the Podcast Moove regarding V2G that they plan to bring their first prototype to the german market in 2026 and then planning to penetrate the market from 2028 onward by offering more widely available bidirectional charging capable EV wallboxes for competitive costs (Moove Podcast 2025). In Droese et al. 2023 the economic viability of V2G is investigated by including battery degradation costs for an EV with a NMC lithium-ion battery. The study comes to the conclusion that V2G is profitable even when the degradation costs for NMC batteries are included. As EVs with LFP batteries tend to have even higher cycle lives the profitability should be even higher. Further, in Straub et al. 2023 the V2G potential was examined through a case study in Berlin and concluded that a wide implementation of V2G would have significant cost savings for the total system costs as well as being highly beneficial for the integration of variable renewables into the system. But as a downside having significant impact on the grid load, stressing the importance of the expansion of distribution grids.

3. Methodology

This chapter outlines the methodological framework applied to investigate the future deployment of battery energy storage systems in Germany. It first introduces the open-source modeling environment PyPSA-DE, which serves as the analytical framework for this study. The model enables comprehensive capacity expansion and dispatch optimization of the German energy system, capturing interactions between electricity generation, storage, and demand across sectors. Building upon the key drivers for utility-scale battery deployment identified in Chapter 2, the following section describes how these drivers are represented within the model. This includes the implementation of flexibility needs, competing technologies, economic parameters and the policy framework. Subsequently, the derivation of the base scenario is presented, detailing assumptions regarding spatial and temporal resolution, the integration of home- and EV battery storage and adjusted technical parameters such as battery lifetime. Finally, the scenario framework developed for this thesis is outlined, including variations in V2G participation, cost assumptions, and renewable generation expansion, which together provide the foundation for the comparative analysis of results in Chapter 4.

3.1. PyPSA-DE Modeling Framework

This thesis uses the open-source energy system modeling tool PyPSA-DE (PyPSA Developers 2025b), which is a Germany-specific implementation of the broader PyPSA-Eur framework (PyPSA Developers 2025c; Hörsch et al. 2018; PyPSA Developers 2025d). PyPSA (Python for Power System Analysis) enables the capacity expansion and dispatch optimization of sector-coupled energy systems with high temporal and spatial resolution, following a myopic foresight approach. A detailed documentation can be found on the official PyPSA website (PyPSA Developers 2025e), on the Ariadne-Project website (Ariadne-Project Website 2025) and in the Github repository (PyPSA Developers 2025a).

PyPSA-DE builds upon the European model PyPSA-Eur, but focuses on Germany with enhanced regional granularity (16 regions in Germany and 13 regions for neighboring countries) and further Germany specific adjustments. A detailed overview of the differences can be found here (PyPSA Developers 2025b). By default, the model operates with a 3-hour temporal resolution. The model performs linear investment optimization to minimize total system costs while simultaneously determining the optimal expansion of generation, storage, and transmission capacities as well as the optimal dispatch across all modeled target years.

PyPSA-DE is a sector-coupled model, integrating the electricity, heating, transport, industry and agriculture sectors. This allows for the endogenous representation of cross-sectoral flexibility and electrification pathways, an essential feature for evaluating the system-level role of different battery storage technologies. The model applies the weather data from the year 2019, obtained from ERA5 and SARA, consistent with the Ariadne scenario framework, to ensure comparability across renewable generation and demand profiles. A CO₂ constraint is implemented to ensure carbon neutrality by 2045, representing Germany's long-term climate target. Renewable generation potentials for wind and solar are derived from the geospatial assessment implemented in PyPSA-Eur, which combines land-use (CORINE), exclusion zones (Natura2000), and technology-specific siting constraints to determine the technically available areas. Combined with technology-specific capacity densities and capacity factors, these potentials define the upper bounds for endogenous renewable capacity expansion in PyPSA-DE.

Storage systems, including battery storage, are modeled as either:

- StorageUnits: characterized by a fixed E2P-ratio
- Stores and Links: allowing a more flexible definition with separate investment variables for charging, discharging, and energy capacity.

Utility-scale batteries are modeled as Stores and Links, enabling endogenous optimization of both energy and power capacity. They receive as input parameters investment costs (EUR/kWh, EUR/kW), lifetime, OPEX, and efficiency, and are connected to the transmission grid. Home PV batteries are modeled analogously but are connected to the distribution grid. Self-consumption optimization and a direct connection to residential rooftop PV is not modeled. V2G is implemented as an optional feature. The number of electric vehicles, their average battery size, and hourly availability profiles are provided exogenously, along with the share of V2G-capable battery electric vehicles (BEVs), defining the total amount of flexible V2G storage capacity available per timestep. PyPSA-DE's open-source nature, structure, and high resolution make it particularly suitable for this thesis, as it allows the endogenous optimization and detailed analysis of stationary utility-scale batteries, home batteries, and V2G-enabled EVs under various scenarios and assumptions.

3.2. Implementation of the Key Deployment Drivers in PyPSA-DE

This section explains how the key drivers described in chapter 2 are implemented and represented in the PyPSA-DE model and in this thesis.

3.2.1. The Need for System Flexibility

Expansion of variable renewables (wind, PV)

In PyPSA-DE, exogenous targets for PV, onshore- and offshore wind capacities are defined up to 2030. From 2035 onward, renewable generation expands endogenously within defined capacity corridors, allowing the model to determine the system-optimal balance between PV and wind. By 2045, the majority of electricity generation is provided by variable renewable energy sources, creating significant short-term flexibility needs for balancing intraday and daily fluctuations.

3.2.2. Competitive Technologies

The development and expansion of other battery types

In this thesis prosumer home batteries are implemented in the model with a minimum exogenous expansion rate and further endogenous expansion. This is further detailed later in this chapter as well as the representation and implementation of EVs, smart charging and V2G.

Other storage options

PyPSA-DE represents the following storage types with distinct temporal characteristics:

- Batteries and pumped hydro storage provide short-term flexibility.
- Thermal storage and hydrogen storage systems serve primarily as long-term storage options.

Other flexibility options

Flexibility on the supply side is provided by dispatchable generation technologies such as gas power plants and, in the transition years, remaining coal plants until their phase-out. Hydrogen-fueled power plants as well as biomass and biogas combined heat and power (CHP) plants are available as low-carbon flexible backup options.

Demand-side flexibility is represented through several modeled sectors and options:

- Electric vehicles (smart charging)
- Electrolyzers (hydrogen DSM)
- Heat pumps (thermal DSM)

Industrial DSM is not yet implemented in the current PyPSA-DE version.

Import/export capacities

Cross-border electricity trade between Germany, neighboring countries and key trading partners is modeled through interconnection links with explicit capacity limits. Within these constraints, the model assumes welfare-optimal electricity exchange, allowing net imports and exports to vary hourly based on relative price levels and system states.

3.2.3. Economic Drivers

Cost developments / trajectories

For the purposes of this thesis, the cost trajectories from the Ariadne Report 2025 are adopted as the baseline investment cost assumptions, as they mostly align with the current projections discussed in Chapter 2. In addition, a sensitivity analysis is conducted by varying battery investment costs to account for uncertainties in future cost developments and to reflect the diversity of trajectories reported in the literature, as summarized in Table 2.3. This approach allows for a more robust assessment of the role of battery storage in Germany's energy transition.

Battery capacities in PyPSA-DE are endogenously optimized. The investment costs in the model are implemented separately as power-related (Euro/kW) and energy-related (Euro/kWh) investment costs, allowing the system to determine the optimal E2P ratio. While this provides flexibility to reflect system-level optimization, it does not directly account for the cost correlation between project-specific E2P ratios (e.g., 2-hour vs. 4-hour systems), where longer durations often reduce cost per kWh (Cole and Karmakar 2023).

Battery markets

The model represents with regard to battery markets an energy-only market, effectively corresponding to the day-ahead market. Consequently, short-term balancing or ancillary service markets are not explicitly modeled. However, their influence is discussed qualitatively in this thesis, given their relevance for current battery deployment and revenue streams. Given the limited size and increasing saturation of ancillary service markets as described in Chapter 2, the day-ahead market is expected to become the primary revenue source for battery storage in the future.

Economic viability & self-cannibalization effects

To assess the system-level profitability of battery storage, price volatility and revenue-relevant indicators are evaluated through daily price spreads and top-bottom spread metrics derived from the model output. These indicators capture self-cannibalization effects that emerge with increasing battery penetration, analogous to the saturation trends already observed in ancillary markets such as FCR and aFRR (Williams 2025). As discussed in Chapter 2, the current revenue streams of utility-scale batteries are only

partially represented in the model, as not all relevant market segments and use-cases are included. The potential implications of this limitation are addressed in Chapter 5.

Technical parameters

In PyPSA-DE, battery storage technologies are characterized by charging and discharging efficiencies, which together determine the round-trip efficiency. Following Luderer et al. 2025, an RTE of approximately 95.5% is applied for utility-scale batteries and home batteries. Based on Schram et al. 2020, an RTE of 88% is implemented for V2G EV batteries. A technical lifetime of 15 years is assumed for utility-scale batteries, consistent with most energy system studies, as shown in Table 2.3. The annual full-cycle count derived from model operation is analyzed to assess degradation relevance and economic implications. The E2P-ratio of utility-scale batteries is endogenously optimized.

3.2.4. Policies and Regulatory Framework

PyPSA-DE does not explicitly model policy instruments such as taxes, levies, or incentive schemes. Nevertheless, the regulatory framework strongly affects the real-world deployment of utility-scale batteries and is therefore conceptually integrated into this thesis.

3.3. Derivation of the Base Scenario

The base scenario is aligned with the net-zero "Technology Mix" pathway from the Ariadne Report 2025 (Luderer et al. 2025). This pathway describes a cost-efficiency-oriented mix of technologies and energy carriers that achieves climate neutrality while broadly meeting the politically mandated expansion targets for renewable electricity. The resulting mixed energy-carrier portfolio assumes medium levels of energy imports. With respect to end-use decarbonization, it strikes a balance between direct electrification and indirect electrification (hydrogen and derivatives), and is therefore positioned between the Ariadne variants "Focus Electrification" and "Focus Hydrogen." To reach the climate targets, the scenario relies on a portfolio of policy instruments rather than a single lever. A medium energy-demand trajectory is assumed. Further details on assumptions, parameterization, and data sources are provided in the Ariadne Report itself and the PyPSA-DE repository (Luderer et al. 2025; PyPSA Developers 2025b).

Related to batteries and other storages and flexibilities, the PyPSA model for the Ariadne Report assumes an endogenous expansion of home storage batteries and utility-scale batteries. While the smart-charging option for BEVs is activated as a flexibility option, the V2G Option is deactivated. This modeling, results in utility-scale batteries being the dominant battery technology with almost no home batteries installed (less than 1 GW and GWh) and EV Batteries not available to discharge.

The base scenario in this thesis has some adjustments to the PyPSA-DE model which was used for the "Technologiemix" scenario. These are summarized here:

- Adjusted spatial resolution
- Adjusted temporal resolution
- Added minimum value for home battery expansion
- Turned on V2G option and adjusted assumptions
- Adjusted battery lifetime assumption

The base scenario includes the target years from 2020 to 2050 in 5 year steps. While the target year for carbon neutrality in Germany is 2045, the European target year is 2050. Therefore, in most of the results, 2050 is also included.

3.3.1. Spatial Resolution

In the Ariadne Model for the Technologiemix scenario, the spatial resolution has 49 nodes in total, representing 30 regions in Germany as well as 19 regions of neighboring countries. Each node has his own electricity price as well as connected generation, transmission and load capacities. This leads to high utility-scale battery deployment as can be seen in Figure 3.4. In this approach batteries will be used to shift electricity temporal (e.g. From Day to Night) as well as spatial (to cover local bottlenecks, shifting due to price differences in the zones).

For the base model in this thesis the spatial resolution is reduced to 15 nodes in total and to only 1 node representing Germany. The other 14 nodes stay the same and represent the following countries: Austria, Belgium, Switzerland, Czech Republic, Denmark, France, United Kingdom, Luxembourg, Netherlands, Norway, Poland, Sweden, Spain and Italy. They include all neighboring countries of Germany as well important electricity trading partners of Germany.

This approach was chosen to reduce the model complexity and to guaranty a uniform electricity price, as this represents the current reality in Germany of the unilateral bidding zone. With this approach grid bottlenecks and electricity arbitrage between neighboring nodes is dismissed and therefore the total installed battery capacity in the model is slightly reduced as can be seen in Figure 3.5 when compared to the results in Luderer et al. 2025 (around 435 GWh in 2045).

3.3.2. Temporal Resolution

To ensure that generation and demand peaks are clearly visible in the model, the temporal resolution is set to hourly in this thesis. The difference in representation in the model from the 3-hourly default to hourly is illustrated in Figure 3.1 for the battery dispatch.

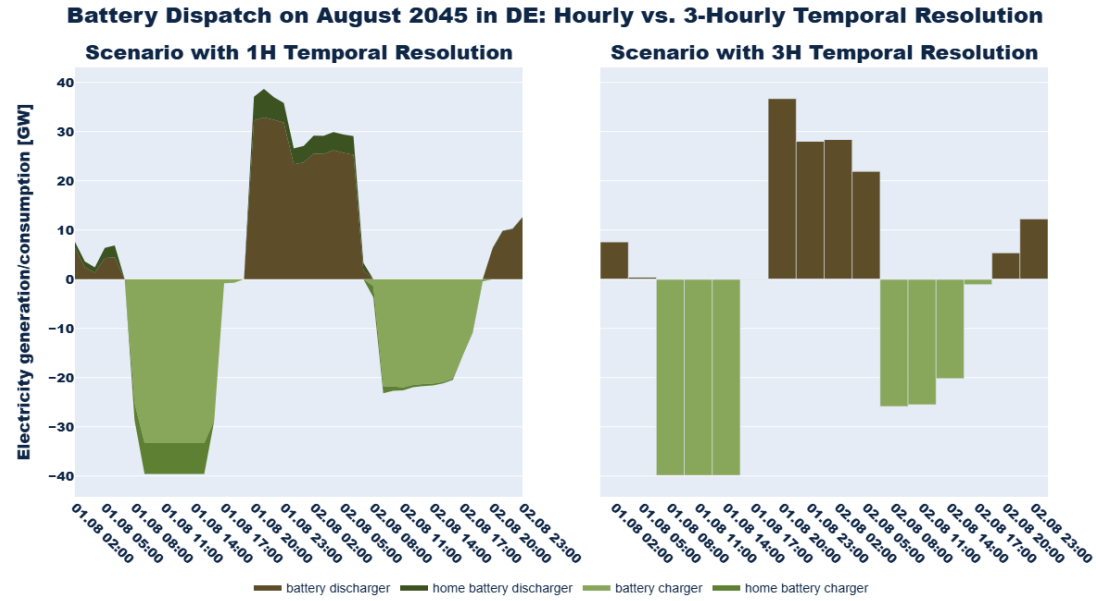


Figure 3.1.: PyPSA-DE scenario comparison of battery dispatch in 3-hourly temporal resolution to hourly resolution

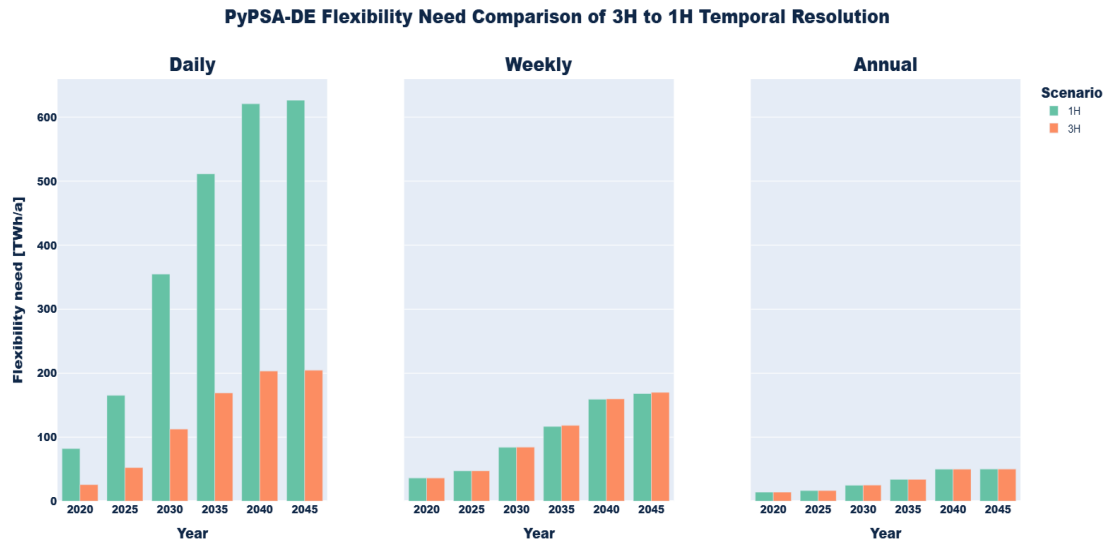


Figure 3.2.: PyPSA-DE flexibility needs comparison of 3-hourly to hourly temporal resolution

The hourly resolution ensures an accurate representation of daily flexibility needs which are an important driver for battery deployment. The daily flexibility needs of the system

in 2045 almost triple when changing the temporal resolution from 3-hourly to hourly as is shown in Figure 3.2. The weekly and annual flexibility needs stay almost the same. The calculation method of the flexibility needs is detailed in the next chapter.

3.3.3. Minimum Value for Home Battery Expansion

In the TechnologiemiX scenario of the Ariadne report almost no additional home batteries are build. This is mainly due to the current modeling approach. Both utility-scale batteries as well as home-batteries are modeled as a store with a charge and a discharge link. But the costs for utility-scale batteries are significantly lower. Therefore the model decides to build almost all batteries as utility-scale batteries. To represent the difference between both battery types, the home batteries are connected to the distribution grid while the utility-scale batteries are connected to the transmission grid. Network losses of 3% are modeled between the transmission and distribution grid. Therefore the model has an incentive to supply the load in the distribution grid with electricity stored in the home batteries. But the cost incentive outweighs this incentive in the TechnologiemiX Scenario. If we change the temporal resolution to hourly, the peaks become more visible and the incentive to build home batteries rises. Therefore in this step, some home batteries are build as can be seen in Figure 3.4. However, an adequate representation of home storage as prosumer systems and their use for self-consumption optimization is currently not sufficiently captured, especially as grid fees are not modeled.

Different approaches exist to address this issue in the model:

1. Introduction of a self-generation constraint for Prosumers: In this novel approach from Guéret, Schill, and Schmidt 2025 to model prosumers in linear capacity expansion models, home batteries are taken into account in the model as "system blind" prosumers by introducing a self-generation constraint to approximate the incentives of prosumers to minimize their electricity bills. Therefore they are focusing on self-consumption optimization like currently most of the home batteries in Germany do. It is assumed that the prosumer households do not have a smart-meter and no dynamic tariff, instead they use a retail tariff and feed-in tariffs for the excess PV generation. They find that for their case study, the optimal battery storage capacity increases by up to 200% if prosumer constraints are included.
2. Transmission to distribution grid expansion costs: Another approach is to explicitly model the costs of expanding transmission grid to distribution grid interconnection capacity. By assigning investment costs to additional transfer capacity between the transmission and distribution grid, the model is incentivized to reduce flows across this boundary. This can indirectly increase the attractiveness of distributed home batteries, as they can alleviate local demand–supply imbalances without relying on costly grid reinforcements.

3. Prosumer node with price signal: A further option is to introduce a dedicated prosumer node, analogous to the existing distribution grid node in PyPSA-DE, to which all prosumers and their respective loads are connected. By assigning a price signal (e.g. around 30 EUR/MWh, representing a grid-fee) to this node, the model internalizes the value of local generation, storage, and consumption. This encourages self-consumption and localized flexibility provision, thereby better reflecting the incentives of prosumers in the current market environment.
4. Input exogenous minimum values for home battery power and energy capacity in each target year based on existing literature.

The last approach is used in this thesis to represent the home battery expansion in the model. Minimum values for installed power and energy capacity are defined for each target year, while an endogenous expansion option is also implemented. The 2025 scenario framework of the NEP provides reference values for decentralized home battery systems in the target years 2037 and 2045 (NEP Szenariorahmenentwurf 2025). These projections are mainly derived from the anticipated expansion of rooftop PV systems, as described in the previous chapter. A fixed E2P ratio of 2 hours is assumed. The NEP distinguishes three scenarios (A, B, and C), with Scenario A reflecting the lowest and Scenario C the highest home battery expansion pathway.

In an intermediate step, the home battery capacities of Scenario A are adopted and linearly interpolated for the intermediate target years. This results in the minimum home battery values shown in Figure 3.4 and Figure 3.5 (bottom-left panels). Since no further endogenous home battery expansion occurs in this scenario, the exogenously implemented values correspond directly to the displayed results. However, this approach neglects the fact that most home batteries currently operate as system-blind prosumers, which means, they optimize self-consumption without responding to market signals or grid conditions. While the use of Scenario A reflects a conservative deployment pathway, it assumes that all home batteries operate as market-oriented prosumers, meaning they actively respond to electricity prices or grid conditions rather than solely maximizing self-consumption. This assumption does not yet reflect current real-world operation.

In the future, a growing share of home batteries is expected to become market-oriented prosumers. Hence, two conceptual operating modes can be distinguished: First, system-blind, self-consumption-oriented batteries, and second, market-oriented batteries that optimize both self-consumption and participation in market operation. To estimate the share of installed home batteries operating in market mode, the NEP 2025 scenario framework provides projected participation rates for “prosumers in market operation” of 35 % in 2037 and 55 % in 2045 (Scenario B). By combining these participation rates with the total home battery deployment of Scenario B and applying linear interpolation for the intermediate years, the resulting minimum values for market-operating home batteries are obtained as summarized in Table 3.1.

Table 3.1.: Deployment of home battery storage capacity (GWh) and charging power (GW) as prosumer in market operation mode in Germany

Year	Home Battery [GWh]	Home Battery Charger [GW]
2020	0.000	0.000
2025	0.656	0.360
2030	11.585	4.962
2035	35.261	14.234
2040	64.547	25.766
2045	96.250	38.500
2050	96.250	38.500

While this approach represents a medium deployment level consistent with Scenario B, it explicitly includes only home batteries operating in market mode. This choice was made for the base scenario, since these prosumer batteries have the greatest influence on the expansion of utility-scale batteries due to their comparable market participation. Consequently, this approach is particularly relevant for analyzing potential substitution effects between home and utility-scale batteries. However, it excludes the system-blind, self-consumption-oriented home batteries and thereby underestimates their potential contribution to system flexibility. Thus, the adopted method represents a conservative estimate regarding total installed home battery capacity.

To comprehensively capture both operating modes, the current approach could be extended in future work by integrating the methodology proposed by Guéret, Schill, and Schmidt 2025, which accounts for both market-oriented and self-consumption-driven home batteries. In the following chapters, home batteries that participate in market operation are referred to as prosumer home batteries.

3.3.4. EV Modeling and V2G Option Adjustments

The original modeling framework for the transport sector in PyPSA-Eur, including the representation of electric vehicles, smart charging, and V2G operation, is described in detail in this paper by Brown et al. 2018. In PyPSA-DE, road transport, specifically passenger cars, is represented as a dedicated carrier bus that is coupled to the electricity bus. The total number of passenger cars enters the model exogenously based on historic data for Germany. For each target year, an electric fleet share is added exogenously, resulting in the corresponding number of EVs shown in Table 3.2.

These values are derived from Germany’s political target of 15 million electric vehicles by 2030 (Deutscher Bundestag 2024). However, as outlined in the last chapter, Germany is currently lagging significantly behind this trajectory, with approximately 1.84 million

Year	Electric Share land transport [%]	Total ber [Mio.]	Num- EVs	Battery Size [kWh (Charger kW)]	Aggregated Power Ca- pacity [GW]	Aggregated Energy Capacity [GWh]
2025	15	7.16		50 (11)	78.76	358.0
2030	30	14.31		50 (11)	157.41	715.5
2035	45	21.47		50 (11)	236.17	1073.5
2040	70	33.40		50 (11)	367.40	1670.0
2045	85	40.56		50 (11)	446.16	2028.0
2050	1	—		50 (11)	—	—

Table 3.2.: Development of EV numbers, battery sizes, and aggregated capacities in the PyPSA-DE model

BEVs corresponding to a total capacity of around 115 GWh. Hence, the model input values represent an optimistic outlook.

Transport demand from EVs is modeled as a load component constructed from driving activity profiles and specific energy consumption values (kWh/km). This load is implemented as a Load component on the same bus as the aggregated EV battery, represented as a Store component, and the BEV Charger link connects this bus to the electricity bus. When the DSM option is enabled, the driving demand is met from the EV store, which is charged via the BEV charger. Otherwise, the load is supplied directly through the charger from the electricity bus. The BEV Charger link component aggregates the total charging power of the EV fleet while the V2G Discharger link models the aggregated discharge capacity of the V2G-enabled fleet.

These two additional flexibility options can be activated in the model: EV-DSM (smart charging) and V2G.

- When the DSM option is enabled, the EV Battery Store component is added to represent the aggregated flexibility from smart charging. The charging and discharging behavior can then vary dynamically, depending on the aggregated energy capacity of the connected fleet.
- When the V2G option is enabled, a second Link component, the V2G Discharger, is added. This component represents bidirectionally charging EVs that can feed energy back into the grid.

A schematic representation of EV charging, discharging, DSM, and V2G integration in the model is shown in Figure 3.3.

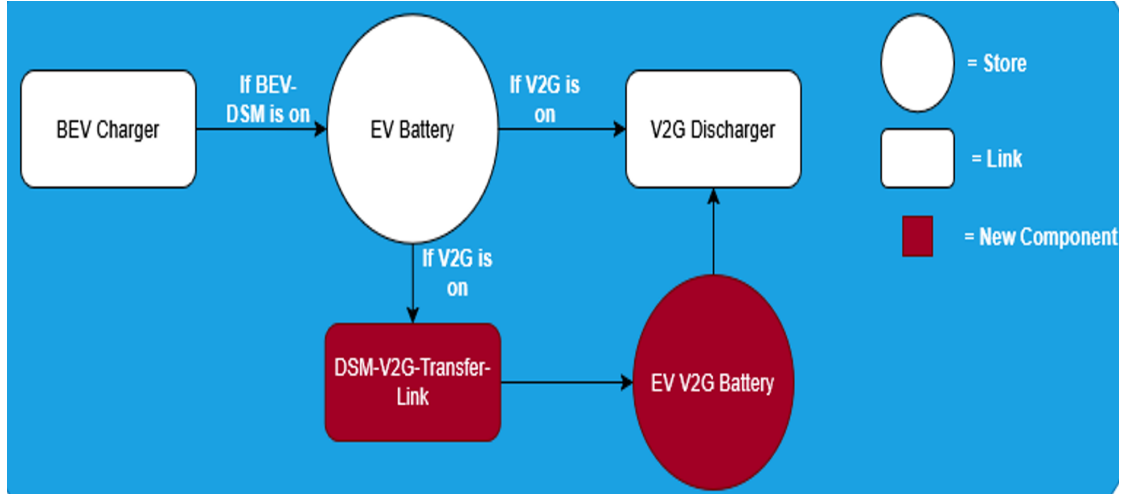


Figure 3.3.: Schematic of the PyPSA-DE modeling framework for EV charging, discharging, DSM and V2G

To enable differentiation between DSM and V2G participation, this thesis extends the PyPSA-DE setup by adding two additional components. Specifically, a second store was introduced to represent the subset of vehicles participating in V2G. Because all V2G-enabled vehicles are also expected to participate in DSM, this second store is modeled as a subset of the first. An additional Link connects the two stores, ensuring a hierarchical relationship between DSM and V2G participation.

The following section provides a detailed overview of the EV-related parameters, highlighting newly added components in red and modified components in blue.

EV Parameters

The EV-DSM participation rate, the V2G participation rate, and the average battery size per vehicle used in the model are listed in Table 3.3.

The availability profile $a(t)$, represents the share of the EV fleet physically plugged in and available for charging or discharging at each time step. It ranges from 0 to 1, corresponding to the available share of the fleet. An average availability of 80% and a peak value of 95% are implemented. Operationally, $a(t)$ scales both the power and energy availability. The DSM profile $d(t)$, defines the lower bound on the EV store's state of charge (SoC) reserve, ensuring that a minimum level of energy remains available for mobility. In this thesis, the profile values have been modified: the minimum SoC in the morning (5 a.m.) was reduced from 75% to 40%, and a baseline minimum SoC of 25% was introduced for all other hours. These adjustments are based on assumptions from the TYNDP 2024 (ENTSO-E and ENTSO-G 2025). Brown et al. 2018 assume an average daily driving distance of 40 km, which is feasible with a 25-40% SoC even for smaller battery sizes.

The implementation of both $d(t)$ and $a(t)$ follows the methodology described in Brown et al. 2018.

BEV Charger

The nominal charging power is defined as the product of the number of EVs, the charging rate per vehicle, and the electric fleet share, as shown in Equation 3.1. The upper bound of the charging power is scaled by the availability profile according to Equation 3.2. A uniform charger power of 11 kW per EV is assumed in all target years.

$$p_{\text{nom}} = N \cdot r_{\text{charge}} \cdot \alpha \quad (3.1)$$

$$p_{\text{max}}(t) = p_{\text{nom}} \cdot a(t) \quad (3.2)$$

EV Battery

The nominal energy capacity of the EV battery store is determined by the number of electric vehicles, their battery size, and the DSM participation rate as shown in Equation 3.4. Both parameters are modeled dynamically for each target year to capture expected increases in average battery size and DSM participation. The maximum available energy capacity is limited by the availability profile $a(t)$, ensuring that only plugged-in vehicles contribute to flexibility (3.3). The minimum energy capacity is constrained by the DSM profile $d(t)$ (3.5), which ensures that a certain share of the stored energy remains available for mobility needs.

$$e_{\text{max}}(t) = 1 \quad \text{adjusted to} \quad e_{\text{max}}(t) = e_{\text{nom}} \cdot a(t) \quad (3.3)$$

$$e_{\text{nom}} = N \cdot \alpha \cdot E_{\text{fix}} \cdot DSM_{\text{fix}} \quad \text{adjusted to} \quad e_{\text{nom}} = N \cdot \alpha \cdot E_{\text{dyn}} \cdot DSM_{\text{rate}} \quad (3.4)$$

$$e_{\text{min}}(t) = e_{\text{nom}} \cdot d(t) \quad (3.5)$$

V2G Discharger

The aggregated nominal discharge power is calculated as the product of the number of EVs, the discharge rate per vehicle, the electric fleet share, and the V2G participation rate, as defined in Equation 3.6. To provide a conservative estimate, the discharge rate is set to 50% of the nominal charging rate, considering that inverters are typically more expensive than rectifiers and thus may result in lower average installed discharge power capacity (Moove Podcast 2025). The maximum available discharge power is further limited by the availability profile (3.7). The V2G participation rate is defined independently from the DSM rate, since smart-charging capability is assumed to become more common than full bidirectional operation. Estimated values for each target year are provided in the V2G scenario section.

$$p_{\text{nom},\text{dis}} = p_{\text{nom}} \cdot DSM_{\text{fix}} \quad \text{adjusted to} \quad p_{\text{nom},\text{dis}} = N \cdot r_{\text{discharge}} \cdot \alpha \cdot V2G_{\text{rate}} \quad (3.6)$$

$$p_{\text{max}}(t) = p_{\text{nom}} \cdot a(t) \quad (3.7)$$

DSM→V2G Transfer (100% efficiency)

A new Link component connects the DSM store with the V2G store to ensure that the latter remains a subset of the former. The transfer operates at 100% efficiency and its nominal and maximum capacities are defined in Equations 3.8 and 3.9.

$$p_{\text{nom},\text{trans}} = p_{\text{nom}} \cdot V2G_{\text{rate}} \quad (3.8)$$

$$p_{\text{max},\text{trans}}(t) = p_{\text{nom}} \cdot a(t) \quad (3.9)$$

EV V2G Battery

The aggregated energy capacity of the EV V2G Battery store is calculated as the number of EVs multiplied by the dynamic battery size per vehicle and the V2G participation rate, as defined in Equation 3.10. The upper and lower energy bounds are constrained by the availability and DSM profiles (Equations 3.3 and 3.12).

$$e_{\text{nom},\text{V2G}} = N \cdot \alpha \cdot E_{\text{dyn}} \cdot V2G_{\text{rate}} \quad (3.10)$$

$$e_{\text{max},\text{V2G}}(t) = e_{\text{nom}} \cdot a(t) \quad (3.11)$$

$$e_{\text{min},\text{V2G}}(t) = e_{\text{nom}} \cdot d(t) \quad (3.12)$$

Legend:

- $a(t)$ = availability profile
- $d(t)$ = dsm profile (min. SoC)
- N = number of cars
- r_{charge} = charge rate per vehicle
- $r_{\text{discharge}}$ = discharge rate per vehicle
- α = electric share

- E_{fix} = battery capacity per vehicle
- E_{dyn} = battery capacity per vehicle in each target year
- $DSM_{\text{fix}} = 0.5$
- DSM_{rate} = EV DSM participation rate
- $V2G_{\text{rate}}$ = EV V2G participation rate

 = adjusted  = new

3.3.5. Adjusted Battery Lifetime

In the Ariadne Report (2025), a battery lifetime of 22.5 years in 2025 and 30 years in 2045 is assumed, as shown in Table 2.3. In a recent interview, the CEO of Ecoster (Geladen Podcast 2025a) mentioned a current lifetime of around 15 years for utility-scale battery projects. This estimate is consistent with the assumptions of several other studies presented in Table 2.3. Consequently, a more conservative approach was adopted in this thesis by assuming a uniform battery lifetime of 15 years across all target years.

3.3.6. Effects of Adjustments

In this section, the results of the battery deployment for utility-scale batteries, prosumer home batteries, and the minimum available EV V2G battery capacity are presented for each derivation step toward the base scenario. The parameters shown include the installed power capacity, installed energy capacity, annual discharge, E2P ratio, and annual full cycles, as illustrated in Figures 3.4, 3.5, and 3.6. The E2P ratios and annual full cycles for each derivation step are provided separately in the appendix (Figures A.1 and A.2).

The first step (top-left) represents the adjustment of the spatial resolution based on the TechnologiemiX scenario of the Ariadne Report and is labeled 1-Zone (DE) in the plots. The second step (top-right) introduces a higher temporal resolution and is labeled 1-Zone (DE), 1H Res. The third step (bottom-left) adds a minimum installed capacity for home batteries and appears as 1-Zone (DE), 1H Res, Min. Home Battery. Finally, the fourth step (bottom-right) activates the V2G functionality and is labeled 1-Zone (DE), 1H Res, Min. Home Battery, V2G.

In Figure 3.4, Step 1 shows that only utility-scale batteries are installed, with around 45 GW of discharge capacity. In Step 2, a small share of prosumer home batteries appears in the model, accompanied by a reduction in utility-scale battery capacity. In Step 3, the implementation of the minimum home-battery capacity causes a significant increase in installed prosumer home storage, while utility-scale capacity decreases accordingly. In Step 4, enabling V2G results in an additional 50 GW of installed discharge capacity from

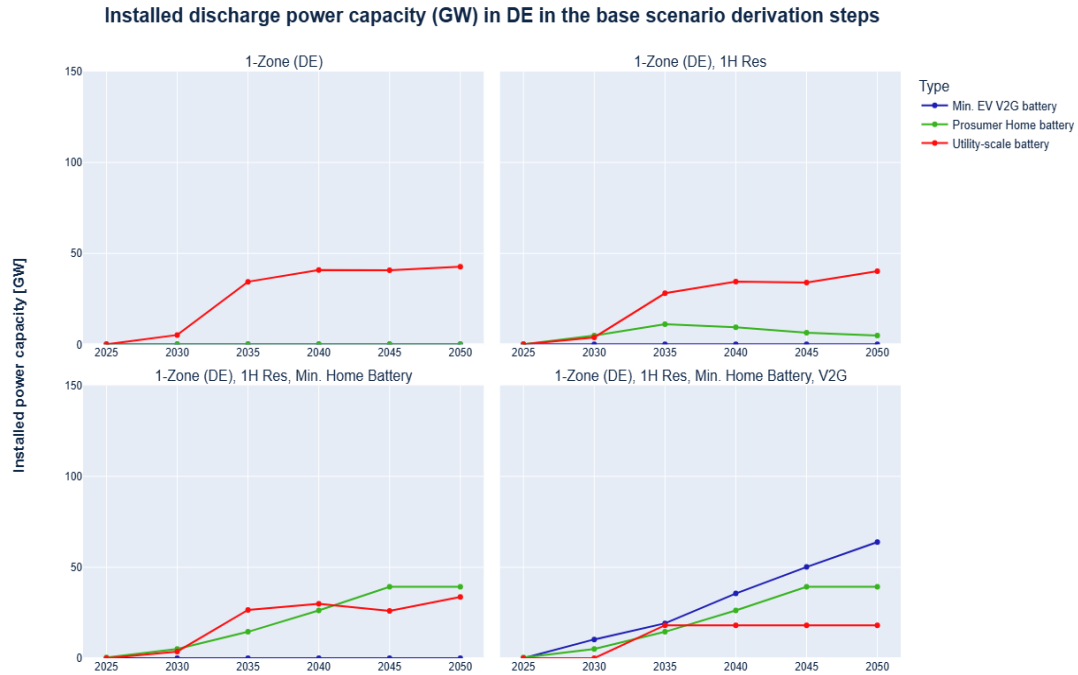


Figure 3.4.: PyPSA-DE results of installed discharge power capacity across target years in the base scenario derivation steps

EV batteries in 2045. The discharge capacity of home batteries reaches approximately 40 GW, while the utility-scale capacity declines further to roughly 20 GW in 2045. These results suggest a direct substitution effect, where increasing shares of prosumer and V2G batteries displace utility-scale storage capacity. A similar pattern can be observed in Figure 3.5 for the installed energy capacity. Starting from around 450 GWh in the Technologiemix scenario for 2045, total energy capacity of utility-scale batteries decreases with each derivation step, reaching approximately 180 GWh in the final base scenario. Step 1 results in around 350 GWh, Step 2 around 320 GWh, Step 3 around 220 GWh, and Step 4 around 180 GWh. In Figure 3.6, the annual discharge follows a comparable trend. As the share of prosumer home batteries and V2G batteries increases, the total annual discharge of utility-scale batteries declines, from about 65 TWh in Step 1 to roughly 35 TWh in Step 4. Nevertheless, utility-scale batteries remain the dominant contributor to total annual discharge until 2045, before being overtaken by V2G batteries by 2050.

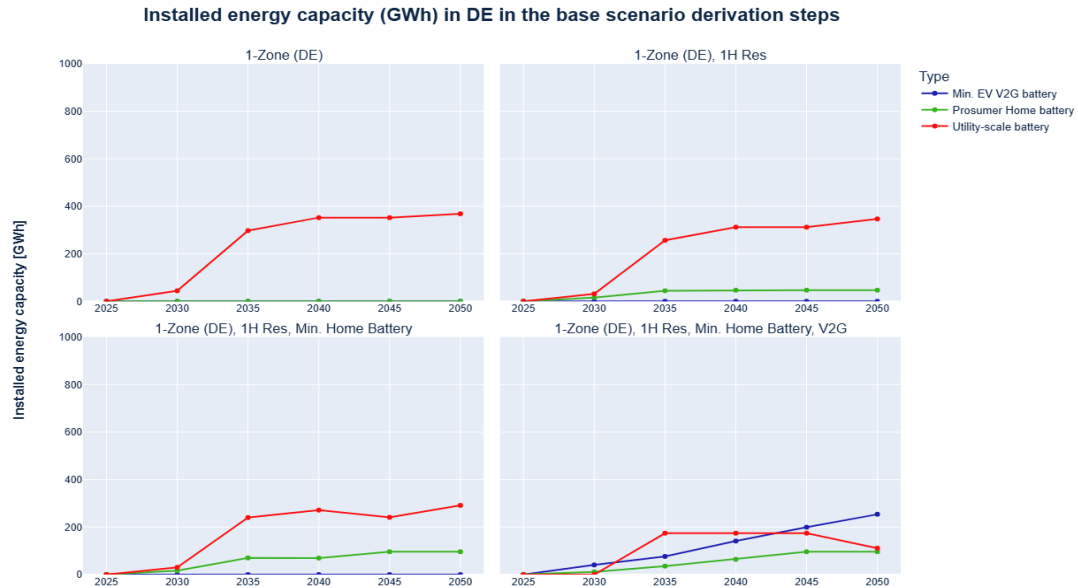


Figure 3.5.: PyPSA-DE results of installed energy capacity across target years in the base scenario derivation steps

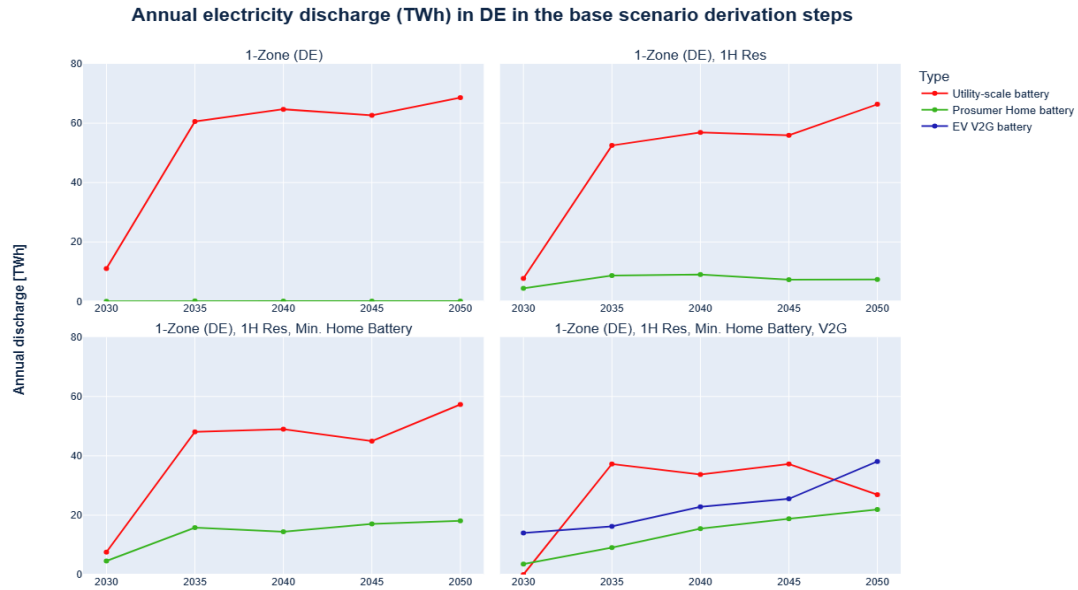


Figure 3.6.: PyPSA-DE results of annual electricity supply by batteries across target years in the base scenario derivation steps

3.4. Scenarios

To evaluate the effects of the key drivers on the expansion of utility-scale battery deployment, three scenarios were designed. As well as to answer the objectives mentioned in the research question subchapter.

Three of the most relevant key drivers on utility-scale battery deployment were picked for this thesis to see the effect by variation of these factors.

These are the following:

- **EV V2G Share Variation:** This could be a technology with direct competition to utility-scale batteries as they could supply some of the flexibility that would be otherwise supplied by utility-scale batteries (substitution-effect).
- **Battery Cost Variation:** If the cost of installed batteries per kW and kWh increases or decreases, this will have a significant impact on the deployment of utility-scale batteries as they get more/less competitive to other flexibility options.
- **RES Variation:** If the amount PV deployment, increases or decreases, the amount of needed utility-scale batteries will change significantly. Since utility-scale batteries are mostly deployed to provide flexibility and shift PV electricity generation from day to evening/night time (Luderer et al. 2025), the amount needed is directly dependent on the amount of installed PV (and Wind) capacity.

3.4.1. Scenario 1: V2G

To examine the potential and the effects of different V2G rates, two different variations are examined:

1. The base scenario but the min. morning SoC value rises to 60% instead of 40%. (Low V2G Scenario)
2. The base scenario but the V2G participation rate is higher as well as the BEV-DSM rate. Also the battery size per vehicle is higher in this scenario. (High V2G Scenario)

The exact assumptions are shown in Table 3.3.

Scenario 1, the low V2G scenario, assumes a higher min. SoC value that the EVs need to comply to. This scenario will show first the significance of the min. SoC parameter in determining the total available V2G capacity and second represent a more conservative scenario with lower relevance and penetration of V2G. Scenario 2, the high V2G scenario, assumes faster, cheaper, and higher penetration of V2G-enabled battery electric vehicles (BEVs) than in the base case, resulting in a significantly larger share of mobile battery storage.

Year	EV-DSM Rate (Base) [%]	V2G Rate (Base) [%]	EV-DSM Rate (High V2G) [%]	V2G Rate (High V2G) [%]	Battery Size (High V2G) [kWh]
2020	0	0	0	0	50
2025	50	0	50	0	50
2030	50	21	75	39	60
2035	50	26	85	65	72
2040	50	31	95	70	83
2045	50	36	95	75	92
2050	50	41	95	75	100

Table 3.3.: Assumptions for EV DSM and V2G participation in the Base and High V2G scenarios.

The assumption for the larger battery size in the high V2G scenario is based on the TYNDP 2024 scenario methodology report assumptions for EV battery size (ENTSO-E and ENTSO-G 2025). The assumption for the higher V2G participation rate in the high V2G scenario is based on a study which is detailed in chapter 3.6.2, which includes numbers for the expected V2G participation rate in the target years (Bonin et al. 2024) while the assumption for the V2G participation rate in the base scenario is based on the TYNDP 2024 scenario methodology report assumption. The DSM participation rate in the base scenario is based on the fixed value of 50% which was used in the Ariadne Report 2025 (Luderer et al. 2025) while the DSM participation rate in the high V2G scenario is more optimistic as it is expecting higher smart-meter and dynamic tariff rollout as it is currently planned by the federal government as well as in the EU directive.

3.4.2. Scenario 2: Cost Variation

To examine the potential of cost variations on utility-scale batteries the following battery costs are varied:

1. A 20% cost decrease of the capital cost per kW and per kWh of the base level in each target year (Low Cost Scenario)
2. A 20% cost increase of the capital cost per kW and per kWh of the base level in each target year (High Cost Scenario)

Further, a cost sensitivity analysis was conducted with more extreme cost variations of +/- 50% and +100% as well as -99%. The cost of home batteries are changed in the same way as utility-scale battery cost are changed in the scenarios.

The values used in the scenarios and sensitivity analysis are shown in Table 3.4.

Year	Battery CAPEX Base [EURO per kWh (and kW)]	+20%	+50%	+100%	-20%	-50%	-99%
2025	198 (228)	237.6 (273.6)	297.0 (342.0)	396.0 (456.0)	158.4 (182.4)	99.0 (114.0)	1.98 (2.28)
2030	150 (169)	180.0 (202.8)	225.0 (253.5)	300.0 (338.0)	120.0 (135.2)	75.0 (84.5)	1.5 (1.69)
2035	125 (138)	150.0 (165.6)	187.5 (207.0)	250.0 (276.0)	100.0 (110.4)	62.5 (69.0)	1.25 (1.38)
2040	100 (106)	120.0 (127.2)	150.0 (159.0)	200.0 (212.0)	80.0 (84.8)	50.0 (53.0)	1.0 (1.06)
2045	89 (85)	106.8 (102.0)	133.5 (127.5)	178.0 (170.0)	71.2 (68.0)	44.5 (42.5)	0.89 (0.85)
2050	80 (64)	96.0 (76.8)	120.0 (96.0)	160.0 (128.0)	64.0 (51.2)	40.0 (32.0)	0.8 (0.64)

Table 3.4.: Battery CAPEX development and sensitivities over time in EURO per kWh and EURO per kW. Upper row: battery-related CAPEX, lower row: charger-related CAPEX.

3.4.3. Scenario 3: PV Variation

In this scenario the amount of PV deployment is varied to examine the effects on battery deployment. In each target year from 2030 onward, the amount of PV deployment is set fix to a +/- 20% compared to the base scenario. The installed capacities in each scenario in each target year are shown in Table 3.5.

1. A 20% decrease of installed PV capacity from the base level in each target year (Low PV Scenario)
2. A 20% increase of installed PV capacity from the base level in each target year (High PV Scenario)

Year	Installed Base [GW]	PV Scenario	Installed PV in High PV Sce- nario [GW]	Installed PV in Low PV Sce- nario [GW]
2025	110		110	110
2030	235		282	188
2035	339		414	276
2040	406		496	330
2045	425		521	348
2050	483		593	395

Table 3.5.: Installed PV capacity in the different scenarios (Base, High PV, Low PV).

4. Model Results

In this chapter, the results from the PyPSA-DE model are presented. First, the rollout of the different battery technologies across the scenarios is shown, followed by their dispatch patterns. Subsequently, the results of the V2G parameters and the flexibility analysis are presented, along with the findings from the cost sensitivity analysis. Afterwards, the system impacts, based on price development, renewable deployment and curtailment, are examined. Finally, the key findings are summarized.

All main results are also available as interactive HTML figures on an online dashboard, accessible via the link provided in the appendix.

4.1. Battery Technology Rollout

4.1.1. Key Parameters

The key parameters of battery deployment presented in this section include the installed discharge power, installed energy capacity, annual battery discharge, E2P ratio, and annual full cycles. Figure 4.1 shows the installed discharge power of batteries until 2050. The vast majority of utility-scale batteries are built by 2035, reaching about 19 GW of installed discharge capacity and approximately 180 GWh of energy capacity. In 2035, the installed discharge power of EV V2G batteries (20 GW), prosumer home batteries (17 GW), and utility-scale batteries are at similar levels, although the installed energy capacity is dominated by utility-scale batteries.

Since EVs are implemented exogenously in the model with a fixed E2P ratio, their installed power and energy capacities are independent of the system's endogenous flexibility requirements. For prosumer home batteries, an exogenous minimum capacity with a fixed E2P ratio is defined, which is only exceeded to a minor extent. Consequently, any additional system flexibility need provided by batteries is predominantly met by utility-scale batteries.

By 2045, the installed power and energy capacity of EV V2G batteries exceed those of utility-scale batteries, as more V2G-capable EVs are deployed on the roads. Prosumer home batteries reach around 100 GWh of energy capacity in 2045, about half of the utility-scale capacity, but surpass them in power capacity with approximately 40 GW. The figure shows the minimum amount of EV capacity available at any given timestep, as explained in more detail in the subchapter on V2G.

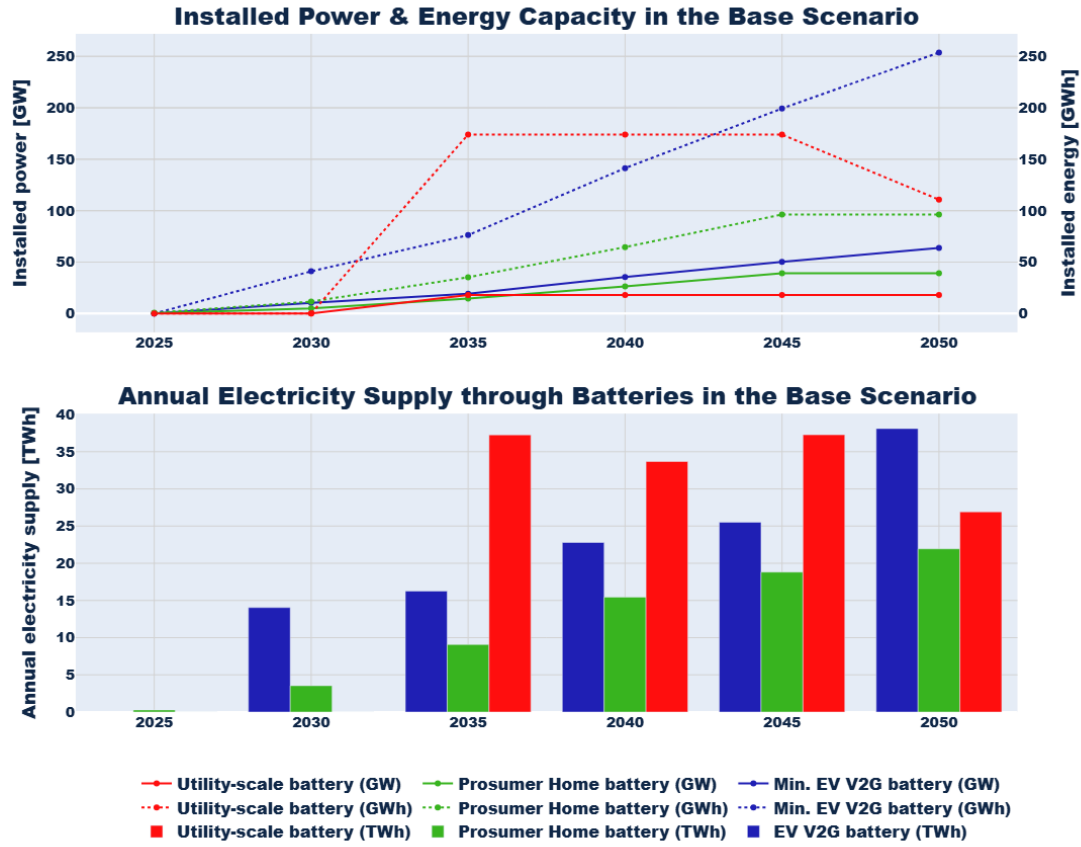


Figure 4.1.: PyPSA-DE Base Scenario results of installed power & energy battery capacity as well as the annual electricity supply from batteries across target years

The same Figure 4.1 also presents the annual electricity supply from batteries until 2050. The total electricity supplied by all battery types in 2045 amounts to roughly 80 TWh, corresponding to about 5-10% of Germany's total annual electricity supply. The largest contribution is provided by utility-scale batteries between 2035 and 2045, followed by a growing share of EV V2G batteries and prosumer home batteries. Although EV V2G batteries exceed utility-scale batteries in installed power and energy capacity by 2045, their share of total electricity supply remains lower. This is because the system prioritizes the full utilization of utility-scale batteries, owing to their higher efficiency and direct connection to the transmission grid, before drawing on the capacity of EV V2G batteries. The E2P ratio for utility-scale batteries remains around 9.5 hours in the base scenario from 2035 to 2045, while the prosumer home battery E2P ratio is approximately 2.4 hours and therefore close to the exogenous ratio implemented as the minimum value. The minimum EV battery E2P ratio is around 4 hours over the same period, based on the exogenous inputs. The annual full cycles of utility-scale batteries are higher in 2035

than in 2045 and, in general, significantly lower than the expected values for currently built utility-scale battery projects (Geladen Podcast 2025a). The corresponding figures for the E2P ratio (Figure A.3) and annual full cycles (Figure A.4) for the base scenario are provided in the appendix.

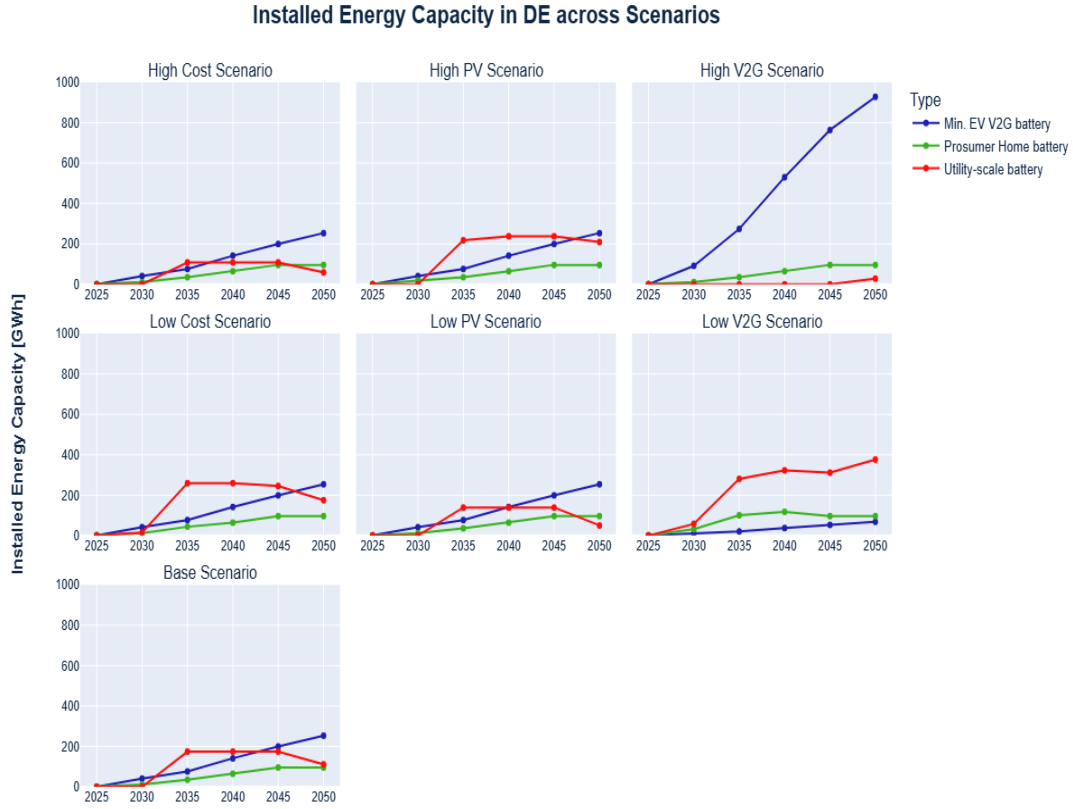


Figure 4.2.: PyPSA-DE results of installed energy capacity across target years across scenarios

In Figure 4.2, the development of installed energy capacity across the different scenarios is illustrated, while the corresponding installed power capacity is provided in the appendix (Figure A.5). The results show that the installed utility-scale battery capacity, both in terms of power and energy, varies considerably between scenarios. In the Low Cost scenario, additional deployment of around +7 GW (+70 GWh) of utility-scale batteries can be observed by 2045, whereas in the High Cost scenario, capacity decreases by approximately −7 GW (−66 GWh). Similarly, the Low PV scenario results in a decline of about −5 GW (−35 GWh), while the High PV scenario leads to an increase of roughly +7 GW (+64 GWh).

The largest impact, however, originates from the assumptions regarding EVs and V2G participation. In the High V2G scenario, utility-scale battery capacity drops sharply by -18 GW (-174 GWh), leaving less than 1 GW of installed capacity in 2045. In contrast, the Low V2G scenario leads to a strong increase of $+16$ GW ($+138$ GWh) in utility-scale batteries. Furthermore, in the Low V2G case, prosumer home battery energy capacity rises significantly, reaching 99 GWh in 2035 and 117 GWh in 2040, thus exceeding the exogenous minimum capacity substantially in these years before returning to around 96 GWh in 2045. This indicates that flexibility requirements at the distribution grid level are met by the exogenous minimum of prosumer home batteries combined with baseline V2G capacities in the other scenarios. In contrast to these shifts, the discharge power of prosumer home batteries remains constant across all scenarios. The figures for the E2P ratios and annual full cycles across the different scenarios are also provided in the appendix (Figures A.7 and A.8). The number of full cycles rises noticeably in 2030 in both the High Cost and Low PV scenarios.

Annual Electricity Supply through Batteries in DE across Scenarios in 2045

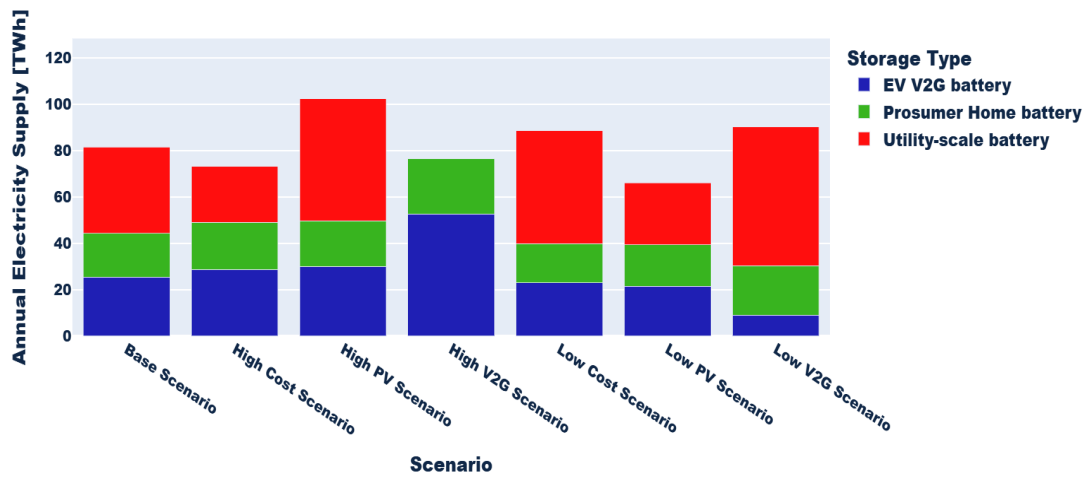


Figure 4.3.: PyPSA-DE results of the annual discharge across target years across scenarios as a stacked bar plot

Figure 4.3 shows the annual electricity supply from batteries in 2045 across all scenarios while Figure A.6 additionally shows the results for all target years. Overall, the total electricity supplied by batteries in 2045 ranges between 65 TWh and 100 TWh across scenarios, with the contribution of individual battery technologies varying substantially. In the Low Cost scenario, most of the electricity is supplied by utility-scale batteries, whereas in the High Cost scenario, the relative contributions of prosumer home batteries and EVs increase slightly, though the total remains below the base case. The Low PV

scenario yields the lowest total electricity supply from batteries, while the High PV scenario achieves the highest contribution. In the Low V2G scenario, total electricity supply is higher than in the base case, mainly due to additional utility-scale batteries. Conversely, in the High V2G scenario, utility-scale batteries are entirely displaced, with most of the supply coming from EVs, resulting in a slightly lower total than in the base scenario.

4.1.2. Dispatch

The total annual dispatch of all technologies in 2045 for the base scenario is shown in Figure A.9 in the appendix, along with the annual dispatch of the individual battery technologies in Figure A.10. The battery dispatch shows a clear seasonal pattern, with significantly higher discharge levels during summer compared to winter, reflecting the strong coupling between battery operation and PV generation. Additional dispatch plots for the year 2045 are provided in the appendix, including an exemplary monthly summer dispatch (Figure A.11), the weekly battery dispatch during the Dunkelflaute period in January (Figure A.12), and exemplary daily dispatch for representative summer (Figure A.13) and winter (Figure A.14) days.

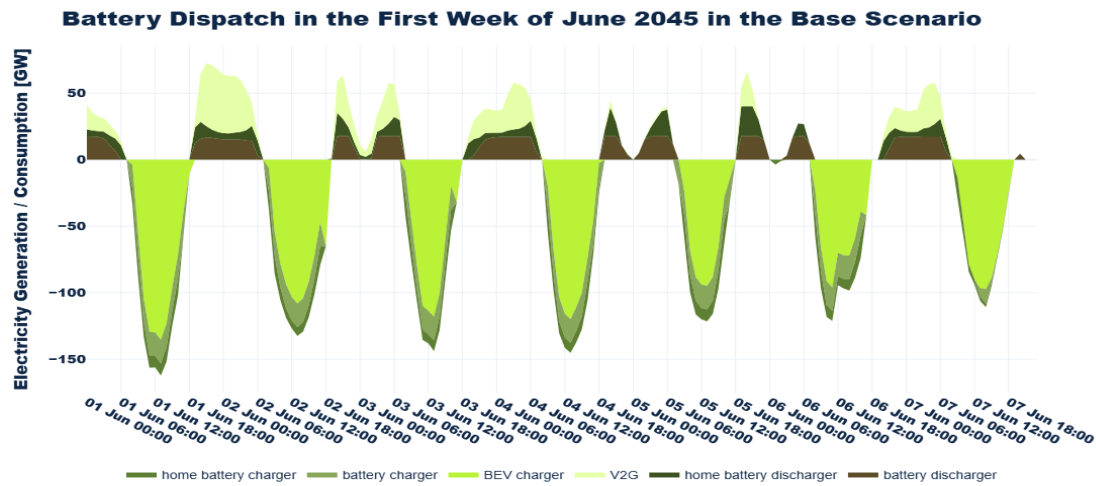


Figure 4.4.: PyPSA-DE results of the battery dispatch in first week of June 2045 in the Base Scenario

Figure 4.4 illustrates the weekly battery dispatch for an exemplary summer week in June 2045 in the base scenario. A distinct day–night pattern can be observed: batteries charge during the daytime when PV generation is high and discharge during the evening and nighttime when PV output declines and demand increases. During discharge, utility-scale batteries typically operate at full capacity first, while home batteries and EVs with V2G functionality are dispatched only when additional flexibility is required. While utility-scale batteries reach their full power capacity regularly for multiple hours in most days, the power capacity limits for prosumer home batteries and V2G batteries are only

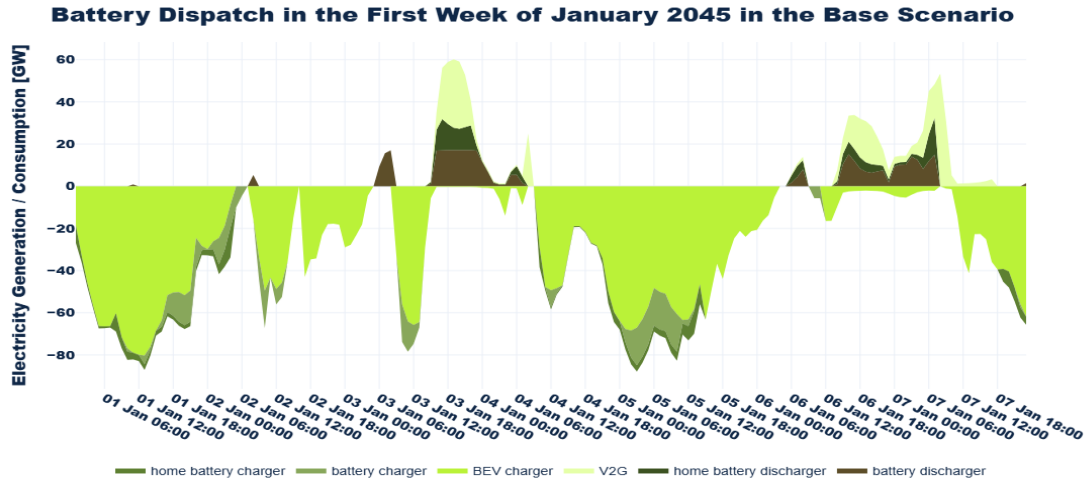


Figure 4.5.: PyPSA-DE results of the battery dispatch in first week of January 2045 in the Base Scenario

reached occasionally. In contrast, Figure 4.5 shows the dispatch pattern for the first week of January 2045, representing a typical winter week. During winter, the dispatch is far less uniform, as battery operation is more closely linked to variations in wind generation. To illustrate this relationship, Figure A.15 in the appendix includes both PV and wind generation alongside the battery dispatch. It becomes evident that batteries are primarily charged during periods of high wind output and discharged during phases of low wind generation. However, due to the intermittent nature of wind power, there are also days when batteries remain inactive, with neither charging nor discharging taking place.

4.2. Vehicle-2-Grid

In Figure 4.6 and Figure 4.7, the discharge power and energy capacity of EV-based V2G batteries are illustrated for the base and both V2G scenarios. The amount of discharge power capacity in the low V2G scenario is identical to that in the base scenario and is therefore fully covered by the corresponding curve in the plot. The available power and energy capacities are substantially higher in the high V2G scenario in 2045, reaching approximately 105–155 GW and 800–2000 GWh, respectively. In comparison, the base scenario reaches about 50–75 GW and 100–500 GWh in 2045, while the low V2G scenario ranges between 50–75 GW and 20–500 GWh. Unlike stationary battery storage, the available power and energy capacities of mobile V2G batteries are time-dependent, as they vary with the number of V2G-enabled EVs currently connected to the grid. To capture this temporal variability, the minimum, mean, and maximum available capacities are shown in the figures. While the available power capacity depends primarily on the share of V2G-enabled EVs connected to the grid at each time step, the available energy capacity is further constrained by the minimum SoC that each vehicle must maintain,

representing a second limiting factor. For a conservative comparison with stationary batteries, the minimum available capacity is used, as it represents the guaranteed capacity that is accessible at all times.

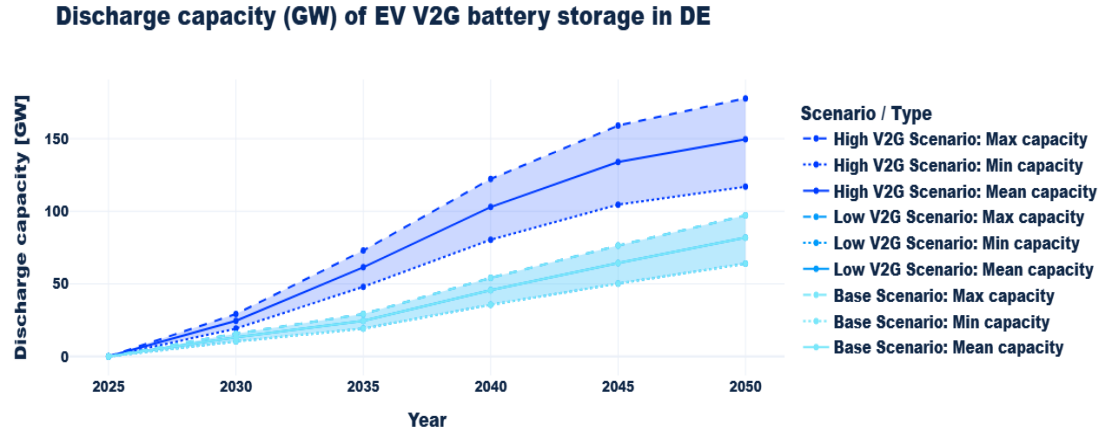


Figure 4.6.: PyPSA-DE results of EV battery discharge capacity values in the Base Scenario compared to the Low and High V2G Scenario

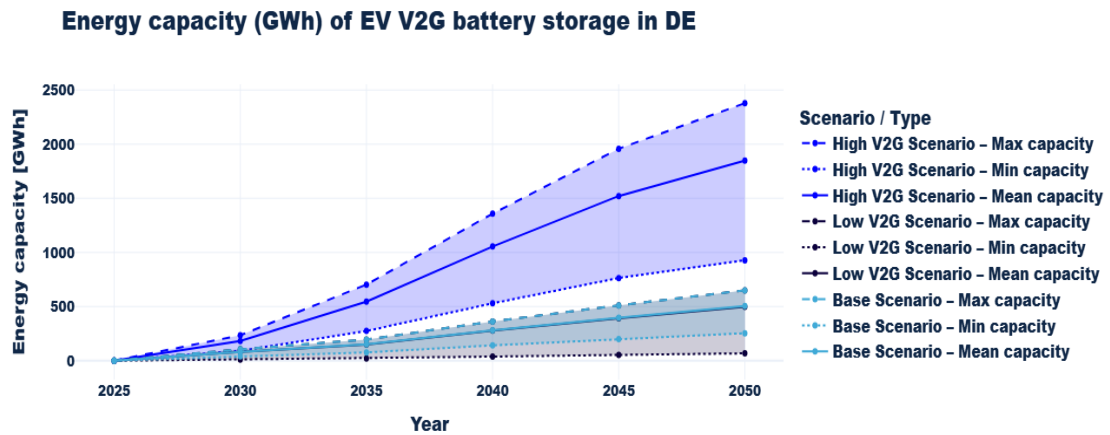


Figure 4.7.: PyPSA-DE results of EV battery energy capacity values in the Base Scenario compared to the Low and High V2G Scenario

In Figure 4.8, the limiting factors of the aggregated EV V2G SoC across the different scenarios are illustrated. The red line represents the maximum amount of energy available, corresponding to the total capacity of EV batteries currently available for V2G, while the green line indicates the minimum SoC reserve that vehicles maintain at all times.

The difference between these two curves defines the potential V2G energy capacity, and the blue line reflects the actual energy stored in the EV batteries.

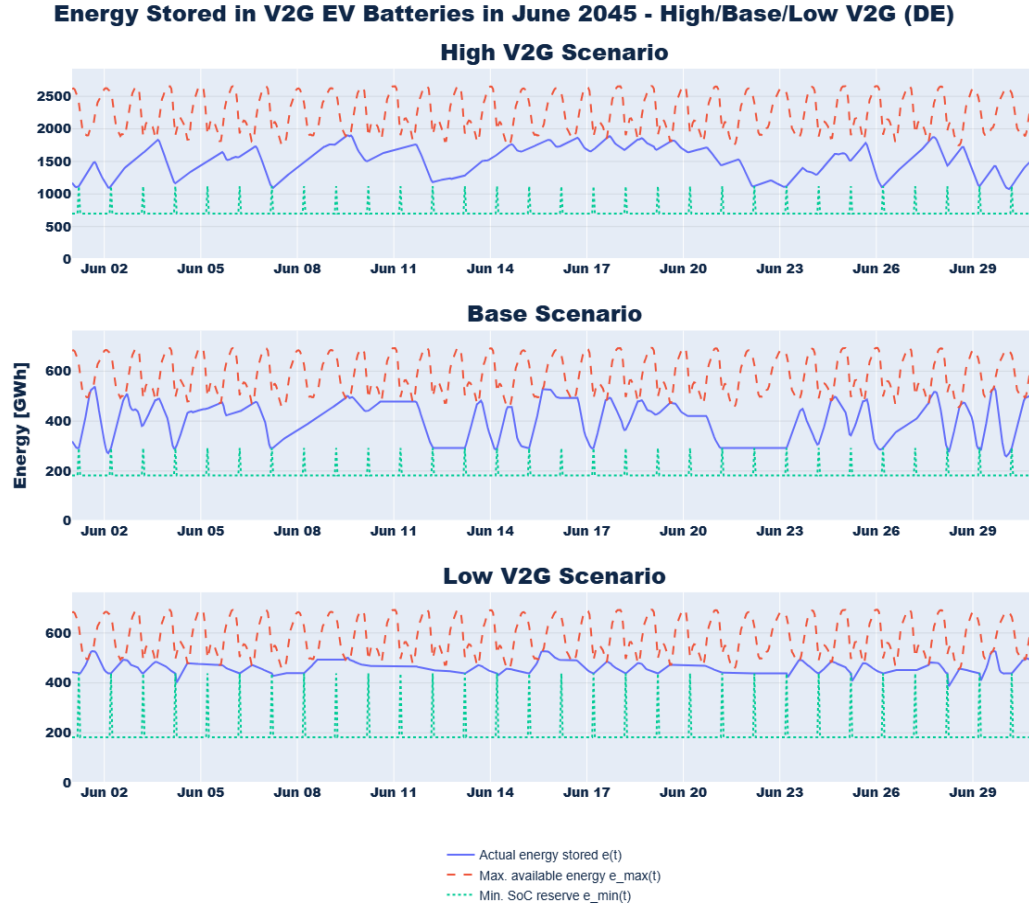


Figure 4.8.: PyPSA-DE results of energy stored in the EV V2G batteries in the High V2G-, Low V2G- and the Base Scenario in June 2045

A comparison between the low V2G and the base scenario reveals that even a moderate increase in the minimum morning SoC reserve, from 40% to 60%, while keeping the minimum SoC reserve for the rest of the day constant, has a substantial impact on the available energy capacity. Although both the upper limit (red line) and the lower limit (green line) fluctuate significantly throughout the day, the actual energy stored (blue line) only rarely approaches the lower bound of the upper limit or the upper bound of the lower limit. Consequently, these two boundaries essentially determine the effective available V2G energy capacity. In quantitative terms, this effective available amount reaches approximately 800 GWh in the high V2G scenario, around 200 GWh in the base scenario, and about 90 GWh in the low V2G scenario. Compared to the base case, the

high V2G scenario shows fewer occurrences of complete discharging down to the minimum SoC, indicating that EV batteries tend to maintain a higher and more stable SoC over longer periods.

Additionally, Figure A.16 in the appendix shows the available power capacity and the actual power dispatch of V2G EVs across scenarios. While the maximum available power capacity is only rarely and briefly reached in the entire month in the base scenario, a complete discharge cycle of the aggregated V2G battery energy capacity occurs almost daily or every few days. This underlines that, in practice, the available energy capacity of V2G batteries is a more critical limiting factor than the available instantaneous power capacity.

4.3. Other Storages and Flexibilities

In Figure 4.9, the development of flexibility needs in the base scenario is illustrated across different target years, differentiated into daily, weekly, and annual components. With the rising share of renewables in the system, overall flexibility requirements increase substantially. While weekly and annual flexibility needs rise only moderately, daily flexibility requirements exhibit by far the strongest growth. By 2045, the daily flexibility requirement reaches approximately 650 TWh, representing a considerable share of the projected total electricity demand in that year. The methodology applied for calculating flexibility needs follows the approach described in RTE - Réseau de Transport d'Électricité (2015) and used in the Ariadne 2025 Report (Luderer et al. 2025). Daily flexibility demand is calculated as the deviation of the residual load from its daily mean value, while weekly flexibility demand is derived from the deviation of daily averages from the weekly mean. This approach ensures that short-term fluctuations are filtered out when analyzing higher time horizons. Annual flexibility demand is determined analogously, focusing on deviations from the monthly mean of the residual load. The differences in flexibility needs across scenarios are shown in Figure A.17 in the appendix. The impact of the PV expansion scenarios on flexibility needs is particularly evident for the daily dimension. Higher PV deployment substantially increases daily flexibility requirements, whereas the effects on weekly and annual flexibility remain negligible. Variations in battery cost and V2G parameters show only minor influences compared to the PV expansion scenarios.

In Figure 4.10, the contributions of different technologies to meeting daily flexibility needs in the base scenario across target years are presented. The contribution of each technology is quantified by iteratively excluding it from the calculation: the flexibility demand is recomputed without the electricity supply or demand of the respective technology, and the difference to the total flexibility demand with all technologies included represents the contribution of the excluded technology (Luderer et al. 2025; RTE - Réseau de Transport d'Électricité 2015). The results indicate that in 2025, battery technologies contribute only marginally to covering daily flexibility needs, while fossil fuels such as coal and gas provide the dominant share. In 2030, fossil fuels, EV flexibility, and cross-border

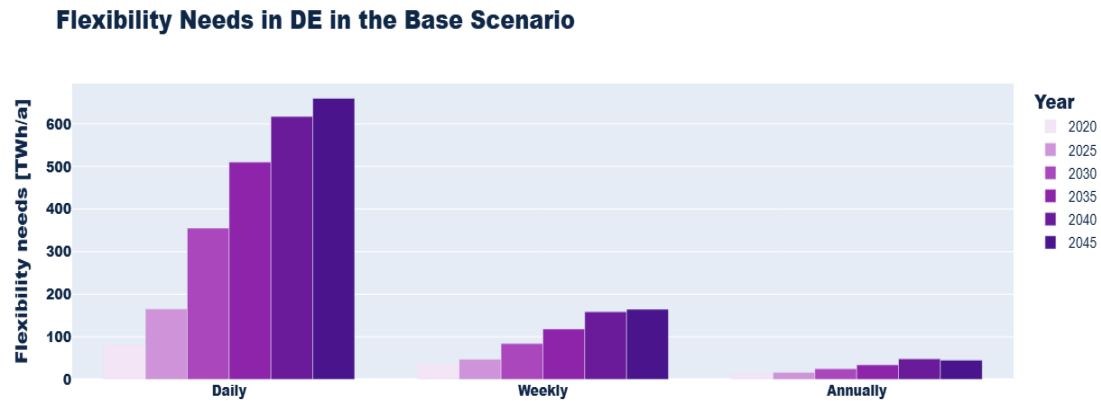


Figure 4.9.: PyPSA-DE flexibility needs in DE in the Base Scenario

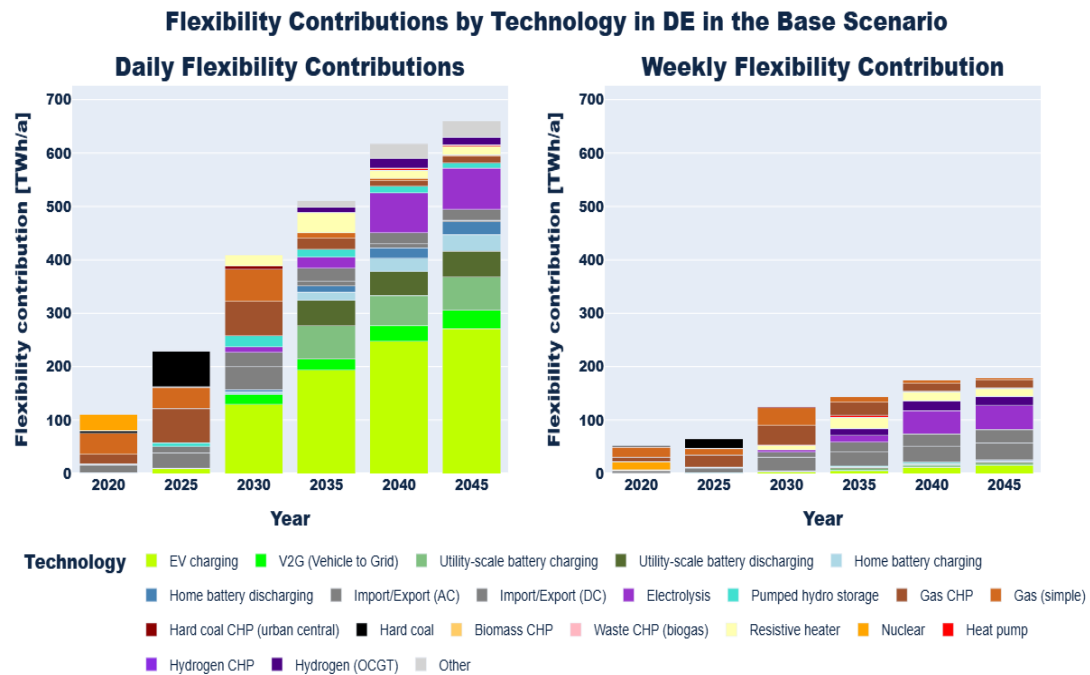


Figure 4.10.: PyPSA-DE flexibility contributions of different technologies in DE in the Base Scenario

electricity exchanges (imports and exports) account for most of the daily flexibility, with pumped hydro storage and resistive heaters contributing smaller shares. By 2035, battery technologies become the most important contributors, collectively providing the major-

ity share, while the share of fossil fuels declines markedly. In 2045, flexible charging of electric vehicles accounts for by far the largest contribution, approximately 260 TWh or around 40% of daily flexibility needs, whereas discharging via V2G plays a smaller role at roughly 25 TWh. The combined charging and discharging of utility-scale batteries and prosumer home batteries form the second-largest contribution, followed by electrolysis. For flexibility requirements on weekly and annual timescales, battery technologies remain of minor importance. Instead, electrolysis and cross-border electricity exchanges provide the largest contributions to balancing system variability in 2040 and 2045.

In Figure 4.11, the contributions of different technologies to daily flexibility needs across scenarios are illustrated.

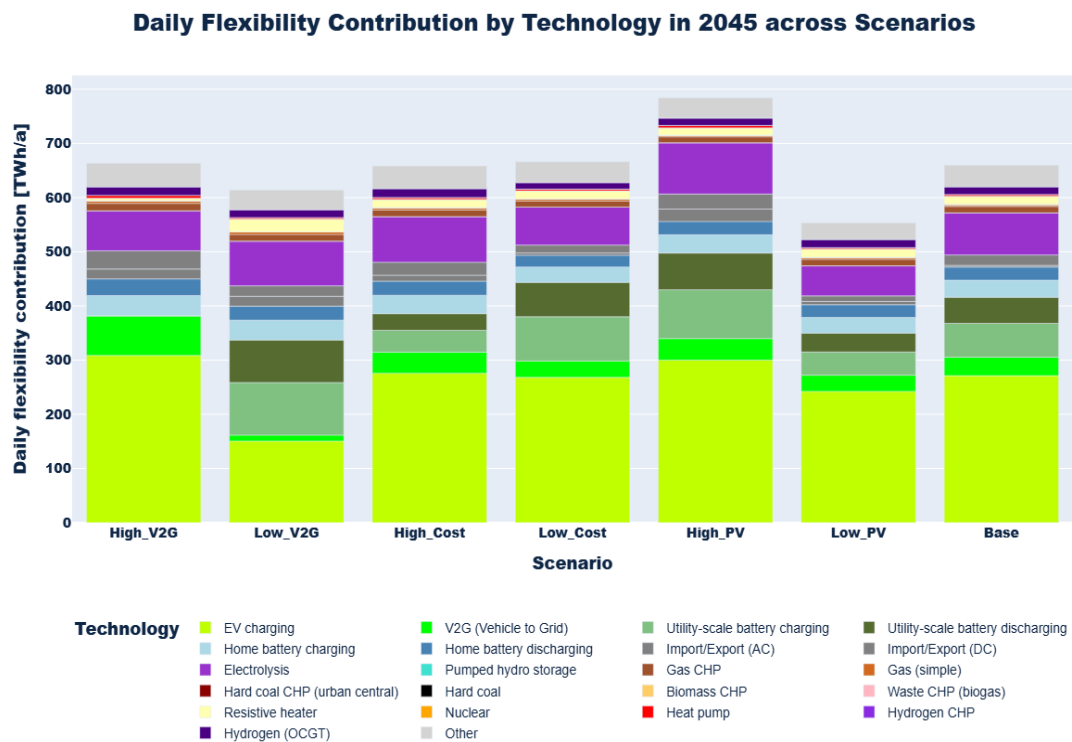


Figure 4.11.: PyPSA-DE daily flexibility contributions of different technologies in DE across scenarios in 2045

In the Low V2G scenario, the reduced availability of V2G resources is directly compensated by higher contributions from other battery technologies, especially from utility-scale batteries, while imports and exports also become more significant. In the High Cost scenario, the utility-scale battery contribution decreases considerably and is primarily replaced by a combination of EV charging, V2G, prosumer home batteries, imports/-exports and electrolysis. Conversely, in the Low Cost scenario, utility-scale batteries

increase their role, largely substituting electrolysis and to a lower degree the other aforementioned technologies. In the High PV scenario, the absolute contribution of utility-scale batteries increases markedly, driven by the higher overall flexibility requirements resulting from the stronger day-night variability of PV generation. Notably, in the High V2G scenario, EV demand-side management and especially V2G discharging increase considerably, replacing most of the contribution provided by utility-scale batteries in the base scenario. The remaining share is compensated by a rise in imports and exports as well as a slight increase in prosumer battery contributions. Overall, the results highlight a clear substitution effect between utility-scale batteries and EV smart charging, V2G, prosumer home batteries, electrolysis and import/export capacities. The effects of the scenarios on weekly and annual flexibility contributions remain relatively minor. The corresponding figures for weekly and annual technology contributions across scenarios in 2045 are provided in the appendix in Figure A.18 and Figure A.19.

4.4. Cost Sensitivity Analysis

In Figure 4.12, the results of the cost sensitivity analysis for the deployment of utility-scale battery power and energy capacity are presented.

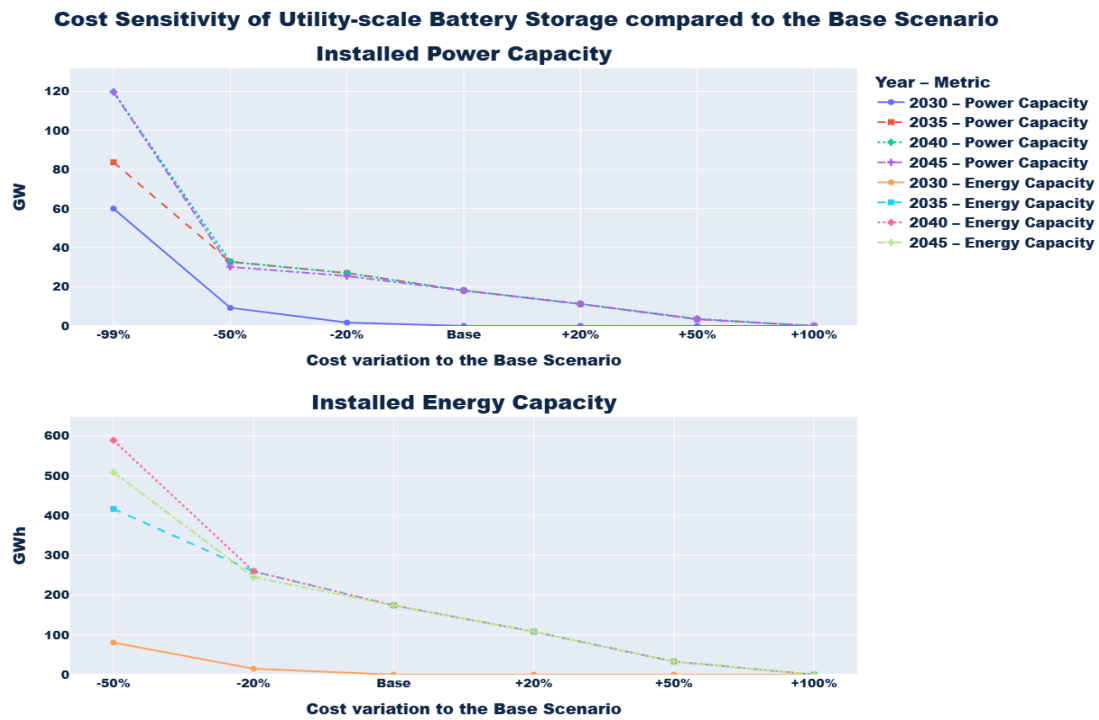


Figure 4.12.: PyPSA-DE utility-scale battery cost sensitivities compared to the base scenario

The analysis clearly demonstrates that variations in investment costs have a pronounced impact on the scale of battery deployment. A doubling of costs results in no deployment of utility-scale batteries at all, as sufficient substitution options, such as other storage technologies, demand-side measures, or imports, are available. This highlights both the sensitivity of battery deployment to potential risks such as supply chain disruptions or trade conflicts, and at the same time, the robustness of the overall energy system due to the presence of alternative flexibility options. Even a 50% cost increase leads to a marked reduction, with only around 40 GWh of energy capacity and 5 GW of power capacity installed between 2035 and 2045, while a 20% increase already reduces the deployment from approximately 175 GWh to 100 GWh (and from 18 GW to 11 GW).

Conversely, cost reductions strongly accelerate the deployment of utility-scale batteries. A 20% cost decrease raises installed energy capacity from roughly 180 GWh to 260 GWh (18 GW to 24 GW). A 50% cost reduction not only increases the total installed capacity but also advances the timing of deployment: around 90 GWh are already commissioned by 2030, increasing to 400 GWh in 2035, 500 GWh in 2040, and approximately 600 GWh by 2045. The effect on installed power capacity is somewhat less pronounced, with about 10 GW in 2030 and 32 GW in 2045. In the extreme case of a 99% cost reduction, deployment reaches approximately 120 GW of power capacity and 18 TWh of energy capacity by 2045, illustrating the transformative potential of drastically lower battery costs.

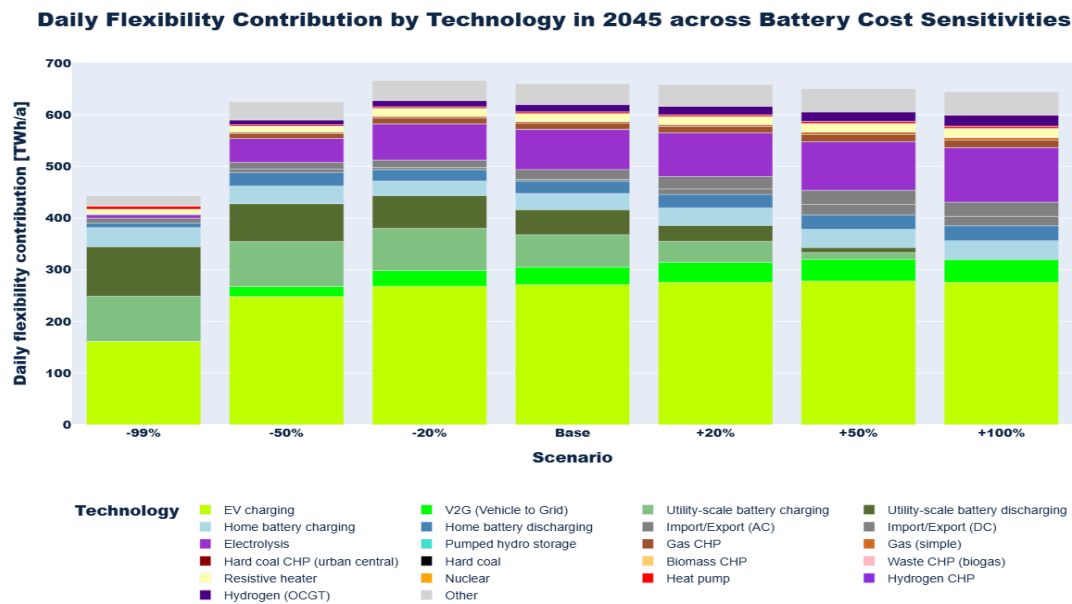


Figure 4.13.: PyPSA-DE impact of battery cost sensitivities on the contribution of different technologies to the daily flexibility needs

In Figure 4.13, the impact of cost variations on the contributions of different technologies to meeting daily flexibility needs is shown. Under a 100% cost increase, no utility-scale batteries are deployed, and the resulting flexibility gap is mainly substituted by electrolysis, cross-border electricity exchanges (imports and exports), and to a lesser extent V2G. A similar pattern is observed with a 50% cost increase, although in this case a small share of utility-scale batteries remains in operation. By contrast, a 20% cost reduction leads to a higher contribution from utility-scale batteries, which primarily substitute electrolysis and imports/exports. At more substantial cost reductions of 50% and beyond, the methodological approach to capture the flexibility contributions appears to underrepresent the full contribution of utility-scale batteries to daily flexibility provision. At a 50% cost reduction, utility-scale batteries begin to contribute meaningfully to weekly flexibility provision, supplying around 10%, while imports/exports and electrolysis remain the dominant sources (Figure A.20 in the appendix). In contrast, under an extreme 99% cost reduction, batteries become the primary flexibility provider, effectively covering the majority of the system's flexibility requirements.

4.5. System Impact

This section analyzes the broader impacts of varying battery deployment on key system indicators, including electricity price volatility, total system costs, renewable generation curtailment, and the expansion of wind and solar capacities.

4.5.1. Price Volatility

In Figure 4.14, the price duration curves for the base scenario are displayed for the target years 2030 to 2045. In 2030, the system shows relatively few very high prices, while a large share of hours lies in the medium-high price range. The year 2035 represents the period of greatest volatility: more than 20% of the time, electricity prices exceed 200 €/MWh, while around 20% of the time prices drop to 0 €/MWh. In the subsequent years, 2040 and 2045, this volatility decreases again, and the price distribution flattens, indicating a stabilization of the market as flexibility options expand while renewable penetration increases.

In Figure 4.15, the annual averages of daily electricity price spreads across scenarios and target years are illustrated. In the base scenario, the spreads increase steadily from around 40 €/MWh in 2020 to approximately 65 €/MWh in 2035, before declining sharply to about 45 €/MWh in 2040 and further to 30 €/MWh by 2045. This trajectory corresponds closely with the findings from the price duration curves, where 2035 also emerges as the year of highest volatility, while the daily spread metric provides complementary insights. The low-cost scenario follows the same pattern but exhibits less pronounced fluctuations. In addition to the presented scenarios, the effect of the battery cost sensitivity analysis on the daily electricity price spreads is shown in Figure A.21 in the appendix. A cost reduction of 50% leads to significantly reduced spreads, remaining

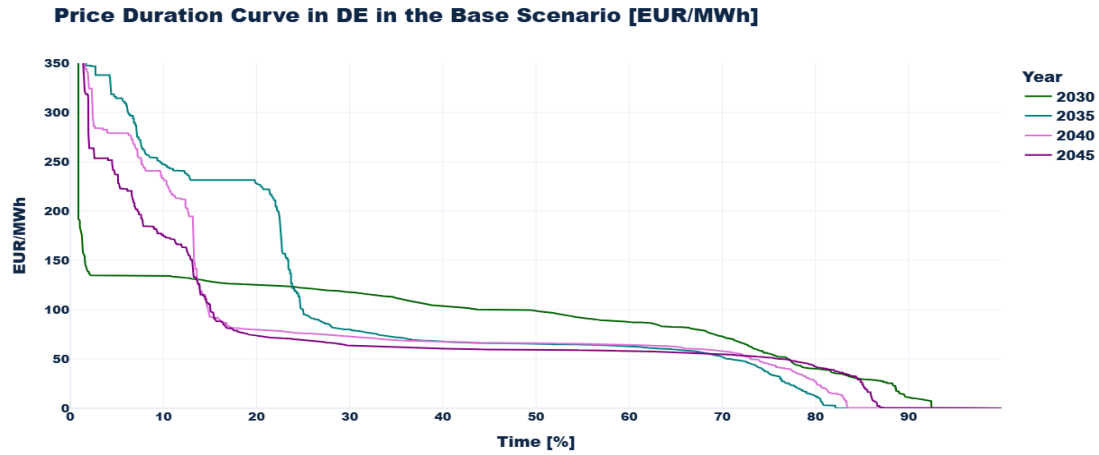


Figure 4.14.: PyPSA-DE price duration curve in DE in the Base Scenario

around 40 €/MWh between 2025 and 2035, and decreasing further to 30 €/MWh in 2040 and 20 €/MWh by 2045.

In contrast, the high-cost scenario results in consistently higher spreads compared to the base case, peaking at around 75 €/MWh in 2035. The more extreme utility-scale battery cost increases of +50% and +100% result in the highest daily spread peaks in 2035 at around 90 €/MWh and 95 €/MWh. The high PV scenario shows slightly higher spreads than the base scenario in 2030, but a less pronounced peak in 2035. The low PV scenario is broadly similar to the base case, though with a somewhat lower spread in 2030. The low V2G scenario mirrors the base case trajectory but at a consistently higher level, with spreads peaking near 70 Euro/MWh in 2035. By comparison, the high V2G scenario leads to substantially lower spreads, closely resembling the pattern of the 50% cost reduction case, thereby highlighting the strong moderating effect of large-scale V2G integration on electricity price volatility. This further shows that utility-scale battery cost increases lead to significant increased electricity price volatility for the whole system. Daily spreads serve as a key indicator for arbitrage revenues of batteries, as they directly determine the daily profitability of charging during low-price periods and discharging at high-price hours.

To complement the price duration curve and daily spread analysis, Figure 4.16 introduces the Top-Bottom (TB) spreads across scenarios for the year 2035. The TB spread is defined as the difference between the highest and lowest hourly electricity prices within a given day, aggregated over the year. It serves as a key benchmark for estimating potential battery revenues from energy arbitrage and is widely used in practice, for example by the BatteryCharts Revenue Index, to assess real-world market opportunities for battery storage systems (Figgenger et al. 2025b). The TB1–TB12 curves in the figure

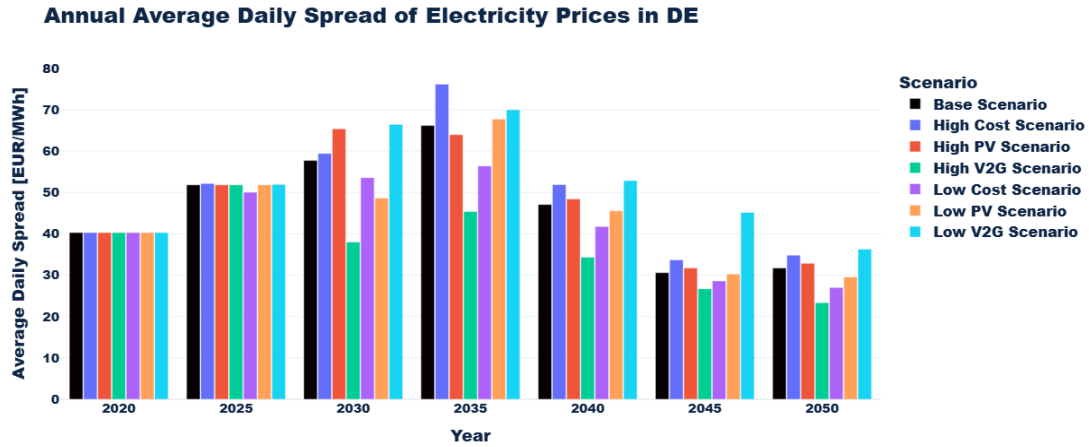


Figure 4.15.: PyPSA-DE annual average daily spread of electricity prices across scenarios

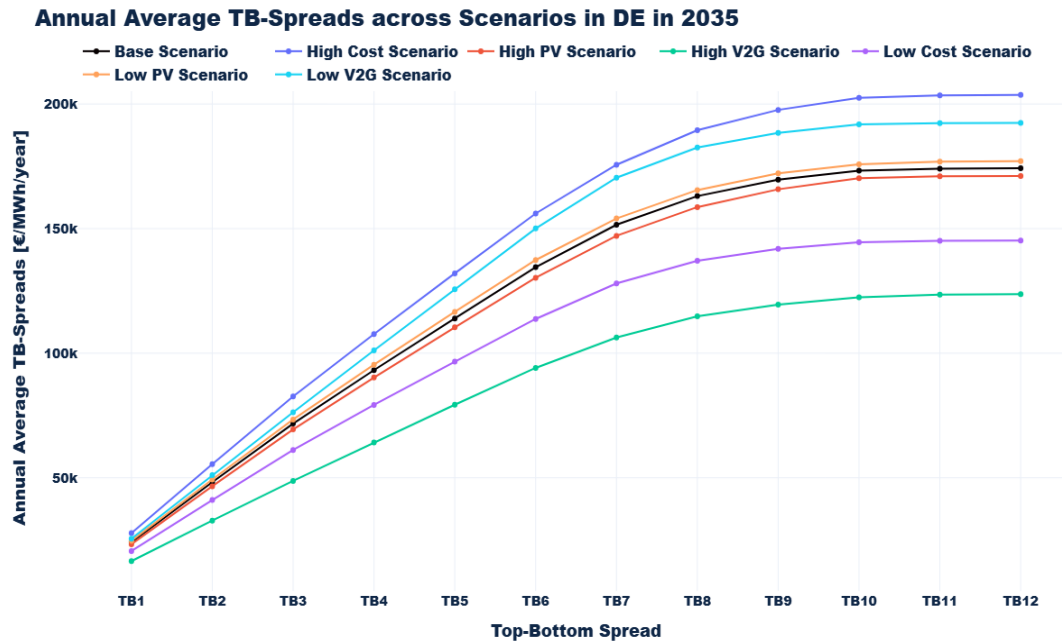


Figure 4.16.: PyPSA-DE annual top-bottom spreads of electricity prices across scenarios in 2035

further refine this analysis by breaking down the spread into increasing cycle durations. TB1 corresponds to the difference between the single highest and lowest hourly price within a day, a suitable proxy for a 1-hour storage system performing one complete charge–discharge cycle. TBn generalizes this concept: TB4 sums the four highest prices

minus the four lowest during a day, representing a 4-hour system, and so forth up to TB12. This methodology mirrors approaches used in battery revenue benchmarks and enables duration-specific assessments of operational profitability.

In this analysis, the TB spread reaches its highest values in 2035, particularly in the High Cost and Low V2G scenarios. This corroborates the findings from the price duration curve and daily spreads, where 2035 also emerged as the year of greatest price volatility. Conversely, scenarios with more favorable conditions for battery deployment, such as the 50% cost reduction or High V2G scenario, exhibit significantly lower TB spreads, highlighting the moderating effect of greater storage capacity and flexibility in the system. In 2045, as shown in Figure A.22 in the appendix, TB spreads decrease markedly across all scenarios. This reduction results from increased deployment of flexibility assets, including utility-scale and prosumer batteries, vehicle-to-grid integration, and sector coupling, leading to flatter price extremes. Notably, despite more ambitious renewable penetration and therefore daily flexibility needs in 2045, the price extremes are dampened compared to 2035. This suggests increasing market saturation and self-cannibalization effects for utility-scale batteries after 2035 on the day-ahead market.

4.5.2. Total System Costs

In Figure A.23, the annual total system costs are presented, encompassing all modeled countries across the different scenarios. For a clearer comparison of scenario effects, Figure 4.17 illustrates the deviations from the base scenario across all scenarios while Figure A.24 in the appendix shows the deviations from the base scenario for the battery cost sensitivity analysis effects on total system costs.

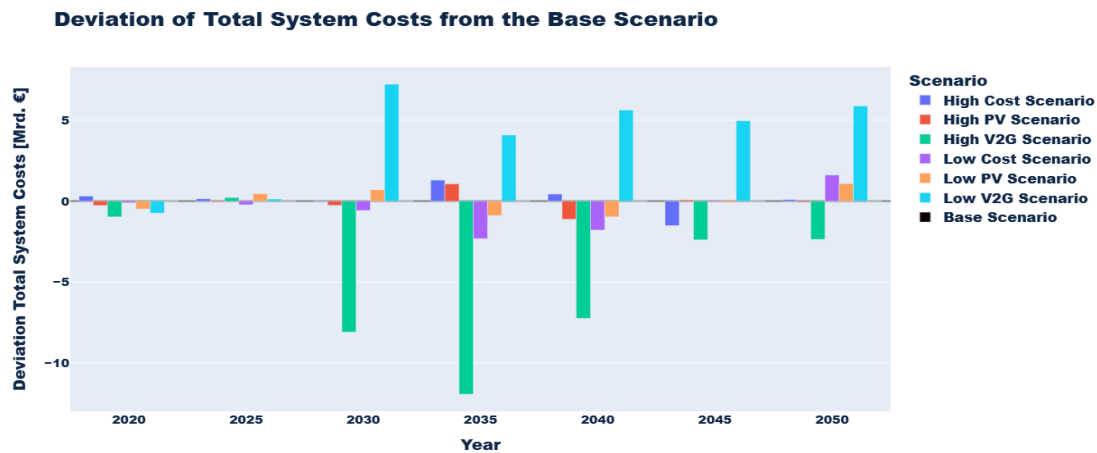


Figure 4.17.: PyPSA-DE deviation of total system costs from the Base Scenario

The High PV, High Cost, and Low PV scenarios exhibit only minor impacts on total system costs, indicating that variations in these parameters exert limited influence on

overall system expenditures. By contrast, the Low Cost scenario leads to a modest reduction in total system costs, while substantially larger decreases are observed in the 50% Cost Reduction and High V2G scenarios. Among these, the High V2G scenario yields the strongest effect, lowering total system costs by up to approximately 12 billion EUR in 2035, followed by the 50% Cost Reduction scenario, which achieves a reduction of around 8 billion EUR in the same year. A notable increase in system costs is observed in the Low V2G scenario, where additional expenditures of roughly 5 billion EUR occur consistently across the target years from 2030 to 2045. A 50% and 100% battery cost increase leads to moderate total system cost increases of 2.5 to 4.5 billion EUR in 2035 and 2040 while small cost reductions of around 2 billion EUR are shown for 2045 and 2050.

4.5.3. PV and Wind Expansion

In Figure A.25 in the Appendix, the installed PV and wind capacities in the base scenario across the target years are shown. By 2045, the base scenario reaches a total variable renewable capacity of approximately 700 GW, consisting of around 420 GW of solar PV, 80 GW of offshore wind, and 200 GW of onshore wind.

Figure 4.18 illustrates the installed PV and wind capacities across scenarios in 2045. Variations in cost assumptions have almost no effect on the expansion of PV and wind capacities.

Installed Generation Capacity (GW) for Solar & Wind across all Scenarios

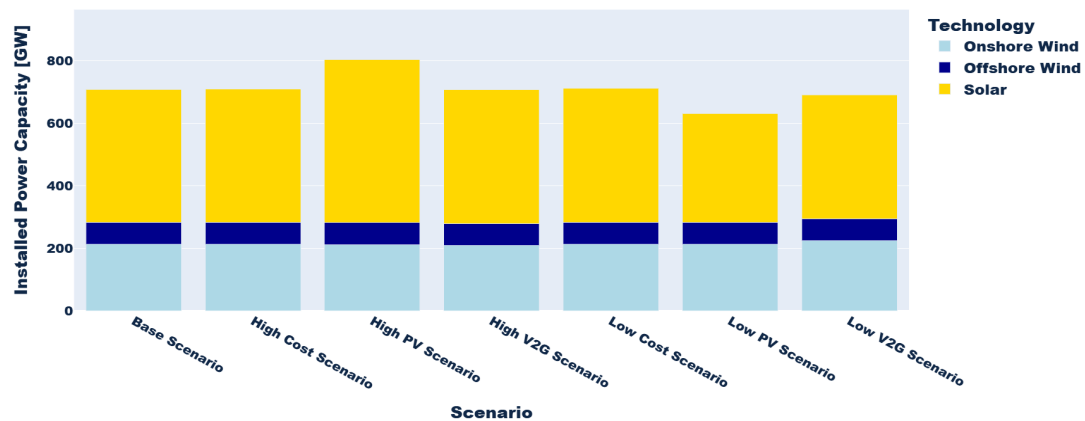


Figure 4.18.: PyPSA-DE installed PV and wind capacities in DE in 2045 across scenarios

Similarly, both the High V2G and Low V2G scenarios exert little influence on renewable capacity build-out. The only notable difference is in the Low V2G scenario, where installed solar capacity is somewhat reduced (396 GW compared to 425 GW in the base case), likely due to the reduced EV flexibility available through smart charging and V2G.

4.5.4. Renewable Curtailment

Figure 4.19 illustrates the total renewable energy curtailment of wind and PV across scenarios in Germany, while Figure A.26 in the appendix shows the impact of battery cost sensitivities on RES curtailment. The results show that only a substantial reduction in battery costs (-50% or higher) has a significant effect on lowering curtailment levels, while a smaller cost decrease (-20%) or a moderate cost increase ($+20\%$) leads to only minor changes. Even a doubling in battery cost leads to just modest impacts on RES curtailment, from around 40 TWh in 2040 in the base scenario to around 47 TWh. The strongest impact is observed in the High PV scenario, where the increased solar capacity results in a pronounced rise in curtailment in absolute terms. Higher V2G penetration rates, on the other hand, tend to slightly reduce curtailment, reflecting the additional flexibility that EVs can provide to the system. RES curtailment peaks in 2040 across scenarios, while curtailment in 2035, 2045 and 2050 is at comparable levels in the base scenario.

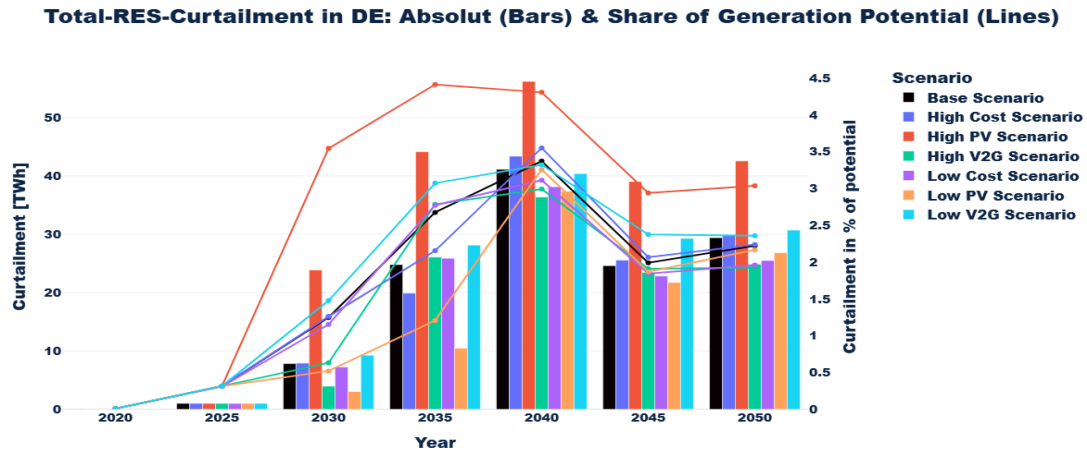


Figure 4.19.: PyPSA-DE RES curtailment in DE across scenarios

4.6. Key Findings

This section summarizes the key findings of the model results, structured into four main themes: technology deployment, flexibilities and substitution effects, battery cost sensitivity and demand elasticity, and overall system impacts.

Technology deployment

- Timing of deployment: Across scenarios, a marked step-change in utility-scale battery deployment occurs between 2030 and 2035. Even though daily flexibility needs

increase already between 2025 and 2030, these needs are mostly covered by fossil fuel power plant and other storage options. Batteries become the cost-optimal substitute as coal is phased out, carbon costs rise, PV share grows, and battery CAPEX declines. In all scenarios, the majority of utility-scale capacity is added in 2035. Thereafter, the daily flexibility needs still grow significantly but the additional needs are increasingly met by prosumer home batteries, EV flexibility (smart charging and V2G) and other flexibilities.

- **Magnitude and role:** In 2035, utility-scale batteries reach around 19 GW and 180 GWh of installed capacity. At that point, utility-scale, prosumer home, and V2G batteries offer similar power capacities, though utility-scale systems dominate in energy terms and serve as the primary flexibility providers. By 2045, rising V2G participation leads to V2G-enabled EV batteries surpassing utility-scale systems in both power and energy capacity, while home batteries exceed utility-scale units in power and reach about half their energy capacity. Despite these shifts in installed capacity, utility-scale batteries remain the main electricity supplier in 2045, delivering around 40 TWh, roughly half of the 80 TWh supplied by all battery technologies. The results highlight that while total battery-supplied electricity in 2045 remains within a consistent range of 65 to 100 TWh across scenarios, the underlying battery technology mix shifts significantly.
- **System-optimal E2P and cycles:** The model indicates a broadly stable system-optimal E2P-ratio of around 9 hours for utility-scale batteries installed in 2035-2045. The annual full cycles tend to be around 200-300 cycles in 2035 to 2045, therefore they tend to be quite a bit lower than the expected annual full cycles for currently build utility-scale battery projects (Geladen Podcast 2025a).
- **Battery Dispatch:** While the battery dispatch in summer follows a clear day-night pattern, the dispatch during winter is less uniform. In summer batteries are mainly used to shift PV generation from day to night while in winter the dispatch is more closely tied to the fluctuations in wind generation. The battery dispatch shows further a clear seasonal pattern, with significantly higher discharge levels during summer compared to winter.

Flexibilities & substitution effects

- **Flexibility mix:** By 2045, battery technologies become the largest contributor to daily flexibility if BEV charging flexibility is included in all scenarios. Smart EV charging provides the dominant share (260 TWh), V2G contributes less (25 TWh), and stationary (utility-scale + prosumer home) batteries form the second-largest block while electrolysis follows as third. In 2025/2030, fossil fuel power plants still cover a large part of daily flexibility but their share decreases sharply afterwards. Weekly/annual flexibility remains the domain of electrolysis and cross-border exchanges mainly in 2045. Batteries play only a minor role at those horizons across scenarios.

- Substitution within battery types: Stationary utility-scale, prosumer home batteries, and mobile V2G batteries compete and substitute each other to a high degree. In the High-V2G scenario, V2G together with home batteries can fully replace utility-scale deployment. Conversely, with low V2G uptake, utility-scale deployment rises significantly. While substitution effects within battery technologies are dominant, other flexibilities, especially electrolysis, import/export and resistive heaters are affected as well.

Battery cost sensitivity & demand elasticity

- High elasticity to CAPEX: Battery build-out is highly sensitive to utility-scale CAPEX. A +20% increase reduces utility-scale energy capacity markedly (e.g., from 175 GWh to 100 GWh, and 18 GW to 11 GW). +50% leads to significant further reduction and +100% essentially halts utility-scale deployment as other options substitute. Conversely, -20% to -50% shifts energy capacity volumes significantly upward (260 GWh and 600 GWh). In the case of -50% cost reduction, significant deployment already starts earlier (2030 instead of 2035).
- PV build-out as demand driver: Higher PV capacities increase daily flexibility needs and raise battery demand: in the High-PV scenario, total battery capacity rises from 475 GWh to 540 GWh, with utility-scale batteries accounting for most of the increase (from 173 GWh to 238 GWh and from 18 GW to 24 GW). Low-PV reduces these needs correspondingly. The renewable mix shift affects E2P only modestly on 2045 level. Timing effects are more pronounced (higher E2P in 2030 with early PV-led deployment).

System impacts

- Curtailment and RES build-out: Changes in total battery deployment changes the amount of renewable curtailment only marginally. Battery cost changes have little impact on endogenously built PV/wind capacities in base sensitivity ranges while the change in PV installations had the biggest effect on curtailment.
- Electricity prices and volatility: Price volatility (captured through average daily spreads, price duration curves and top-bottom spreads) reaches its peak around 2035 across all scenarios. Higher battery integration mitigates extreme price hours and narrows daily spreads, indicating pronounced self-cannibalization effects for utility-scale batteries. In the base scenario, market saturation becomes evident after 2035: by 2045, average daily spreads are roughly halved, even though overall flexibility demand continues to rise. Scenarios with extensive battery deployment, such as the 50% cost reduction case, exhibit a markedly smoother price profile without a pronounced volatility spike in 2035. Similarly, high V2G participation significantly dampens price fluctuations. In contrast, scenarios with increased utility-scale battery costs show substantially higher volatility, underscoring that while alternative

flexibility options can partially compensate for reduced battery deployment, the system must endure higher price volatility as a consequence.

- Total system costs: The High PV, High Cost, and Low PV scenarios show only minor effects on overall system costs, indicating that variations in these dimensions have limited influence on the total expenditures of the energy system. The Low Cost scenario, by contrast, results in a modest reduction in system costs. Substantially larger reductions are achieved in the 50% battery cost reduction scenario and the High V2G scenario. Among these, the High V2G scenario has the most pronounced effect, lowering total system costs by up to around 12 billion euros in 2035.

5. Discussion

Building on the key findings presented in Chapter 4, this chapter discusses the implications of the model results within a broader academic and practical context. The focus lies on the systemic role of utility-scale batteries and the alignment of model outcomes with real-world deployment trends and State-of-the-Art studies. Finally, this chapter reflects on the limitations of the work.

5.1. Ancillary Services Potential

As outlined in Chapter 2, many utility-scale batteries in Germany currently follow a cross-market strategy that combines revenue from ancillary services and energy arbitrage. However, ancillary service markets are not represented in the present model. The combined tendered capacity for FCR, aFRR, and mFRR amounted to approximately 5.4 GW in 2023. Batteries already dominate the FCR market, which is increasingly saturated, and are gaining an increasing share in aFRR. Given their technical capabilities, they could potentially expand further into the mFRR market. Beyond reserve markets, additional flexibility contributions are conceivable, such as black-start capability, virtual inertia provision, redispatch support, and acting as “grid boosters.” Participation in these ancillary markets significantly improves the business case for utility-scale batteries and currently leads to earlier and more profitable deployment than would be expected under an energy-only revenue framework. Recent analysis confirms that the highest returns can be achieved through ancillary services and intraday trading, whereas day-ahead trading yields the lowest profits (Figgenger et al. 2025b). Due to the limited size of these markets, they are expected to become increasingly saturated in the near future, leading to diminishing revenues for battery operators. However, if further mechanisms such as capacity markets or bidding zone divisions are introduced in Germany, new revenue streams could emerge, thereby strengthening deployment incentives for utility-scale batteries.

5.2. Comparison to State-of-the-Art Studies & Current Trends

In Figures 5.1 and 5.2, the results from this study for utility-scale battery deployment (power capacity in GW and energy capacity in GWh) are compared to the results from the Ariadne Reports (Kopernikus-Projekt Ariadne 2021; Luderer et al. 2025) and the German grid development plan. Additionally, the current grid-connection requests for utility-scale batteries are shown and discussed.

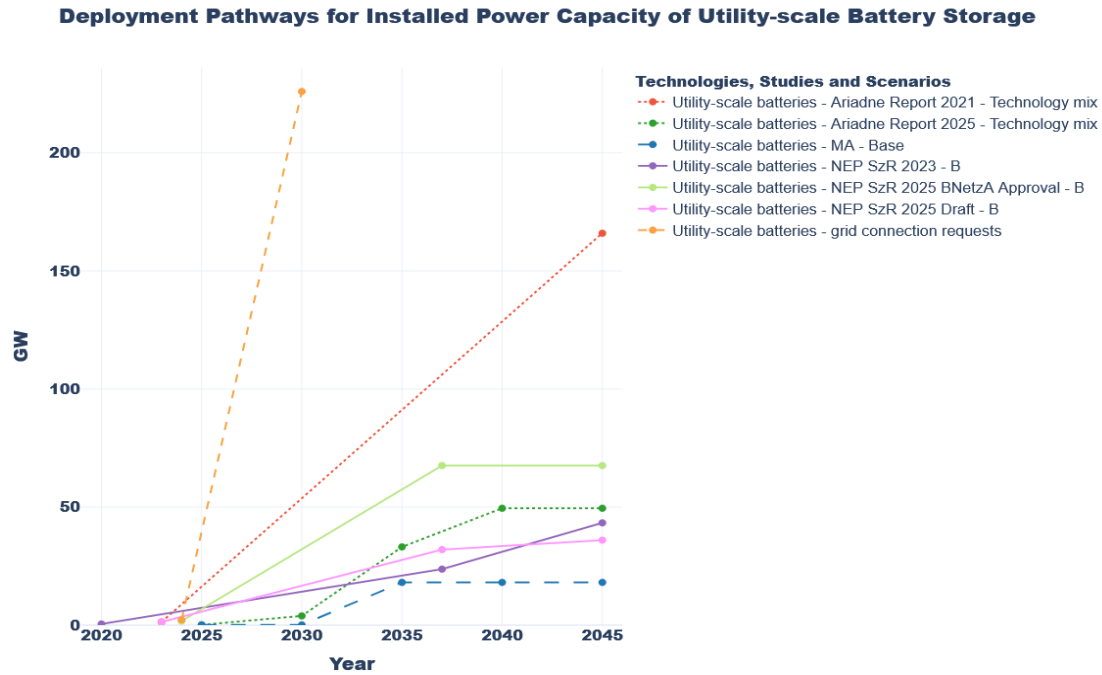


Figure 5.1.: Results comparison of utility-scale battery deployment to findings in other studies for installed power capacity

While this thesis reports the lowest installed power capacity among the studies, the installed energy capacity is comparatively high. The primary reason is the different treatment of the expected E2P-ratio. The scenario frameworks in the grid development plans and the Ariadne 2021 report typically derive energy capacity by applying a fixed E2P-ratio to installed power capacity (NEP Szenariorahmenentwurf 2025; NEP Szenariorahmen 2023; Kopernikus-Projekt Ariadne 2021), whereas in this thesis the system-optimal E2P ratio is determined endogenously by the model. The divergence in E2P ratios from current battery projects and the model optimized E2P ratio arises from the different revenue environments: in practice, battery operators optimize for high cycling frequencies by leveraging top-bottom spreads in ancillary and intraday markets, which favor shorter storage durations. In contrast, the model assumes a day-ahead energy-only setting, where longer storage durations are optimal due to the lower frequency and magnitude of arbitrage opportunities. This can be seen when comparing the TB-spreads in Figure 4.16 with current TB-spreads as shown in Figgner et al. 2025b. The high E2P ratio in the model is resulting in significantly lower full annual cycles compared to the expectations for currently installed battery projects (Geladen Podcast 2025a). The Ariadne 2025 report follows a similar endogenous approach and arrives at a comparable E2P ratio, yet projects significantly higher installed capacity overall. This divergence is mainly attributable to differences in model scope, particularly the explicit representation

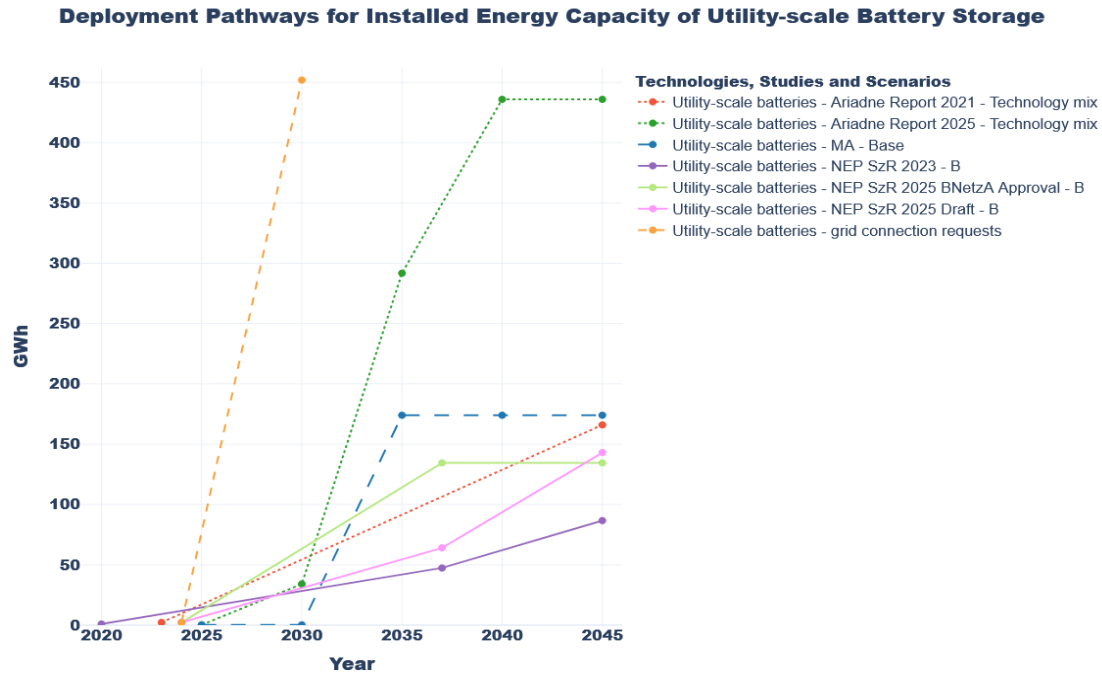


Figure 5.2.: Results comparison of utility-scale battery deployment to findings in other studies for installed energy capacity

of V2G-capable EVs and market-oriented prosumer home batteries, as well as the higher spatial resolution used in Luderer et al. 2025. In this thesis, most utility-scale batteries are installed between 2030 and 2035, while several external studies anticipate continued growth beyond 2035. An exception is the BNetzA approval of the NEP 2025 scenario framework case, which implies a flatter post-2037 trajectory (NEP Szenariorahmenentwurf 2025).

The current wave of grid-connection requests exceeds the system-optimal utility-scale build-out by a wide margin as well as the deployment trajectory occurs much earlier in time. For comparability, these requests were normalized to an assumed E2P of 2 hours, consistent with the prevailing configuration of today's projects (NEP Szenariorahmenentwurf 2025). The fact that the model concentrates installations around 2035, whereas connection requests surge already from now to 2030, can be explained to some degree by the differing revenue stacks mentioned above. As a result, the current profitability and operating strategies of utility-scale batteries differ from the modeled world, in which only day-ahead arbitrage revenues are captured. Given that energy-arbitrage value is likely to grow in importance as ancillary-services markets saturate, the model may better reflect the business case in later years. Moreover, daily spreads and top-bottom spreads of electricity prices peak around 2035 in the model and decline thereafter. As these

are indicators for possible energy arbitrage revenue for utility-scale batteries, rational operators may anticipate this and bring forward investments and battery installations. In practice, today's cross-market projects appear highly profitable, with reported expected payback times on the order of six to seven years (Geladen Podcast 2025a). At the same time, market-oriented prosumer storage and V2G are not yet present at scale, so current utility-scale projects face limited displacement from these competing options. Utility-scale battery project deployment is, however, constrained by grid-access bottlenecks (permitting and limited suitable connection points), which can shape the pace of near-term build-out (Grimm 2025).

Further, even under an extreme cost-reduction sensitivity (-99% battery CAPEX), the model builds only about 120 GW of utility-scale battery power by 2045. This suggests an effective upper bound on the system need for battery power capacity in the analyzed framework. Beyond that level, diminishing marginal system value limit further expansion.

5.3. Limitations of This Work

While the PyPSA-DE framework enables a detailed capacity expansion and dispatch analysis, several simplifications and assumptions remain that affect the interpretation of the results. First, ancillary service and capacity markets were excluded, potentially underestimating the current deployment incentives for flexible technologies, as already mentioned above. Second, the spatial resolution is aggregated to a single German bidding zone, which simplifies congestion and redispatch dynamics. Moreover, the large-scale deployment of utility-scale batteries may in practice be constrained by technical and administrative factors that are not represented in the model. These include limitations in grid connection availability and lengthy permitting procedures. Such real-world barriers could significantly slow down the pace of battery deployment (Grimm 2025). Moreover, behavioral aspects of prosumers and EV users remain highly uncertain. Real-world adoption rates and operating behavior may diverge significantly from the modeled assumptions. The uptake of home batteries and V2G-capable vehicles depends strongly on the future policy framework, market design, and consumer acceptance, all of which shape the economic viability of these technologies.

Another limitation concerns the current modeling of prosumer home batteries. The incentive for self-consumption optimization is very limited, leading to an underestimation of prosumer battery installations. To approximate a more realistic uptake, an exogenous minimum capacity for market-oriented prosumer home batteries was introduced. This simplification serves as a proxy to represent the expected diffusion of prosumer batteries and is further discussed in Chapter 3. With respect to electric vehicles, two additional simplifications were made in the model that slightly distort the results. First, the lower bound for the aggregated EV SoC, representing the energy reserve that remains constantly charged, is set too high in the base scenario. This restricts the available energy

capacity in the model. The reason lies in the definition of the EV V2G battery store component (see Equation 3.12), which does not yet account for the hourly availability profile. The correct implementation would multiply Equation 3.12 by the availability profile. A sensitivity run applying this correction was conducted, and the results are shown in Figures A.27 and A.28 in the appendix, indicating a moderately higher effective available energy capacity for EVs. Second, the current implementation of the smart charging option likely overestimates the flexibility potential of EVs. Although the EV DSM participation rate is limited to 50% of the fleet in the base scenario, the aggregated model representation allows shifting almost all EV charging demand. This effect arises from the aggregation of EV capacity: 50% of flexible capacity is sufficient to shift total charging demand to selected hours, leaving other hours with no charging activity at all. A more accurate representation could involve introducing an additional inflexible EV charging load that remains unaffected by the DSM option. This component would directly charge through the BEV charger link without intermediate storage in the EV battery. The total EV load would then be divided into flexible and inflexible portions, based on the DSM participation share parameter. Despite these simplifications, the presented results provide a transparent and robust assessment of the potential role of utility-scale batteries in the German energy transition.

6. Conclusion

The findings of this thesis highlight the central role of utility-scale battery energy storage systems in enabling Germany's transition toward a climate-neutral energy system by 2045. Using the open-source PyPSA-DE framework, this study systematically analyzed the deployment of different battery technologies under various scenarios. The results provide a differentiated understanding of how utility-scale batteries interact with other flexibility options such as prosumer home batteries and Vehicle-to-Grid systems, and how these interactions shape the optimal design of a future renewable-dominated power system. The model results demonstrate that batteries will become a core short-term flexibility technology, balancing daily fluctuations in renewable generation and contributing to price stabilization in the electricity spot markets. While utility-scale batteries will become the dominant battery technology in the short- and medium term, it is likely that prosumer home batteries and V2G enabled EVs become more relevant in the longer term. For the scenario with high V2G penetration, almost no utility-scale get installed as the daily flexibility provided by batteries is supplied by V2G and prosumer home batteries. The model suggests a significant uptake in utility-scale battery installations between 2030 and 2035, reaching a combined total of approximately 400–500 GWh of installed stationary storage capacity and effective available V2G battery capacity by 2045 across most scenarios. The share of this capacity contributed by utility-scale batteries varies substantially depending on the respective scenario assumptions. As the model captures the full business case of batteries only limited, their current profitability as well as the high current grid-connection requests suggests an already earlier uptake of utility-scale batteries. The cost sensitivity analysis reveals a strong elasticity of battery deployment with respect to capital cost changes. A 50% cost decrease nearly doubles installed energy capacity, confirming that the pace of technological cost decline will remain one of the most decisive factors for large-scale adoption. Conversely, a 100% cost increase from the base scenario projections leads to almost no deployment of utility-scale batteries and significantly higher price volatility in 2035 and 2040. This elasticity indicates that other flexibility options can effectively substitute utility-scale battery storage for daily balancing needs, although this comes at the cost of increased price peaks and volatility. The primary substitutes in this context are other battery types, electrolysis, and electricity imports and exports. Total system costs can be especially reduced in the High V2G scenario as well as in the 50% battery cost reduction case. The High PV and Low PV scenarios, as well as the High Cost scenario, result in only minor reductions in total system costs.

6.1. Further Research Topics

Future research could enhance the representation of home batteries in the model by explicitly distinguishing between market-oriented and system-blind self-consumption-oriented operating modes. As outlined in Chapter 3, the current approach focuses solely on prosumer batteries participating in market operation, which provides valuable insights into substitution effects with utility-scale storage but neglects the flexibility potential of system-blind, self-consumption-driven batteries. A refined modeling framework could therefore integrate both behavioral types, for example by adopting the dual-mode methodology proposed by Guéret, Schill, and Schmidt 2025. This would enable a more realistic assessment of the aggregate flexibility contribution of prosumer home battery storage systems and their role in balancing renewable generation at different spatial and temporal scales.

Moreover, further development of the smart charging implementation, as described in the limitations section in Chapter 5, could improve the representation of EV flexibility. Dividing the EV load into a flexible and an inflexible share would allow a more detailed depiction of charging behavior and the associated demand-side management potential. In addition, the current model aggregates all EVs into a single national fleet, which introduces simplifications regarding spatial and temporal charging availability. Disaggregating the fleet into multiple groups with differentiated driving and parking profiles could capture regional and behavioral heterogeneity more accurately, thereby improving the estimation of available flexibility from the transport sector.

While this study focused on the “when” and “how much” of battery deployment, it did not explicitly address the “where.” Future analyzes could investigate the spatial distribution of utility-scale batteries within Germany, taking into account grid constraints, connection availability, and locational value for system stability. Such spatially resolved modeling would provide valuable insights for grid planning and investment prioritization. Finally, future work could explore market segments and use-cases for utility-scale batteries that are not represented in the current framework, as discussed in Chapter 3 and 5. This includes, for instance, participation in ancillary service markets, capacity mechanisms, or grid support applications, which could further enhance the economic viability and deployment potential of large-scale battery systems.

Bibliography

50Hertz Transmission GmbH et al. (2023). *Szenariorahmen zum Netzentwicklungsplan Strom 2037 mit Ausblick 2045, Version 2023*. Übertragungsnetzbetreiber (ÜNB). URL: https://www.netzentwicklungsplan.de/sites/default/files/2023-01/Szenariorahmen_2037_Genehmigung.pdf (visited on 04/17/2025).

50Hertz Transmission GmbH et al. (2025). *Szenariorahmen zum Netzentwicklungsplan Strom 2037/2045, Version 2025*. Übertragungsnetzbetreiber (ÜNB). URL: https://www.netzentwicklungsplan.de/sites/default/files/2024-07/Szenariorahmenentwurf_NE_P2037_2025_1.pdf (visited on 04/17/2025).

Agora Energiewende (2021). *Klimaneutrales Deutschland 2045 - Wie Deutschland seine Klimaziele schon vor 2050 erreichen kann*. URL: <https://www.agora-energiewende.de/publikationen/klimaneutrales-deutschland-2045-1> (visited on 04/17/2025).

Agora Energiewende (2024). *Klimaneutrales Deutschland - Von der Zielsetzung zur Umsetzung*. URL: <https://www.agora-energiewende.de/publikationen/klimaneutrales-deutschland-studie> (visited on 04/17/2025).

Antunes, Rafael (2024). *Current trends on Ancillary Services Market for Batteries in Germany*. URL: <https://synertics.io/blog/151/current-trends-on-ancillary-services-market-for-batteries-in-germany> (visited on 09/24/2025).

Ariadne-Project Website (2025). *Model documentation: PyPSA*. URL: <https://ariadneprojekt.de/modell-dokumentation-pypsa/> (visited on 04/22/2025).

BloombergNEF (2025). *Global Energy Storage Growth Upheld by New Markets*. URL: <https://about.bnef.com/insights/clean-energy/global-energy-storage-growth-upheld-by-new-markets/> (visited on 09/24/2025).

Bonin, Michael von et al. (2024). *Kurzstudie: Ladeprofile von elektrischen Fahrzeugen – Modellierung für den Szenariorahmen des Netzentwicklungsplans 2025*. Abschlussbericht. Fraunhofer-Institut für Energiewirtschaft und Energiesystemtechnik IEE. URL: <https://www.netzentwicklungsplan.de/en/nep-aktuell/netzentwicklungsplan-20372045-2025> (visited on 04/24/2025).

Brown, T. et al. (2018). “Synergies of sector coupling and transmission reinforcement in a cost-optimised, highly renewable European energy system”. In: *Energy* 160, pp. 720–

739. DOI: 10.1016/j.energy.2018.06.222. URL: <https://doi.org/10.1016/j.energy.2018.06.222> (visited on 09/24/2025).

Bundesministerium für Wirtschaft und Energie (2023). *Stromspeicher-Strategie: Handlungsfelder und Maßnahmen für eine anhaltende Ausbaudynamik und optimale Systemintegration von Stromspeichern*. Stromspeicher-Strategie. Bundesministerium für Wirtschaft und Energie. URL: <https://www.bmwk.de/Redaktion/DE/Artikel/Energie/Energiespeicher/stromspeicher-strategie.html> (visited on 04/24/2025).

Bundesministerium für Wirtschaft und Energie (2024). *Strommarktdesign der Zukunft: Optionen für ein sicheres, bezahlbares und nachhaltiges Stromsystem*. URL: <https://www.bundeswirtschaftsministerium.de/Redaktion/DE/Publikationen/Energie/20240801-strommarktdesign-der-zukunft.pdf> (visited on 09/24/2025).

Bundesnetzagentur (2023). *Festlegungsverfahren zur Integration von steuerbaren Verbrauchseinrichtungen und steuerbaren Netzanschlüssen nach § 14a Energiewirtschaftsgesetz*. URL: https://www.bundesnetzagentur.de/DE/Beschlusskammern/BK06/BK6_83_Zug_Mess/841_SteuVE/BK6_SteuVE_node.html (visited on 11/11/2025).

Bundesnetzagentur (2025). *Genehmigung des Szenariorahmens für den Netzentwicklungsplan Strom 2025-2037/2045*. Genehmigung des Szenariorahmens. Bundesnetzagentur. URL: <https://www.netzentwicklungsplan.de/en/nep-aktuell/netzentwicklungsplan-20372045-2025> (visited on 04/24/2025).

Bundesnetzagentur and Bundeskartellamt (2024). *Monitoring Report 2024: Developments in the Electricity and Gas Markets*. URL: https://www.bundesnetzagentur.de/SharedDocs/Downloads/EN/Areas/ElectricityGas/Monitoring/key_monitoringreport2024.pdf (visited on 09/24/2025).

CDU/CSU and SPD (2025). *Koalitionsvertrag - Verantwortung für Deutschland (barrierefreie Version)*. URL: <https://www.cdu.de/app/uploads/2025/04/Koalitionsvertrag-%E2%80%93-barrierefreie-Version.pdf> (visited on 11/11/2025).

Cebulla, Felix et al. (2018). “How much electrical energy storage do we need? A synthesis for the U.S., Europe, and Germany”. In: *Journal of Cleaner Production* 181, pp. 449–459. DOI: 10.1016/j.jclepro.2018.01.144. URL: <https://doi.org/10.1016/j.jclepro.2018.01.144> (visited on 04/17/2025).

Claire, Arthur (2025). *Battery Cell Chemistry in BESS: LFP vs. NMC - Which Is Better?* URL: <https://sinovoltaics.com/energy-storage/storage/battery-cell-chemistry-in-bess-lfp-vs-nmc-which-is-better> (visited on 11/11/2025).

- Cole, Wesley and Akash Karmakar (2023). *Cost Projections for Utility-Scale Battery Storage: 2023 Update*. Technical Report NREL/TP-6A40-85332. National Renewable Energy Laboratory (NREL). URL: <https://www.nrel.gov/docs/fy23osti/85332.pdf> (visited on 11/11/2025).
- Deutscher Bundestag (2023). *Gesetz zum Neustart der Digitalisierung der Energiewende (GNDEW)*. URL: <https://dip.bundestag.de/vorgang/gesetz-zum-neustart-der-digitalisierung-der-energiewende/296240> (visited on 11/11/2025).
- Deutscher Bundestag (2024). *15 Millionen E-Autos in Deutschland bis 2030 als Ziel*. URL: <https://www.bundestag.de/presse/hib/kurzmeldungen-997098> (visited on 11/11/2025).
- Droese, Dominik et al. (2023). “Economical Comparison of Smart Bi-directional Charging Principles”. In: *Energy Storage Conference 2023 (ESC 2023)*. IET Conference Publications. The Institution of Engineering and Technology. DOI: 10.1049/icp.2023.3094. URL: <https://doi.org/10.1049/icp.2023.3094> (visited on 04/24/2025).
- e.venture (2024). *Stromverbrauch bis 2045 – Weniger Stromverbrauch, mehr Zeit für den Netzausbau*. e.venture. URL: https://e-vc.org/wp-content/uploads/2501_Stromverbrauchsstudie_Abstract-3.pdf (visited on 04/17/2025).
- Energiefinanzierungsgesetz (2025). *§21 EnFG - Umlageerhebung bei Stromspeichern und Verlustenergie*. URL: https://www.gesetze-im-internet.de/enfg/__21.html (visited on 11/11/2025).
- Energiewirtschaftsgesetz (2024). *§14a EnWG - Netzorientierte Steuerung von steuerbaren Verbrauchseinrichtungen und steuerbaren Netzanschlüssen*. URL: https://www.gesetze-im-internet.de/enwg_2005/__14a.html (visited on 11/11/2025).
- Enkhardt, Sandra (2025). *Übertragungsnetzbetreibern liegen zum Jahreswechsel 650 Anschlussanfragen für große Batteriespeicher mit 226 Gigawatt vor*. URL: <https://www.pv-magazine.de/2025/01/13/uebertragungsnetzbetreibern-liegen-zum-jahreswechsel-650-anschlussanfragen-fuer-grosse-batteriespeicher-mit-226-gigawatt-vor/> (visited on 02/20/2025).
- ENTSO-E and ENTSO-G (2025). *TYNDP 2024 Scenarios Methodology Report – Final Version*. Tech. rep. ENTSO-E / ENTSG. URL: https://2024.entsoe-tyndp-scenarios.eu/wp-content/uploads/2025/01/TYNDP_2024_Scenarios_Methodology_Report_Final_Version_250128.pdf (visited on 09/24/2025).
- EUPD Group (2025). *Germany to Transform Energy Usage with Dynamic Tariffs and Smart Meters Starting 2025*. URL: <https://eupd-group.com/germany-to-transform>

- energy- usage- with- dynamic- tariffs- and- smart- meters- starting- 2025/ (visited on 11/11/2025).
- European Commission (2025). *Capacity mechanisms*. URL: https://energy.ec.europa.eu/topics/markets-and-consumers/capacity-mechanisms_en (visited on 09/24/2025).
- FfE - Forschungsstelle für Energiewirtschaft e. V. (2025). *How is the German Smart Meter Rollout Progressing?* URL: <https://www.ffe.de/en/publications/how-is-the-german-smart-meter-rollout-progressing/> (visited on 11/11/2025).
- Figgenger, Jan et al. (2025a). *Battery Charts*. URL: <https://battery-charts.de/battery-charts/> (visited on 09/24/2025).
- Figgenger, Jan et al. (2025b). *Battery Charts - Revenue Index*. URL: <https://battery-charts.de/revenue-index/> (visited on 09/24/2025).
- flexa GmbH (2025). *The Virtual Power Plant*. URL: <https://www.flexa.energy/> (visited on 09/24/2025).
- Fraunhofer ISE (2021). *Wege zu einem klimaneutralen Energiesystem – Version 2020/21*. Fraunhofer-Institut für Solare Energiesysteme ISE. URL: <https://www.ise.fraunhofer.de/de/veroeffentlichungen/studien/wege-zu-einem-klimaneutralen-energiesystem-version-2020-21.html> (visited on 04/17/2025).
- Fraunhofer ISE (2024). *Wege zu einem klimaneutralen Energiesystem - Version 2024*. Fraunhofer-Institut für Solare Energiesysteme ISE. URL: <https://www.ise.fraunhofer.de/de/veroeffentlichungen/studien/wege-zu-einem-klimaneutralen-energiesystem.html> (visited on 04/17/2025).
- Fraunhofer ISI et al. (2024). *Langfristszenarien für die Transformation des Energiesystems in Deutschland*. URL: <https://langfristszenarien.de/enertile-explorer-de/dokumente/> (visited on 04/17/2025).
- Geladen Podcast (2025a). *340GW Großbatterien! Was bringen XXL-Akkus wirklich? H.Urban U.Bürger (EcoStor)*. Podcast episode on YouTube. URL: <https://www.youtube.com/watch?v=W6RYeD3aFfA> (visited on 09/24/2025).
- Geladen Podcast (2025b). *Max Fichtner - Batterie Forschung von A bis Z – AKKU Wissen dass du WIRKLICH brauchst!* Podcast episode on YouTube. URL: <https://www.youtube.com/watch?v=fHm8gBQeiv4&t=101s> (visited on 09/24/2025).
- Greitemeier, Tim et al. (2025). “China’s hold on the lithium-ion battery supply chain: Prospects for competitive growth and sovereign control”. In: *Journal of Power Sources*

- Advances* 32, p. 100173. DOI: 10.1016/j.powera.2025.100173. URL: <https://www.sciencedirect.com/science/article/pii/S2666248525000071> (visited on 09/24/2025).
- Grimm, Rico (2024a). *Das Killerfeature der neuen Batterie-Großspeicher*. URL: <https://www.cleantech.ing/p/das-killerfeature-der-neuen-batteriegrospeicher> (visited on 09/24/2025).
- Grimm, Rico (2024b). *Wie viel Speicherkapazität Deutschland brauchen wird*. URL: <https://www.cleantech.ing/p/wie-viel-speicherkapazitt-deutschland-brauchen-wird> (visited on 09/24/2025).
- Grimm, Rico (2025). *Anfragentsunami - so viele Batteriespeicher werden wirklich kommen*. Cleantech Ing. URL: <https://www.cleantech.ing/p/anfragentsunami-so-viele-batteriespeicher-werden-wirklich-kommen> (visited on 11/11/2025).
- Guéret, Adeline, Wolf-Peter Schill, and Felix Schmidt (2025). *Effects of system-blind prosumers in energy models*. URL: <https://arxiv.org/abs/2505.14186> (visited on 09/24/2025).
- Hörsch, Jonas et al. (2018). “PyPSA-Eur: An open optimisation model of the European transmission system”. In: *Energy Strategy Reviews* 22, pp. 207–215. DOI: 10.1016/j.esr.2018.08.012. URL: <https://doi.org/10.1016/j.esr.2018.08.012> (visited on 04/17/2025).
- International Energy Agency (2025). *Share of Electric Vehicle Sales by Battery Chemistry in Selected Regions, 2022-2024*. URL: <https://www.iea.org/data-and-statistics/charts/share-of-electric-vehicle-sales-by-battery-chemistry-in-selected-regions-2022-2024> (visited on 11/11/2025).
- Jürgens, Patrick et al. (2024). *Flexibilität im deutschen Energiesystem bis 2045: Beitrag verschiedener Technologien auf dem Weg zur Klimaneutralität*. Ariadne-Analyse. Kopernikus-Projekt Ariadne, Potsdam-Institut für Klimafolgenforschung (PIK). DOI: 10.48485/pik.2024.003. URL: <https://doi.org/10.48485/pik.2024.003> (visited on 04/24/2025).
- Kopernikus-Projekt Ariadne (2021). *Deutschland auf dem Weg zur Klimaneutralität 2045: Szenarien und Pfade im Modellvergleich*. Kopernikus-Projekt Ariadne, Potsdam-Institut für Klimafolgenforschung (PIK). DOI: 10.48485/pik.2021.006. URL: <https://doi.org/10.48485/pik.2021.006> (visited on 04/17/2025).
- Lichner, Cornelia (2025). *Marktübersicht für Groß- und Gewerbespeicher zeigt 390 Produkte und Preisindikation*. URL: <https://www.pv-magazine.de/2025/02/19/pv-magazine-marktuebersicht-fuer-gross-und-gewerbespeicher-zeigt-390-produkte/> (visited on 04/17/2025).

- Luderer, Gunnar et al. (2025). *Die Energiewende kosteneffizient gestalten: Szenarien zur Klimaneutralität 2045*. Tech. rep. Kopernikus-Projekt Ariadne, Potsdam-Institut für Klimafolgenforschung (PIK). DOI: 10.48485/pik.2025.003. URL: <https://doi.org/10.48485/pik.2025.003> (visited on 04/17/2025).
- Maisch, Marija (2024). *Sodium-ion batteries – a viable alternative to lithium?* URL: <http://www.pv-magazine.com/2024/03/22/sodium-ion-batteries-a-viable-alternative-to-lithium/> (visited on 09/24/2025).
- Moove Podcast (2025). *Wann liefert das E-Auto endlich Strom für die Wärmepumpe? - Manfred Przybilla, Ambibox (194)*. Podcast episode on YouTube. URL: <https://www.youtube.com/watch?v=C1kSCRrTw-4&t=40s> (visited on 09/24/2025).
- Murray, Cameron (2025). *‘It is not technology-agnostic’: Will the Capacity Market in Germany help or hurt BESS?* URL: <https://www.energy-storage.news/it-is-not-technology-agnostic-will-capacity-market-in-germany-help-or-hurt-bess/> (visited on 09/24/2025).
- Next Kraftwerke (2025a). *How does Day-Ahead Trading of Electricity work?* URL: <https://www.next-kraftwerke.be/knowledge-hub/day-ahead-trading> (visited on 09/24/2025).
- Next Kraftwerke (2025b). *Intraday Trading of Electricity*. URL: <https://www.next-kraftwerke.com/knowledge/intraday-trading> (visited on 09/24/2025).
- Octopus Electric Vehicles (2025). *Power Pack Bundle*. URL: <https://octopusev.com/power-pack-bundle> (visited on 09/24/2025).
- Pant, Prashant, Thomas Hamacher, and Vedran S. Perić (2025). “Ancillary Services in Germany: Present, Future and the Role of Battery Storage Systems”. In: *TechRxiv*. DOI: 10.36227/techrxiv.171177336.66870344/v2. URL: <https://www.techrxiv.org/users/714191/articles/734392-ancillary-services-in-germany-present-future-and-the-role-of-battery-storage-systems> (visited on 09/24/2025).
- PyPSA Developers (2025a). *Github PyPSA*. URL: <https://github.com/PyPSA/pypsa> (visited on 04/22/2025).
- PyPSA Developers (2025b). *Github PyPSA-DE*. URL: <https://github.com/PyPSA/pypsa-de> (visited on 04/22/2025).
- PyPSA Developers (2025c). *Github PyPSA-EUR*. URL: <https://github.com/PyPSA/pypsa-eur> (visited on 04/22/2025).

- PyPSA Developers (2025d). *PyPSA-Eur: A Sector-Coupled Open Optimisation Model of the European Energy System*. URL: <https://pypsa-eur.readthedocs.io/en/latest/> (visited on 04/22/2025).
- PyPSA Developers (2025e). *PyPSA: Python for Power System Analysis*. URL: <https://pypsa.readthedocs.io/en/latest/> (visited on 04/22/2025).
- Regelleistung.net (2025). *Welche Arten der Regelreserve gibt es?* URL: <https://www.regelleistung.net/de-de/Grundlagen/Welche-Arten-der-Regelreserve-gibt-es> (visited on 09/24/2025).
- RTE - Réseau de Transport d'Électricité (2015). *Bilan Prévisionnel - de l'équilibre offre-demande d'électricité en France – Édition 2015*. Tech. rep. RTE – Réseau de Transport d'Électricité. URL: https://assets.rte-france.com/prod/public/2024-07/bp2015_0.pdf (visited on 09/24/2025).
- Rühs, Marc (2025). *Capacity Market: Future-proof Design Instead of a Step Back into Old Structures*. URL: <https://www.next-kraftwerke.com/energy-blog/capacity-market-future-proof-design> (visited on 09/24/2025).
- Schill, Wolf-Peter et al. (2025). *Welche Rolle spielen Speicher in der Energiewende?* Ariadne background report. Kopernikus-Projekt Ariadne, Potsdam-Institut für Klimafolgenforschung (PIK). DOI: 10.48485/pik.2025.004. URL: <https://doi.org/10.48485/pik.2025.004> (visited on 04/24/2025).
- Schram, Wouter et al. (2020). “Empirical Evaluation of V2G Round-trip Efficiency”. In: *3rd International Conference on Smart Energy Systems and Technologies (SEST 2020)*, pp. 1–6. DOI: 10.1109/SEST48500.2020.9203459. URL: https://ssc-fleet.nl/wp-content/uploads/sites/358/2023/10/2020_Empirical_Evaluation_of_V2G_Round-trip_Efficiency.pdf (visited on 11/11/2025).
- Shaw, Vincent (2025). *China launches world's first grid-forming sodium-ion battery storage plant*. URL: <https://www.pv-magazine.com/2025/06/03/china-launches-worlds-first-grid-forming-sodium-ion-battery-storage-plant/> (visited on 09/24/2025).
- Solarserver (2024). *Enpal startet neues virtuelles Kraftwerk*. URL: <https://www.solarserver.de/2024/11/05/enpal-startet-neues-virtuelles-kraftwerk/> (visited on 09/24/2025).
- Straub, Florian et al. (2023). “Sector Coupling through Vehicle to Grid: A Case Study for Electric Vehicles and Households in Berlin, Germany”. In: *World Electric Vehicle Journal* 14.3, p. 77. DOI: 10.3390/wevj14030077. URL: <https://www.mdpi.com/2032-6653/14/3/77> (visited on 09/24/2025).

- Stromsteuergesetz (2022). *StromStG - § 5 Entstehung der Steuer, Steuerschuldner*. URL: https://www.gesetze-im-internet.de/stromstg/___5.html (visited on 11/11/2025).
- Thimet, P. J. and G. Mavromatidis (2023). “What-where-when: Investigating the role of storage for the German electricity system transition”. In: *Applied Energy* 351, p. 121764. DOI: 10.1016/j.apenergy.2023.121764. URL: <https://www.sciencedirect.com/science/article/pii/S0306261923011285> (visited on 04/17/2025).
- Westerheide, Carla (2025). *Octopus Energy and BYD launch UK’s first V2G bundle*. URL: <https://www.electrive.com/2025/06/24/octopus-energy-and-byd-launch-uks-first-v2g-bundle/> (visited on 09/24/2025).
- Williams, Zach (2025). *Germany: Europe’s biggest merchant market – but why isn’t investment pouring in?* URL: <https://modoenergy.com/research/germany-bess-merchant-case-investment-bottlenecks-june-2025> (visited on 09/24/2025).
- Zhang, Cong et al. (2025). “Economic Viability of Vehicle-to-Grid (V2G) Reassessed: A Degradation Cost Integrated Life-Cycle Analysis”. In: *Sustainability* 17.12, p. 5626. DOI: 10.3390/su17125626. URL: <https://www.mdpi.com/2071-1050/17/12/5626> (visited on 11/11/2025).

A. Appendix

This section presents all additional referenced figures from Chapters 3 and 4, illustrating the model results and providing further insights and supplementary information.

An interactive online dashboard containing all main results and visualizations of this thesis is available at:

<https://pypsa-de-9rkxgibub4ajnid82uvsli.streamlit.app/>

The version of the PyPSA-DE model used for this thesis, including all modifications, is publicly accessible at:

<https://github.com/HeydersHub/PyPSA-DE-MA-Model.git>

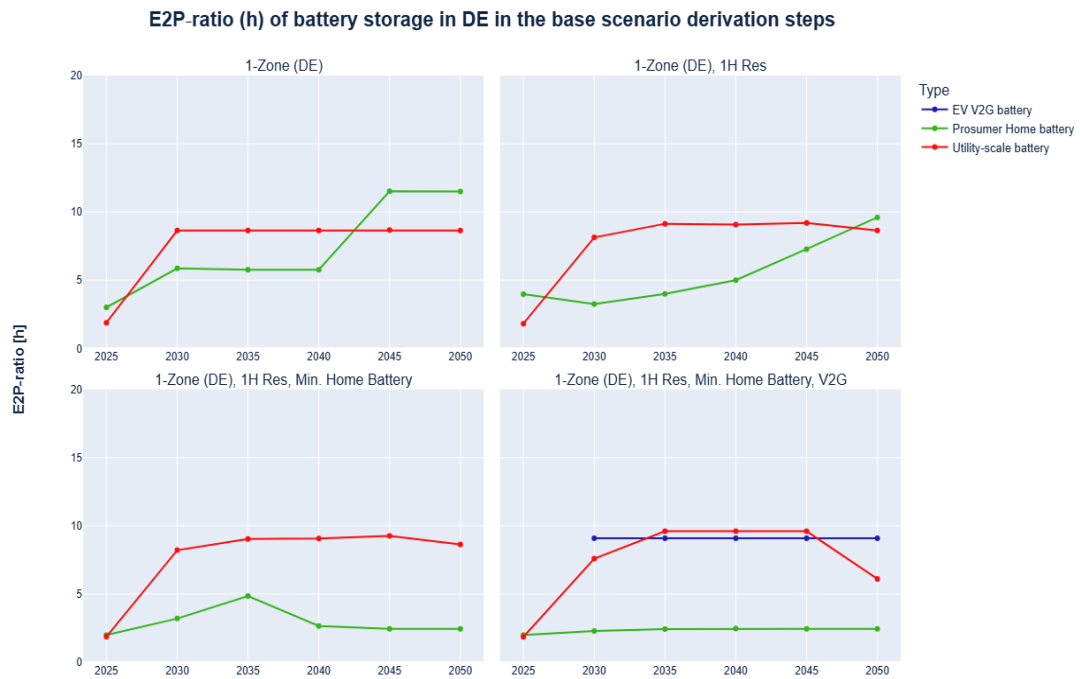


Figure A.1.: PyPSA-DE results of the E2P ratios of batteries across target years in the Base Scenario derivation steps

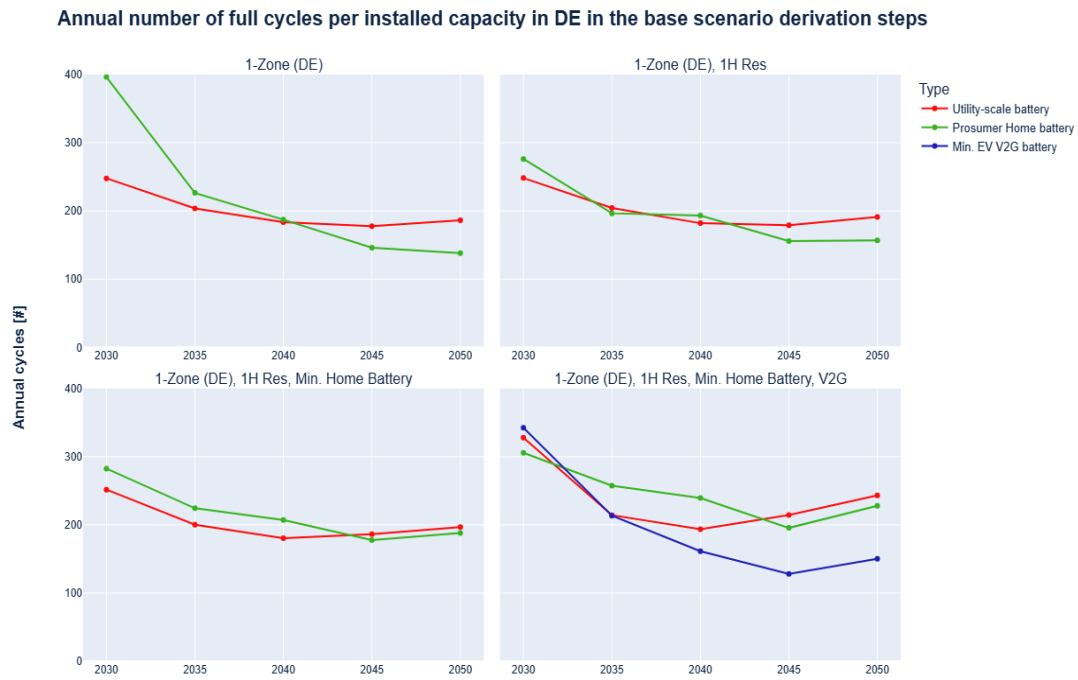


Figure A.2.: PyPSA-DE results of the annual full cycles of batteries across target years in the Base Scenario derivation steps

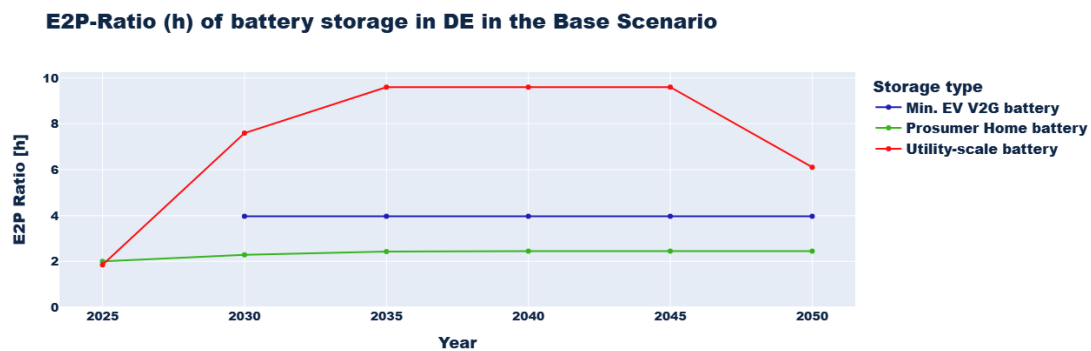


Figure A.3.: PyPSA-DE Base Scenario results of the E2P-ratio across target years

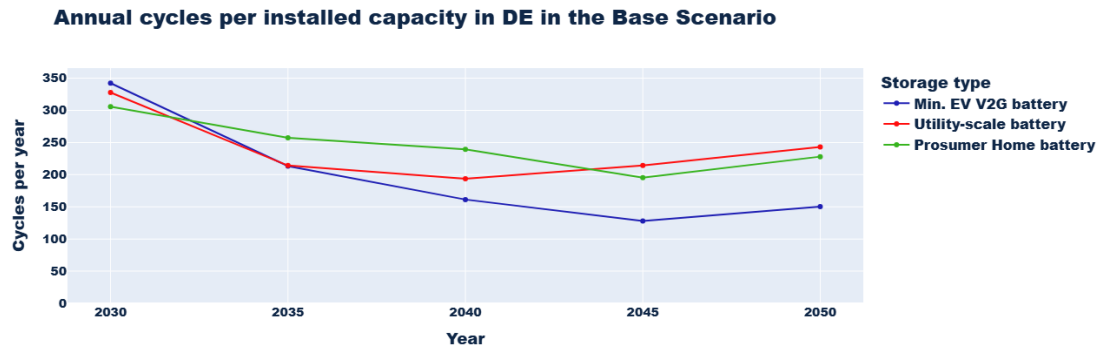


Figure A.4.: PyPSA-DE Base Scenario results of full annual cycles across target years

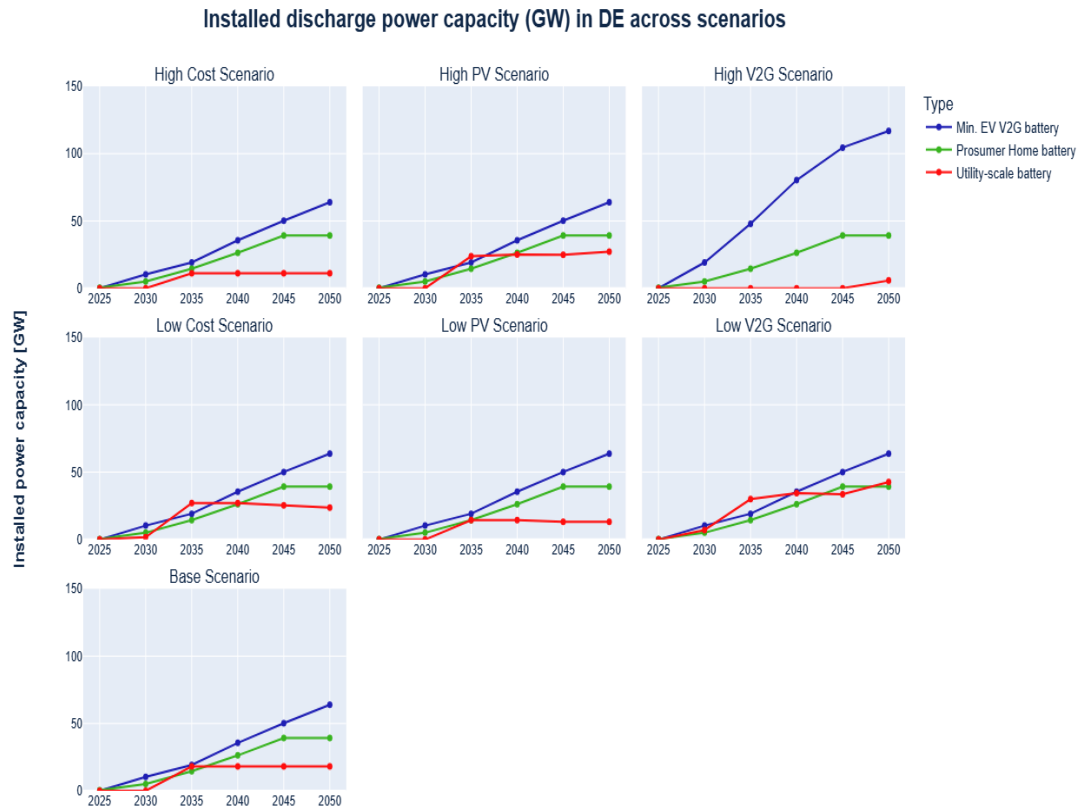


Figure A.5.: PyPSA-DE results of installed discharge power capacity across target years across scenarios

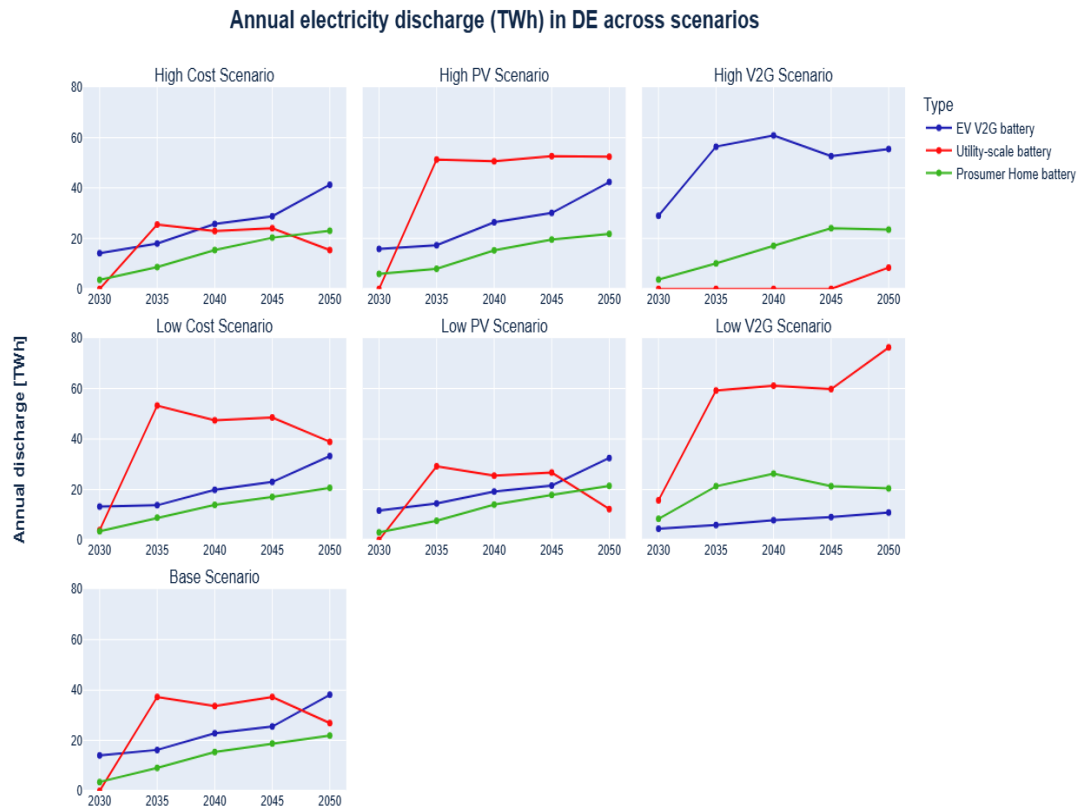


Figure A.6.: PyPSA-DE results of annual electricity supply across target years across scenarios



Figure A.7.: PyPSA-DE results of the E2P ratio across target years across scenarios



Figure A.8.: PyPSA-DE results of the annual full cycles across target years across scenarios

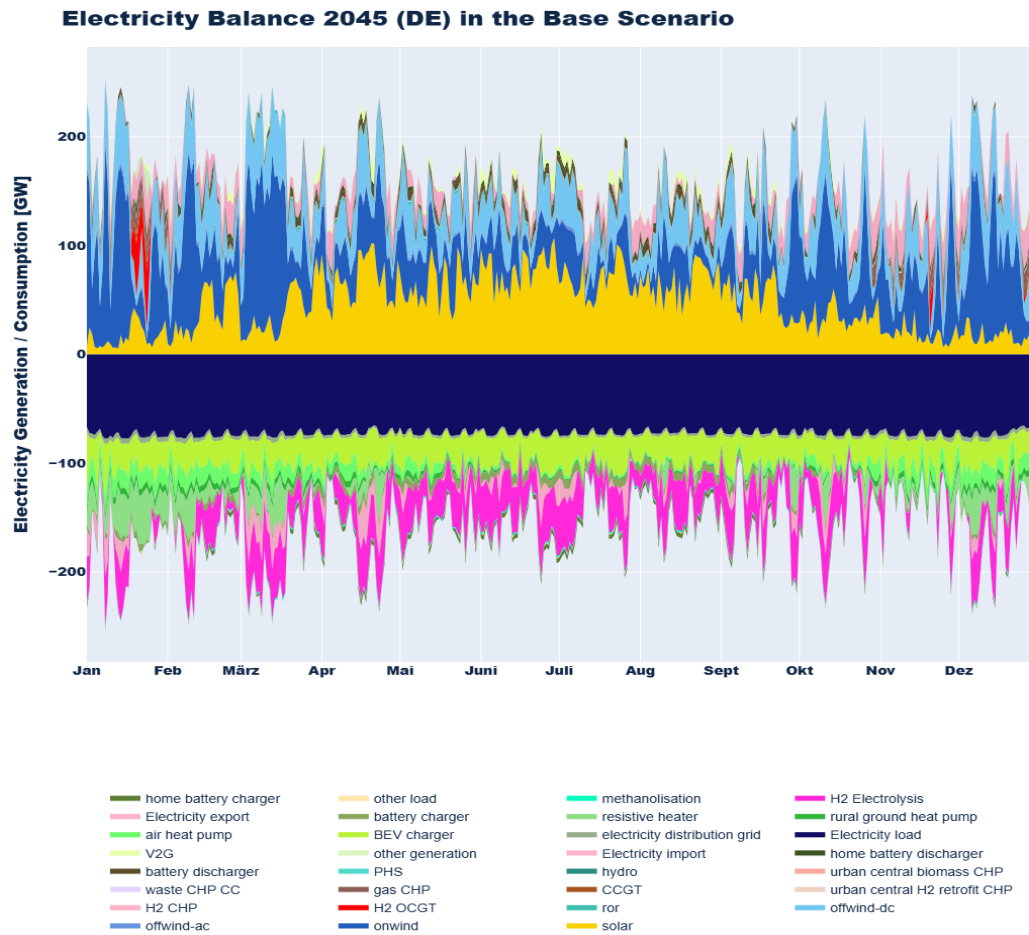


Figure A.9.: PyPSA-DE results of the annual electricity balance in DE in 2045 in the Base Scenario

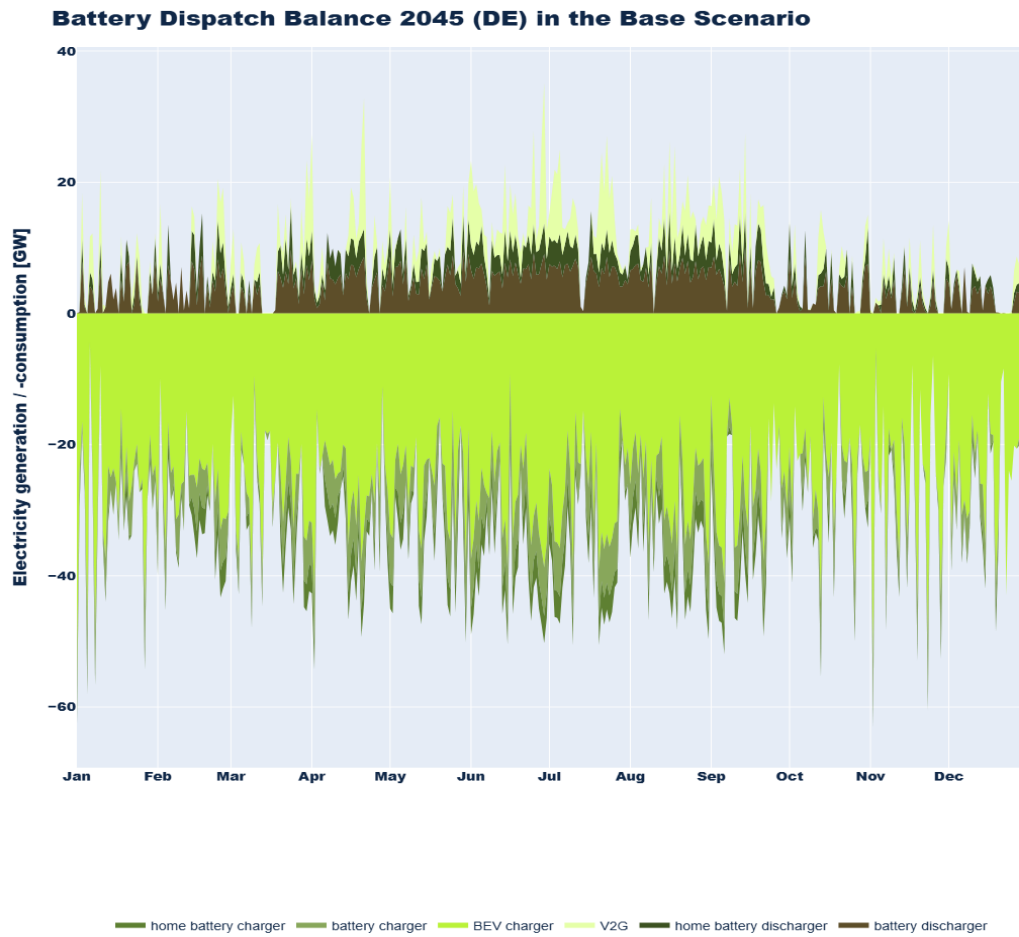


Figure A.10.: PyPSA-DE results of the annual battery dispatch 2045 in the Base Scenario

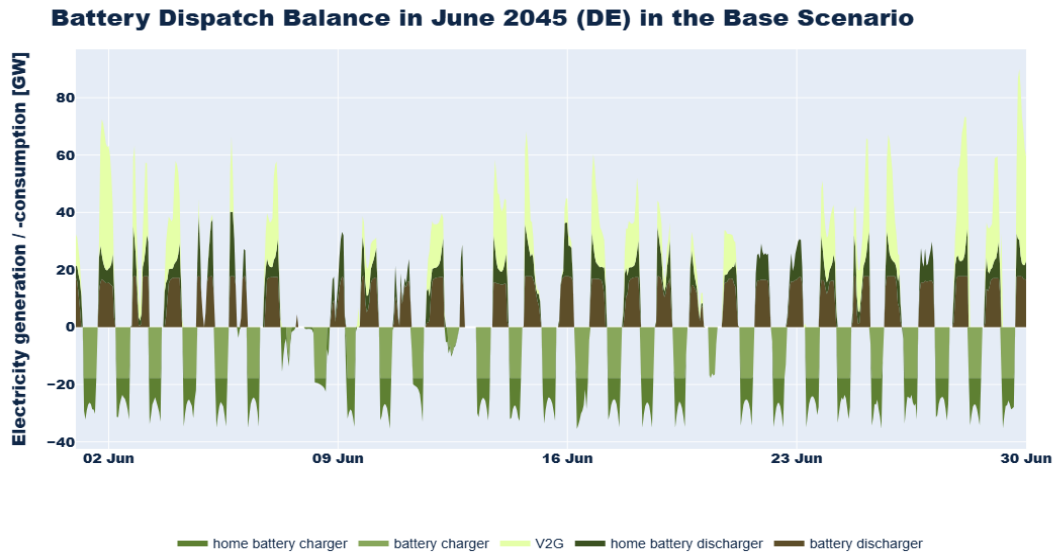


Figure A.11.: PyPSA-DE results of the monthly battery dispatch in June 2045 in the Base Scenario (without BEV charger)

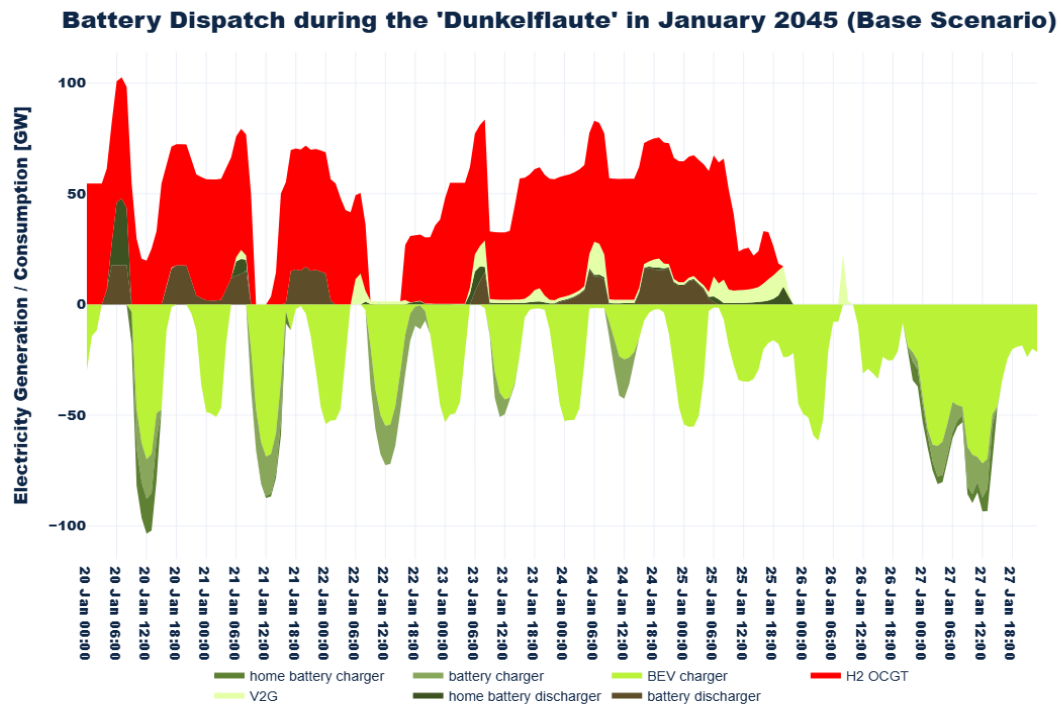


Figure A.12.: PyPSA-DE results of the battery dispatch during the 'Dunkelflaute' in January 2045 in the Base Scenario

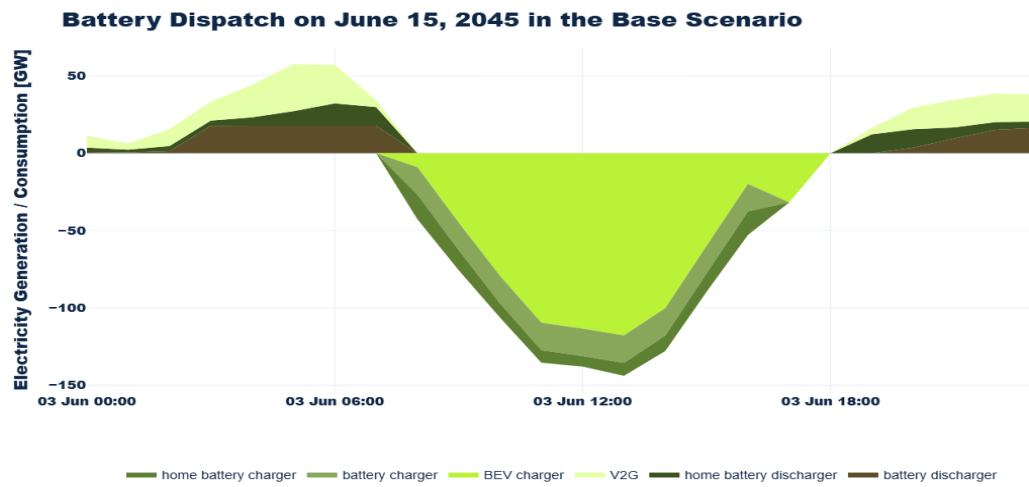


Figure A.13.: PyPSA-DE results of the battery dispatch in June 15, 2045, in the Base Scenario

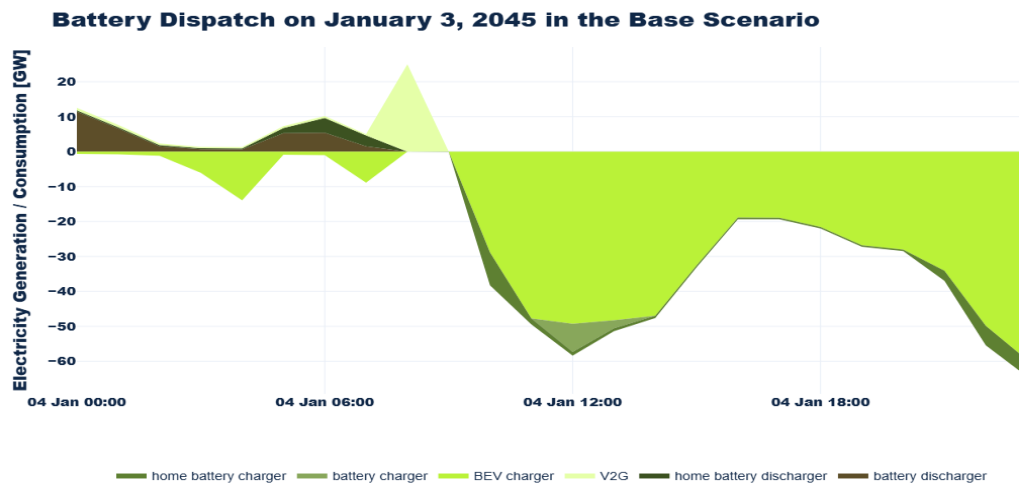


Figure A.14.: PyPSA-DE results of the battery dispatch in January 3, 2045, in the Base Scenario

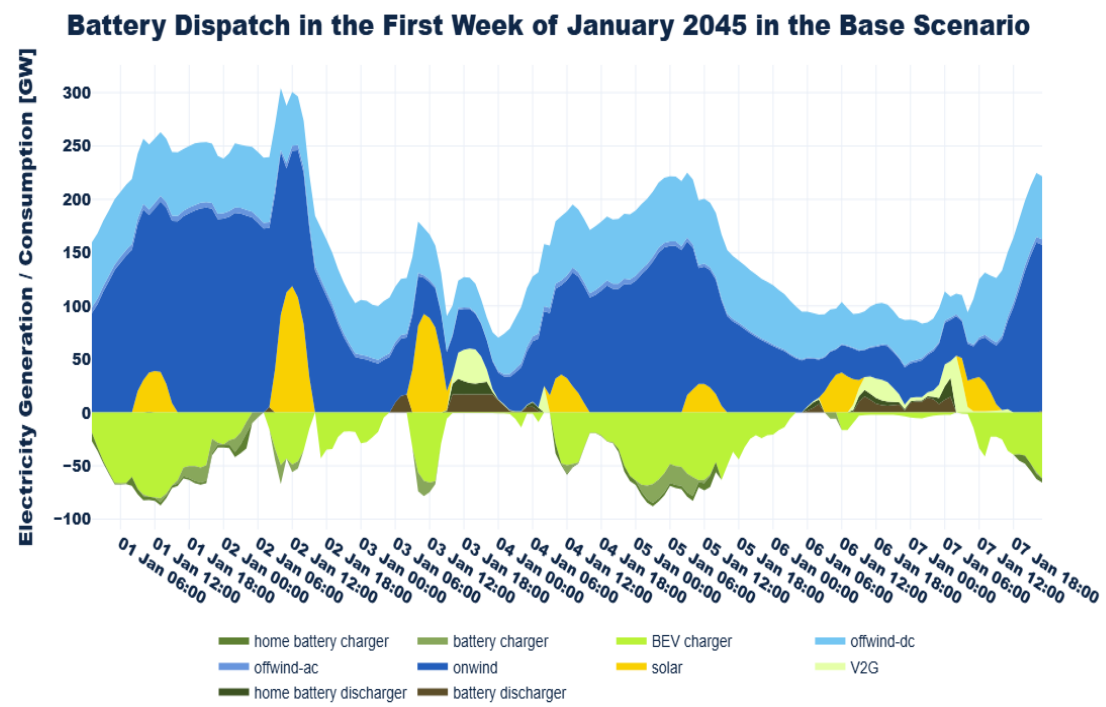


Figure A.15.: PyPSA-DE results of the battery, PV and wind dispatch in the first week of January, 2045, in the Base Scenario

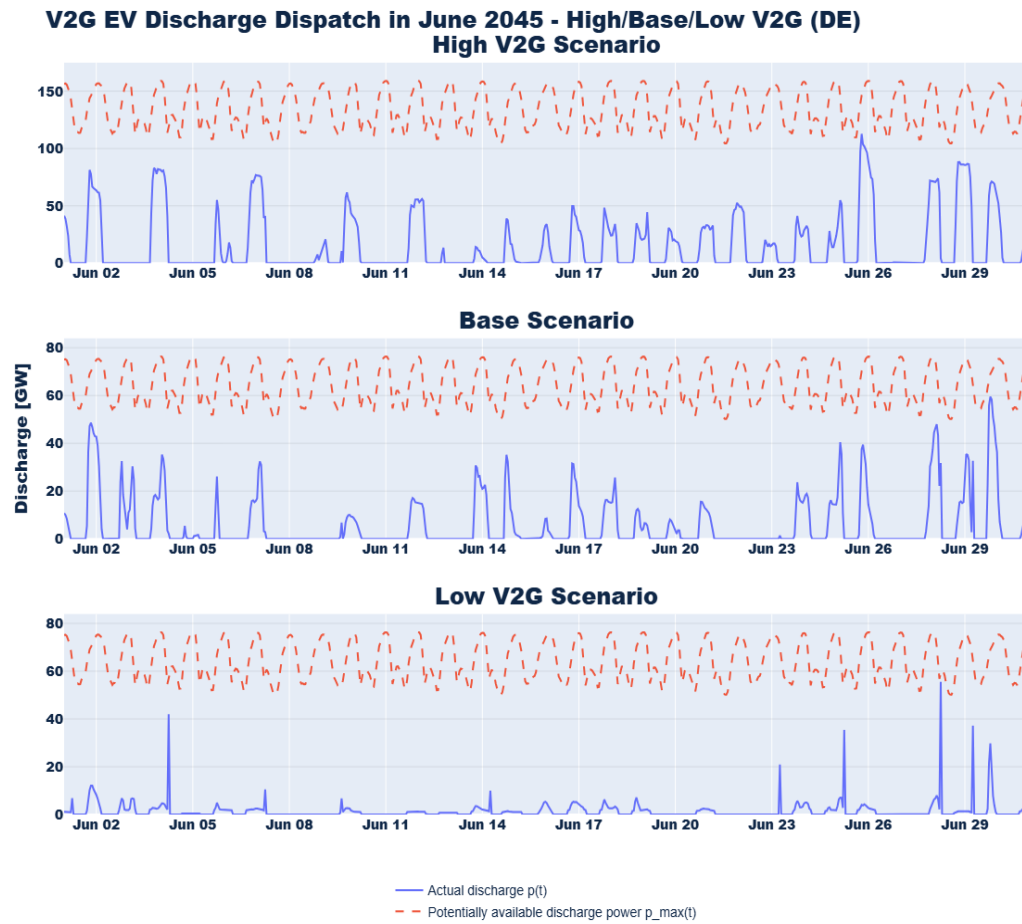


Figure A.16.: PyPSA-DE results of V2G discharge dispatch in the High V2G-, Low V2G-, and Base Scenario in June 2045

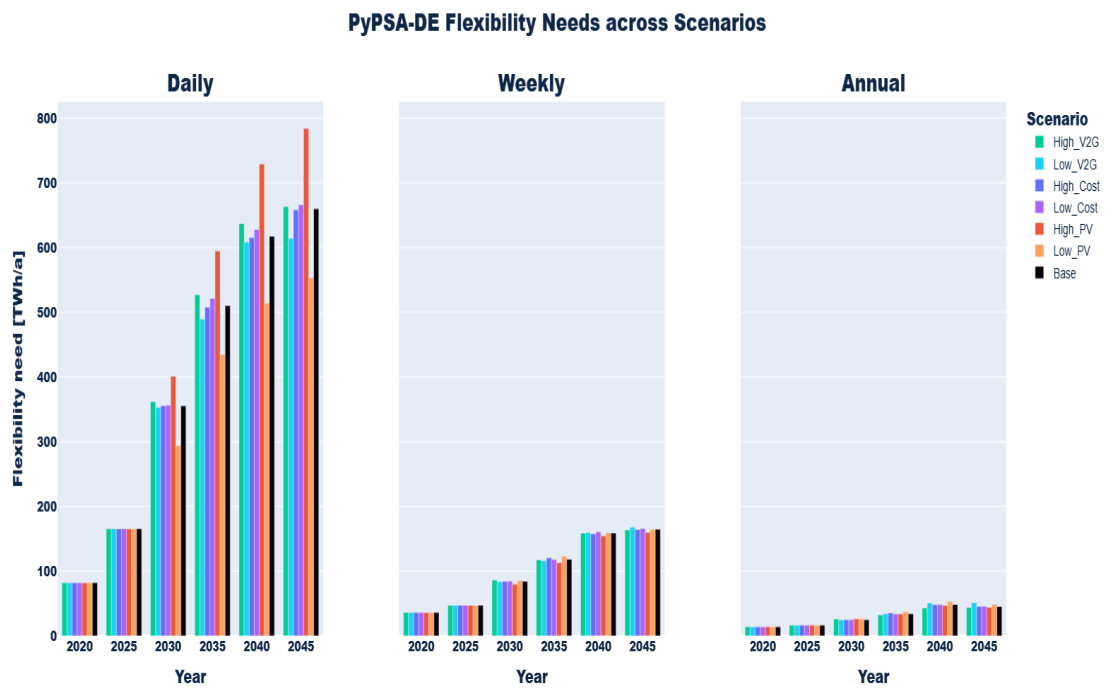


Figure A.17.: PyPSA-DE flexibility needs of different timescales in DE across scenarios

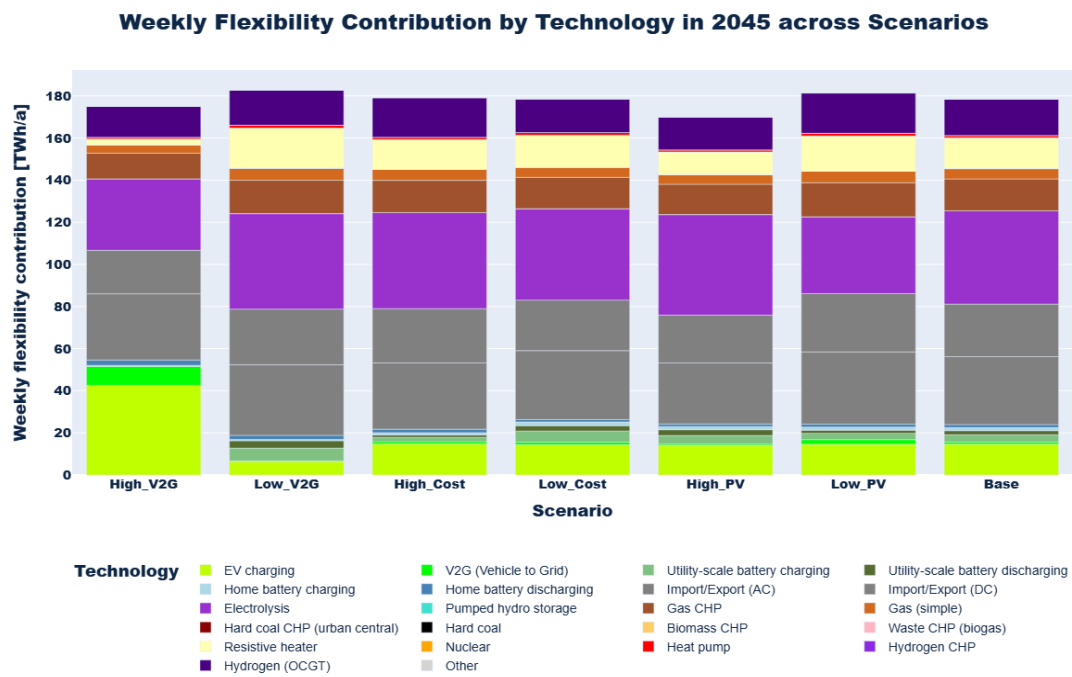


Figure A.18.: PyPSA-DE Weekly flexibility contributions of different technologies in DE across Scenarios in 2045

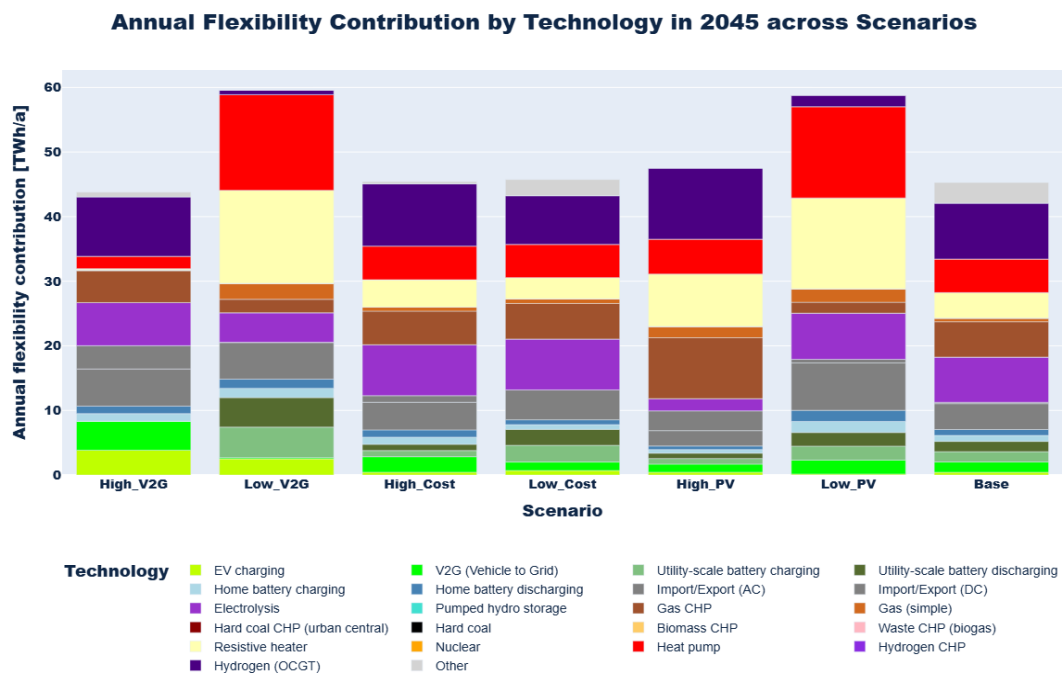


Figure A.19.: PyPSA-DE annual flexibility contributions of different technologies in DE across scenarios in 2045

Weekly Flexibility Contribution by Technology in 2045 across Battery Cost Sensitivities

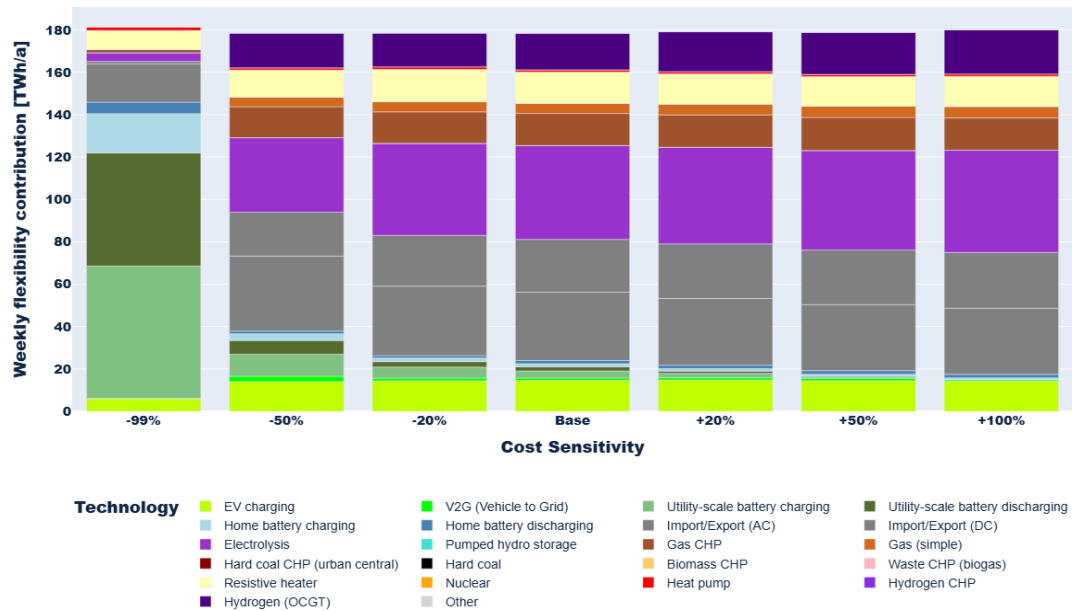


Figure A.20.: PyPSA-DE contributions of different technologies to the weekly flexibility needs across battery cost sensitivities in DE in 2045

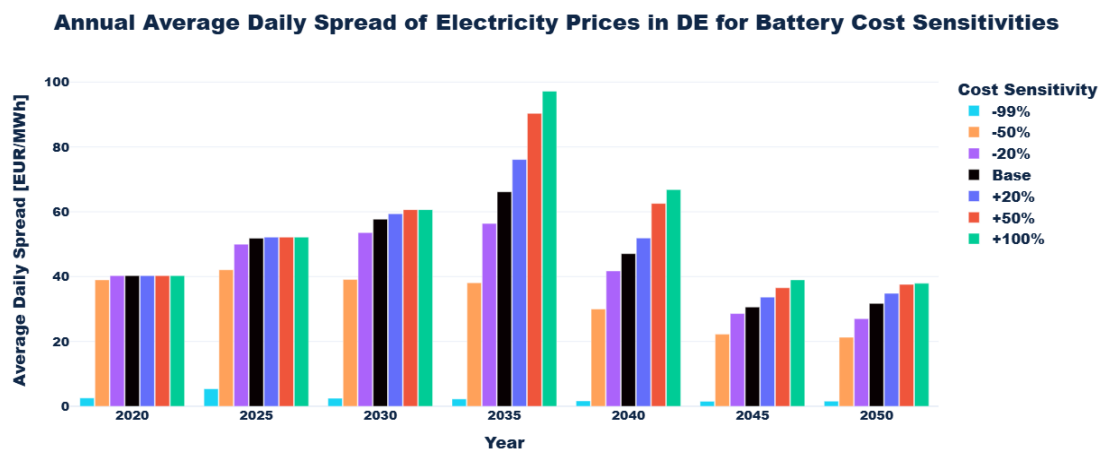


Figure A.21.: PyPSA-DE electricity price daily spreads across battery cost sensitivities

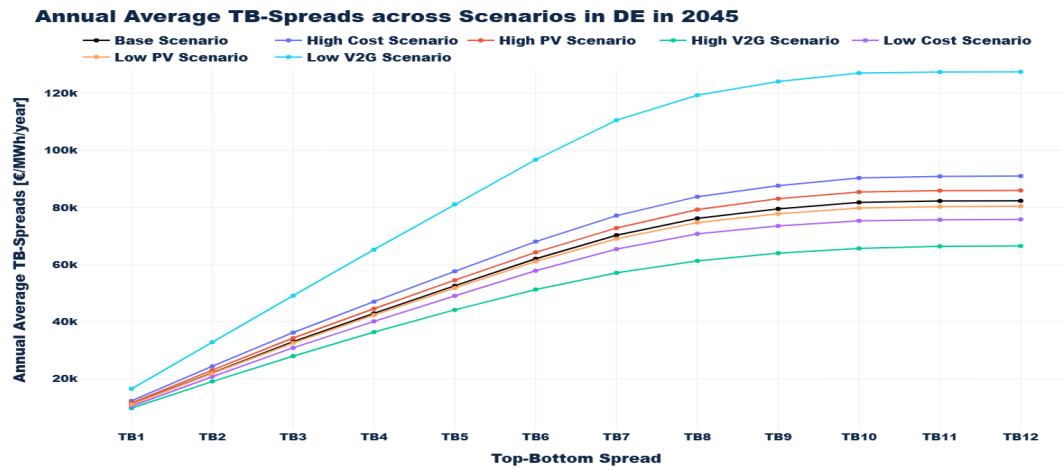


Figure A.22.: PyPSA-DE annual top-bottom spreads of electricity prices across scenarios in 2045

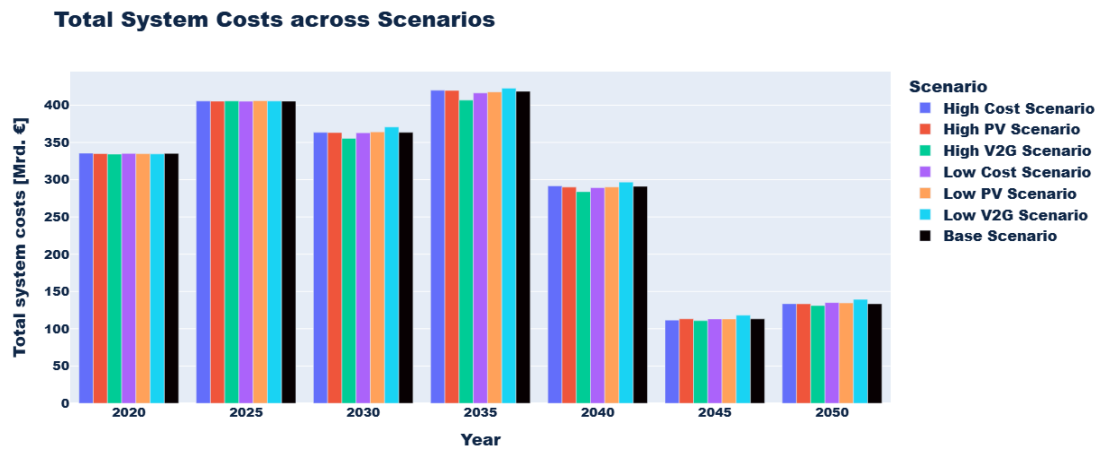


Figure A.23.: PyPSA-DE annual total system costs in DE across scenarios

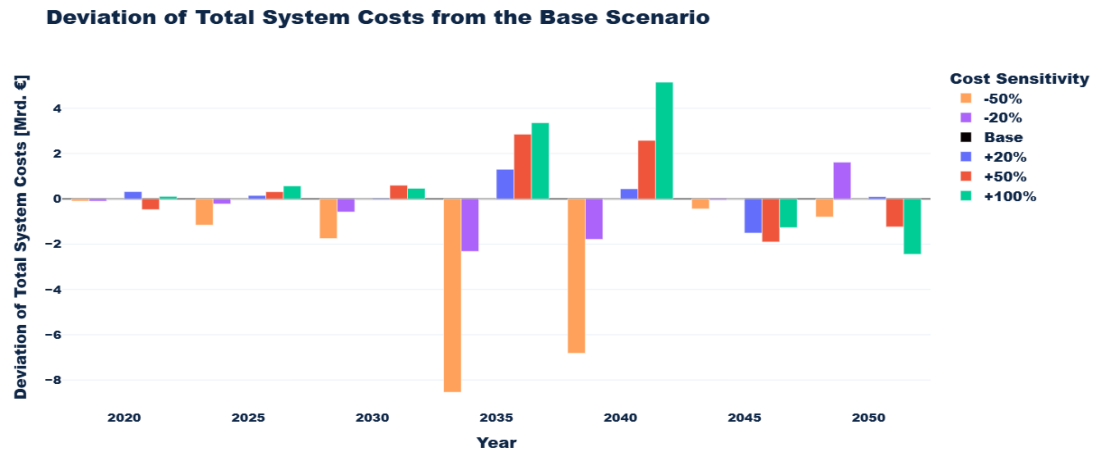


Figure A.24.: PyPSA-DE deviations of total system costs from the base scenario across battery cost sensitivities

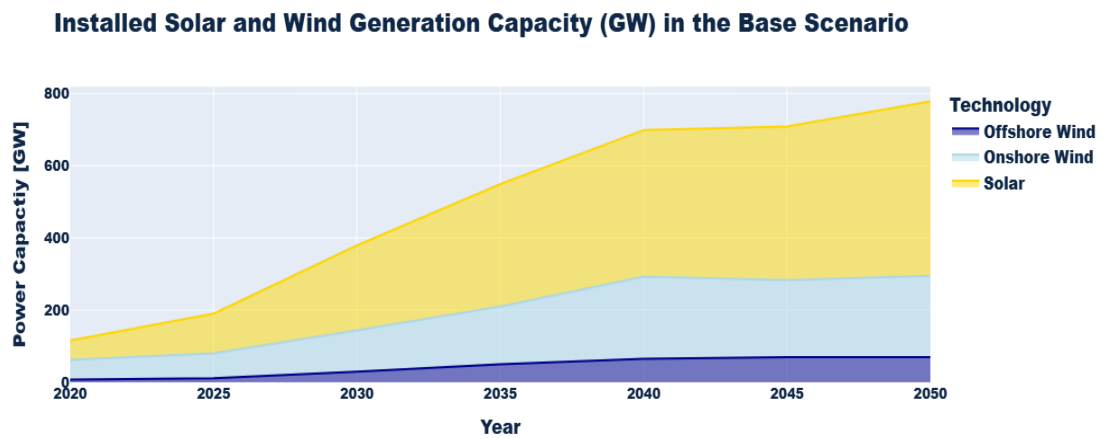


Figure A.25.: PyPSA-DE installed PV and wind capacities in DE in the Base Scenario

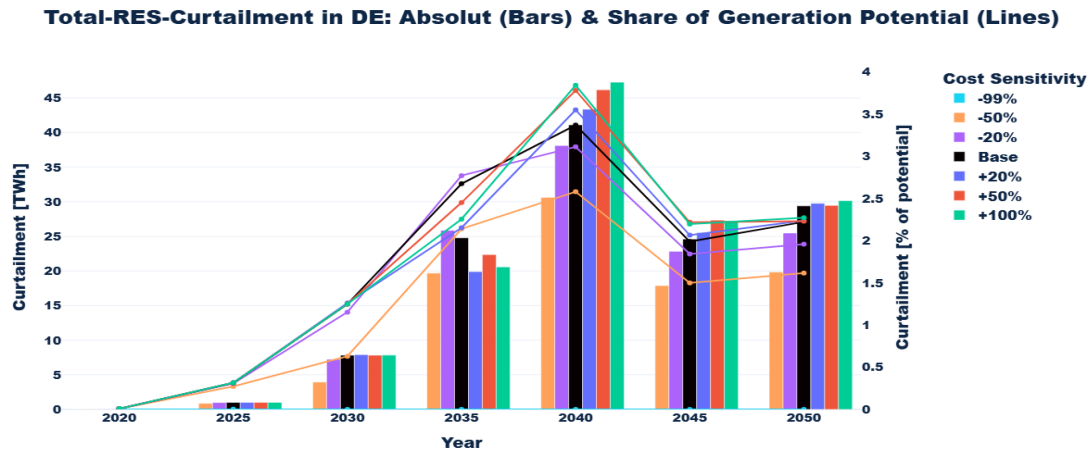


Figure A.26.: PyPSA-DE RES curtailment across battery cost sensitivities

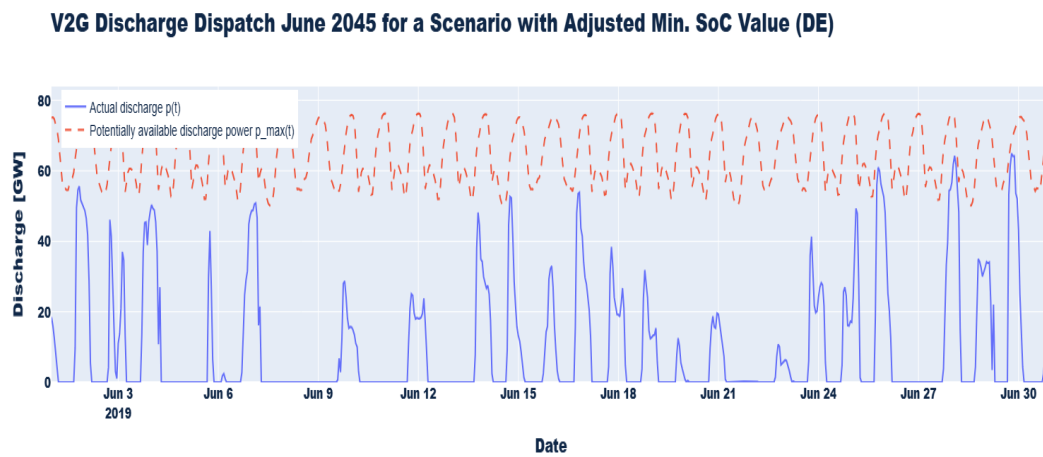


Figure A.27.: PyPSA-DE results of V2G discharge dispatch in the adjusted min. SoC scenario in June 2045

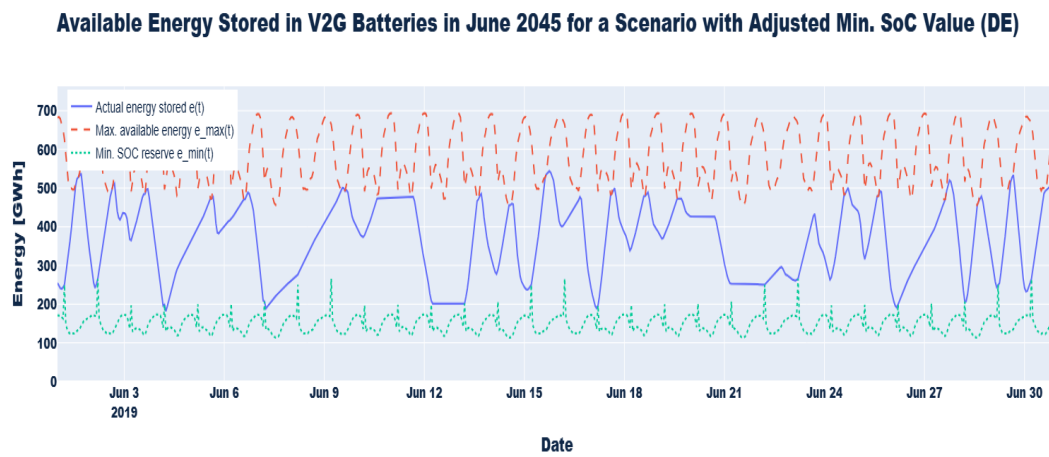


Figure A.28.: PyPSA-DE results of energy stored in the V2G battery in the adjusted min. SoC scenario in June 2045